BASUCHANDRA'S

BIOLOGY FOR ENGINEERS

As per new Scheme of
Visvesvaraya Technological University (VTU)
Effective from the academic year 2021-22

Dr. Chandra Shekara B. M.
M.Sc., Ph.D

Dr. Basavaraju B. C.
M.Sc., Ph.D

For IV Semester BE students of all branches

BASUCHANDRA'S BIOLOGY FOR ENGINEERS

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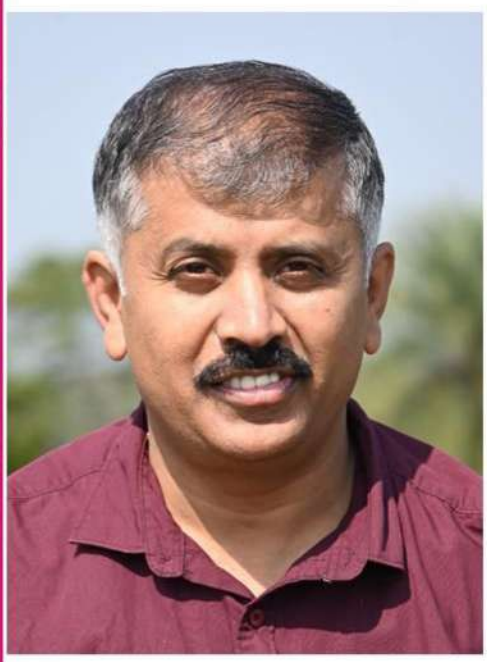

Dr. Chandra Shekara B.M. obtained M.Sc degree with specialization in inorganic chemistry in 1998 from University of Mysore. He obtained Ph.D degree in 2012 from Visveswaraya Technological University in the field of Heterogeneous catalysis. He has published number of research papers in reputed national and international journals. He has been working in Bangalore Institute of Technology since 1998. He is having teaching experience of 24 years.

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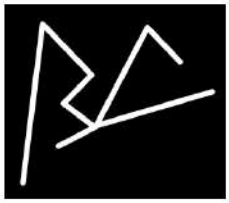

BASUCHANDRA'S

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For IV Semester Engineering Students of EEE, Mechanical, Civil, CSE and allied branches

As per new Scheme of

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Effective from the academic year 2022-23.

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BASUCHANDRA'SBIOLOGY FOR ENGINEERS, **for IV Semester Engineering Students of EEE, Mechanical, Civil, CSE and allied branches,** As per new Scheme of **Visvesvaraya Technological University (VTU)**, Effective from the academic year 2022-23.

By Dr. Chandra Shekara B. M.and Dr.Basavaraju. B. C.

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SYLLABUS

BIOLOGY FOR ENGINEERS

(Covers the syllabus of EEE, Civil, Mechanical, CSE and allied stream branches of engineering)

As per new Scheme of **Visvesvaraya Technological University (VTU)**,

(Effective from the academic year 2022-23)

MODULE- I

INTRODUCTION TO BIOLOGY

CELL: BASIC UNIT OF LIFE

Introduction. The cell: the basic unit of life, Structure and functions of a cell. The Plant Cell and animal cell, Prokaryotic and Eukaryotic cell, Stem cells and their application.

BIOMOLECULES

Properties and functions of Carbohydrates, Nucleic acids, proteins, lipids. Importance of special biomolecules; Enzymes, Classification with one example each. Properties and functions of enzymes, vitamins and hormones.

(8 hours)

MODULE- II

BIOMOLECULES AND THEIR APPLICATIONS (QUALITATIVE)

Carbohydrates in cellulose-based water filters production, PHA and PLA in bioplastics production, Nucleic acids in vaccines and diagnosis (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting), , Proteins in food production (Proteins as food – whey protein and meat analogs, Plant based proteins), , lipids in biodiesel and cleaning agents/detergents production, Enzymes in biosensors fabrication (glucose-oxidase in biosensors) Enzymes in textile processing (lignolytic enzyme in bio-bleaching). Enzymes in food processing, and detergent formulation.

(8 hours)

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MODULE- III

ADAPTATION OF ANATOMICAL PRINCIPLES OF HUMAN ORGAN SYSTEMS FOR BIOENGINEERING DESIGN (QUALITATIVE)

Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for Parkinson's disease). Eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye). Heart as a pump system (architecture, electrical signaling - ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pace makers, defibrillators). Lungs as purification system (architecture, gas exchange mechanisms, spirometry, abnormal lung physiology - COPD, Ventilators, Heart-lung machine). Kidney as a filtration system (architecture, mechanism of filtration, CKD, dialysis systems). (8 hours)

MODULE- IV

NATURE-BIOINSPIRED MATERIALS AND MECHANISMS (QUALITATIVE):

Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swim suits), Kingfisher beak (Bullet train). Human Blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perflourocarbons (PFCs). (8 hours)

MODULE- V

TRENDS IN BIOENGINEERING (QUALITATIVE)

Muscular and Skeletal Systems as scaffolds (architecture, mechanisms, bioengineering solutions for muscular dystrophy and osteoporosis), scaffolds and tissue engineering, Bioprinting techniques and materials, 3D printing of ear, bone and skin. 3D printed foods. Electrical tongue and electrical nose in food science, DNA

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origami and Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis. Self- healing Bioconcrete (based on bacillus spores, calcium lactate nutrients and biomineralization processes) and Bioremediation and Biomining via microbial surface adsorption (removal of heavy metals like Lead, Cadmium, Mercury, Arsenic). (8 hours)

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MODULE- I

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CELL BASIC UNIT OF LIFE

Introduction. The cell: the basic unit of life, Structure and functions of a cell. The Plant Cell and animal cell. Prokaryotic and Eukaryotic cell. Stem cells and their application.

BIOMOLECULES

Properties and functions of Carbohydrates, Nucleic acids, proteins, lipids. Importance of special biomolecules; Enzymes, Classification with one example each. Properties and functions of enzymes, vitamins and hormones.

(8 hours)

THE CELL: THE BASIC UNIT OF LIFE

INTRODUCTION

The cell is the fundamental building block of all living beings starting from the simplest single-celled bacteria to the complex multicellular organisms like plants and animals. It carries out the essential processes necessary for the growth, development, survival and functioning of all living organisms on this planet.

In this chapter, we will briefly explore the general structure and functions of plant and animal cells and their unique characteristics. We shall also learn about prokaryotic and eukaryotic cells. Prokaryotic cells are simple and lack a nucleus found in bacteria and archaea. Eukaryotic cells are more complex containing membrane-bound organelles and found in plants, animals, fungi, and protists.

Finally, beyond these basic cell types, we will also explore the fascinating world of stem cells. These are unique cells with the ability to develop into different cell types. Due to this unique ability, stem cells are becoming invaluable in medical applications such as regenerative medicine, tissue engineering, and the treatment of diseases. Stem cells

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has become a fascinating and rapidly advancing area of research in medicine and biotechnology.

PROKARYOTIC AND EUKARYOTIC CELL

Prokaryotic and eukaryotic cells are the two fundamental types of cells found in living organisms. Prokaryotic cells are simple, single-celled organisms that lack a nucleus and membrane-bound organelles. Examples include bacteria and archaea. Eukaryotic Cells are complex cells that contain a nucleus and membrane-bound organelles. Examples include plants, animals, fungi, and protists.

Prokaryotic Cells are smaller in size typically in the range of 1–10 µm. Eukaryotic Cells are larger in size typically in the range of 10–100 µm. True nucleus is absent in Prokaryotic Cells. Genetic material (DNA) is present in a region called the nucleoid. Eukaryotic Cells have a well-defined nucleus enclosed by a nuclear membrane. Prokaryotic Cells lack membrane-bound organelles. Eukaryotic Cells contain membrane-bound organelles that perform specialized functions.

In Prokaryotic Cells, DNA is circular and not associated with proteins (no histones). It is found in the nucleoid region. In Eukaryotic Cells, DNA is linear and associated with histone proteins to form chromosomes. It is located inside the nucleus.

Prokaryotic Cells have smaller 70S ribosomes (50S + 30S subunits). Eukaryotic Cells have larger 80S ribosomes (60S + 40S subunits).

In Prokaryotic Cells, cell wall is present and made of peptidoglycan in bacteria or pseudopeptidoglycan in archaea. In Eukaryotic Cells, cell wall is present in plants and it is made of cellulose. In animal cells, cell wall is absent.

Prokaryotic Cells reproduce asexually through binary fission whereas Eukaryotic Cells reproduce both asexually through mitosis and sexually through meiosis. Prokaryotic Cells are found in Bacteria (e.g., E. coli), and Archaea (e.g., Methanogens). Eukaryotic Cells are present in plants, animals, fungi, and protists.

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Comparison between Prokaryotic and Eukaryotic Cells

While both Prokaryotic and Eukaryotic Cells are essential for life, they have significant differences in their structure, complexity, and functions. A brief comparison between two cell types is given below:

Feature	Prokaryotic Cell	Eukaryotic Cell
Nucleus	Absent – genetic material (DNA) is found in the nucleoid (an open region in the cytoplasm)	Present – DNA is enclosed within a membrane-bound nucleus
Size	Small (0.1–5 µm)	Larger (10–100 µm)
Complexity	Simple structure	Complex structure with specialized organelles
Membrane-bound Organelles	Absent	Present (e.g., mitochondria, endoplasmic reticulum, Golgi apparatus)
Cell Type	Mostly unicellular (e.g., bacteria, archaea)	Can be unicellular or multicellular (e.g., plants, animals, fungi, protists)
Cell Division	Binary fission (a simple and rapid process)	Mitosis or meiosis (more complex processes)
Ribosomes	Smaller (70S)	Larger (80S)
Cell Wall	Present in most (made of peptidoglycan in bacteria)	Present in plant and fungal cells (made of cellulose or chitin), absent in animal cells
DNA Structure	Circular, with fewer associated proteins	Linear, packed with histone proteins
Examples	Bacteria, archaea	Plants, animals, fungi, protists.

THE PLANT CELL AND ANIMAL CELL

Both plant and animal cells are eukaryotic cells. They have a nucleus and their organelles are bound by a membrane. However, there are several key differences between them, primarily due to their different functions and lifestyles. Here is a comparative account of plant and animal cells:

1. **Cell Wall:**

Plant Cells have a cell membrane which is covered by a rigid cell wall made of cellulose. Rigid cell wall provides structural support and protection.

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Animal Cells do not have a cell wall. Instead, they have only a cell membrane which is not rigid but flexible. Due to flexible cell wall, animal cells exhibit variety of cell shapes and movements.

2. Shape:

Plant Cells typically have a fixed, rectangular shape due to the rigid cell wall. Animal Cells usually have a more rounded and irregular shape.

3. Chloroplasts:

Plant Cells contain chloroplasts, which are organelles that perform photosynthesis, converting light energy into chemical energy (glucose).

Animal Cells do not have chloroplasts and hence they cannot perform photosynthesis.

4. Vacuoles:

Plant Cells have a large central vacuole that stores water, nutrients, and waste products. It also helps maintain turgor pressure against the cell wall.

Animal Cells have more number of smaller vacuoles. These vacuoles are used for storage and transport of substances.

5.Lysosomes: Lysosomes are organelles that digest cellular waste and debris using hydrolytic enzymes.

Animal Cells contain lysosomes but in Plant Cells lysosomes are less common and are not as prominent. The vacuole often takes on the role of breaking down waste materials.

6. Centrioles: Centrioles are organelles that are involved in organizing the microtubules during cell division (mitosis and meiosis).

Centrioles are present in Animal Cells but they are absent in Plant Cells. Instead, Plant Cells have other structures that help in cell division.

8. Storage of Energy:

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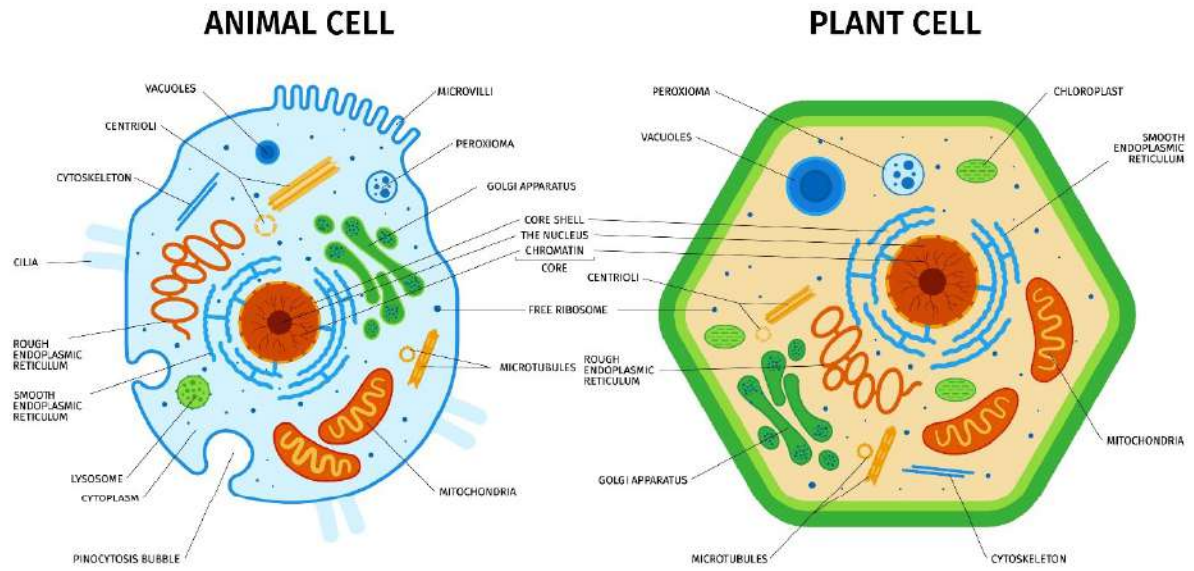
Plant Cells store energy in the form of starch. Animal Cells store energy in the form of glycogen.

9. Plastids: Plant Cells contain plastids like chloroplasts, chromoplasts, and leucoplasts which are involved in storage and photosynthesis. Animal Cells do not have plastids.



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Comparison between Plant and Animal Cells

Feature	Plant Cell	Animal Cell
Cell Wall	Present (made of cellulose) – provides structure and support.	Absent – shape maintained by cytoskeleton
Shape	Usually rectangular or square due to the rigid cell wall.	Usually round or irregular
Chloroplasts	Present – used for photosynthesis.	Absent. Hence no photosynthesis. Animals obtain energy from food.
Vacuole	Large central vacuole – stores water,	Small and numerous vacuoles –

Feature	Plant Cell	Animal Cell
	nutrients, and waste; helps maintain cell shape.	used mainly for storage.
Centrioles	Absent in most plant cells.	Present – help in cell division.
Lysosomes	Rare – functions handled by vacuole and other organelles.	Present – help break down waste and digest materials.
Plastids	Present	Absent
Mode of Nutrition	Autotrophic – makes its own food through photosynthesis.	Heterotrophic – depends on consuming food for energy.
Energy Storage	Store energy in the form of starch	Store energy in the form of glycogen

Stem cells and their application.

Stem cells are undifferentiated cells found in the body which have the unique, remarkable ability to develop into many different specialized types of cells. They serve as raw materials for repairing of body. This is because they can divide, renew and replenish themselves over long periods. Also, they have the potential to transform into specialized cells needed for growth, repair, or maintenance of body.

There are two main types of stem cells:

a) **Embryonic stem cells:** They come from early-stage embryos. They are incredibly versatile and can turn into any cell type in the body.
b) **Adult stem cells:** They are found in tissues like bone marrow, skin, brain. They are more limited and can turn into a limited range of cell types related to their tissue of origin.

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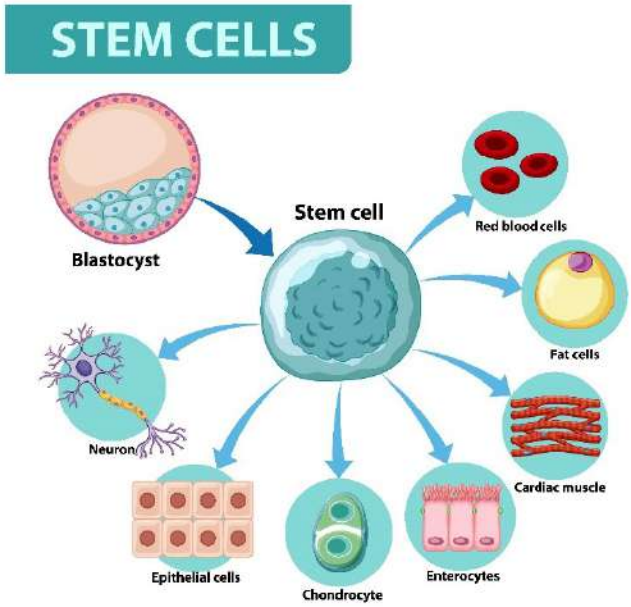

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Applications of Stem Cells

Stem cells have potential applications in medicine, research, and biotechnology mainly due to their ability to:

i) Divide, renew and replenish themselves over long periods (Regenerative capabilities)
ii) Turn into any other specialized cell type in the body (Differentiation capabilities)

Few of the key applications of stem cells are given below:

1. Stem cells find applications in Regenerative Medicine due to their Regenerative capabilities.
   a) Stem cells can regenerate damaged tissues, such as in heart disease, spinal cord injuries, or cartilage damage. They are used for tissue repair.
   b) Research is ongoing to grow organs (e.g., kidneys, liver) in labs using stem cells which can be used for Organ Transplants. This reduces the need for donor organs.
2. Stem cells from bone marrow or umbilical cord blood are used to treat Blood Disorders like leukemia, lymphoma, and sickle cell anemia.
3. Stem cells are being studied to treat Neurodegenerative Diseases like Parkinson's, Alzheimer's, and ALS by replacing damaged neurons.
4. Stem cells may help regenerate insulin-producing cells in the pancreas and thus can be used in treating Diabetes.
5. Stem cells are used Drug Development and Testing. They are used to create disease models for testing new drugs, reducing reliance on animal testing. They help study drug toxicity and efficacy in specific cell types.
6. Stem cells are used in Cosmetic and Anti-Aging Applications mainly in skincare products and treatments to promote tissue regeneration and reduce signs of aging.

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7. Stem cells are used in Cancer Research. They help researchers understand how cancer develops and spreads. They are used to study cancer stem cells, which are thought to drive tumor growth.

BIOMOLECULES

Biomolecules are organic molecules made up of carbon, hydrogen, oxygen, nitrogen, and, small amount of phosphorus and sulfur. They play a major role in functioning of living organisms. The four main types of biomolecules are carbohydrates, lipids, proteins, and nucleic acids. Carbohydrates, such as glucose and glycogen, are primary sources of energy for living organisms. Polysaccharides like cellulose provide structural support in plant cell walls. Lipids, such as phospholipids, form the basic structure of cell membranes. Lipids, like triglycerides, serve as an efficient way to store energy in the body. Certain lipids, such as vegetable oils and biodiesel, are used as bio-fuels. Proteins provide structural support in tissues and organs. Enzymes, which act as catalysts in biochemical reactions are basically proteins. Antibodies responsible for immune system's defense against pathogens are also proteins. Deoxyribonucleic acid (DNA) carries genetic information and serves as a blueprint for protein synthesis. Ribonucleic acid (RNA) plays a role in protein synthesis and gene expression. Nucleic acids are also used in genetic engineering techniques such as polymerase chain reaction, gene cloning, and genome editing. Brief description of these biomolecules and their applications are given here.

PROPERTIES AND FUNCTIONS OF CARBOHYDRATES

Carbohydrates are a diverse group of organic compounds made of carbon, hydrogen, and oxygen atoms. They are also known as saccharides. Carbohydrates are a primary source of energy for the body. They also play an important role in cell structure, and numerous biological functions in organisms.

Thus, carbohydrates play a crucial role in the human diet. Carbohydrates are one of the three macronutrients, along with proteins and fats required to meet the body's nutritional needs. Carbohydrates also include dietary fiber, which is essential for

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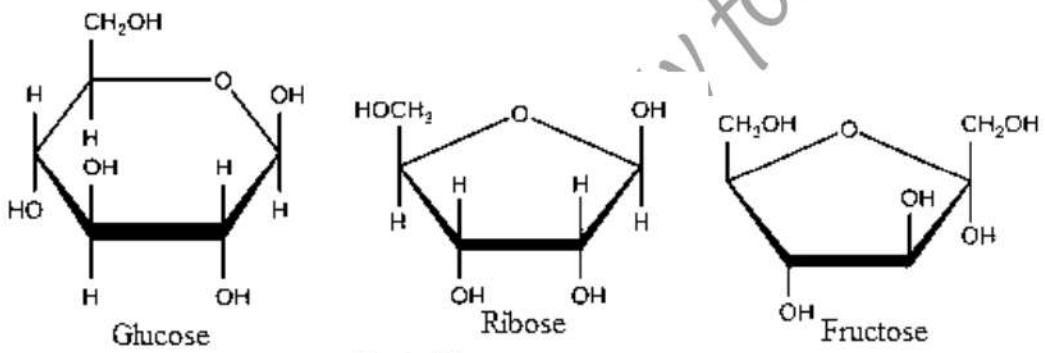
maintaining a healthy digestive system. A healthy diet must contain complex natural carbohydrates, such as whole grains, legumes, vegetables, fruits and nuts which are rich in fiber, vitamin, minerals, and can provide more sustained energy. It is advisable to minimize the consumption of processed carbohydrates which contain additives, preservatives, refined sugars and very less fiber.

Carbohydrates can be classified into several types based on their chemical structure and properties. The main types of carbohydrates are:

1. **Monosaccharides:** Monosaccharides are the simple sugar molecules. Examples of monosaccharides include glucose, fructose, and galactose. These sugars serve as energy sources for cells and are essential for various biological processes.



2. **Disaccharides:** In disaccharides, two monosaccharide units are joined together by a glycosidic bond. A glycosidic bond is formed by condensation of –OH groups of two monosaccharide units with elimination of water. Common examples of disaccharides include sucrose (glucose + fructose), lactose (glucose + galactose), and maltose (glucose + glucose). Disaccharides are commonly found in sweeteners and are broken down into monosaccharides during digestion.

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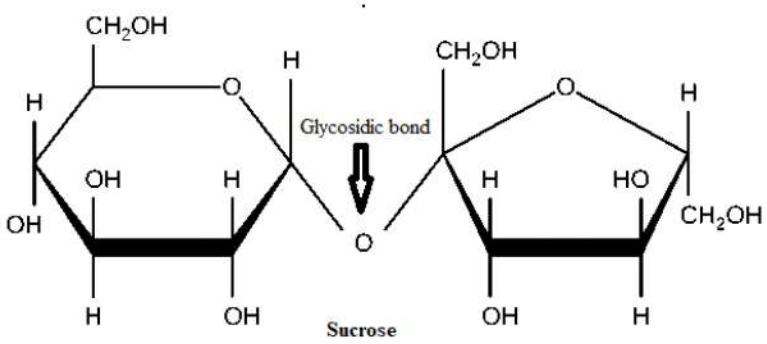
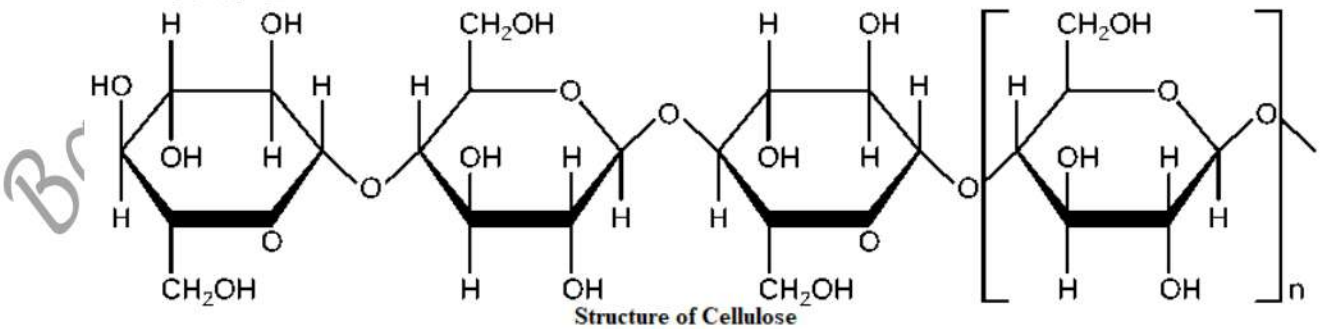


Sucrose

3. **Oligosaccharides:** Oligosaccharides consist of a small number of monosaccharide units (usually 3 to 10) linked together by glycosidic bonds. They are found in certain foods and can serve as prebiotics, providing nourishment for beneficial gut bacteria. Examples of oligosaccharides include raffinose and fructo-oligosaccharides.
4. **Polysaccharides:** Polysaccharides are complex carbohydrates composed of long chains of monosaccharide units. They can be branched or unbranched and can contain hundreds to thousands of sugar molecules. Polysaccharides are important for energy storage and structural support in living organisms. Examples of polysaccharides include starch (found in plants), glycogen (stored in animals) and cellulose.

Cellulose is a polysaccharide composed of repeating units of simple sugar molecules. In cellulose, glucose units are linked together by beta-1,4-glycosidic bonds. These glucose units are connected in a linear chain, forming a long, fibrous structure. This is called as cellulose Fibre.



Structure of Cellulose

Cellulose is a major structural component of plant cell walls. It provides rigidity and strength to plant cells, allowing them to maintain their shape and withstand

mechanical stress. Wood, cotton, and other plant-based materials are rich sources of cellulose.

Cellulose finds various industrial applications. It is the primary component of paper and cardboard, and it is also used to produce materials like rayon, cellophane, and cellulose acetate. Additionally, cellulose fiber is also used in water purification.

PROPERTIES AND FUNCTIONS OF NUCLEIC ACIDS

Nucleic acids are macro biomolecules. They play a fundamental role in the storage, transmission, and expression of genetic information in living organisms.

Nucleic acids are composed of monomeric units called nucleotides. A nucleotide consists of three components:

a) a nitrogenous base (adenine, thymine, cytosine, guanine, or uracil),
b) a pentose sugar. deoxyribose is the sugar in DNA and ribose is the sugar molecule present in RNA.
c) a phosphate group.

The sequence of nucleotides along the DNA or RNA strand forms the genetic code, which determines the information necessary for protein synthesis and other cellular processes.

There are two primary types of nucleic acids:

1. **Deoxyribonucleic acid (DNA):** DNA (Deoxyribonucleic Acid): DNA is a biomolecule with double-stranded helical structure. It carries the genetic instructions used in the development, functioning, and reproduction of all known organisms. DNA is made of four nucleotides: adenine (A) with thymine (T) and cytosine (C) with guanine (G). There are two strands of nucleotides connected by hydrogen bonds. DNA is found primarily in the nucleus of cells and contains the complete set of genetic information for an organism.
2. **Ribonucleic acid (RNA):** RNA is a biomolecule with single-stranded structure. It is involved in various biological processes. It is synthesized from DNA through

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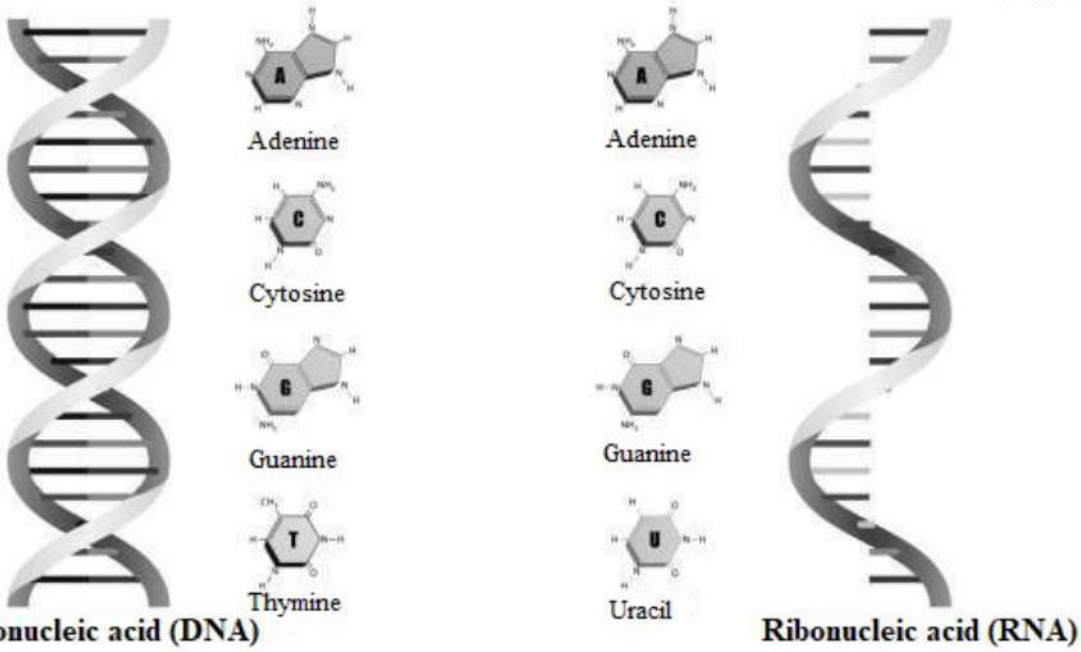
a process called transcription. RNA consists of nucleotides similar to DNA, but it uses uracil (U) instead of thymine. There are several types of RNA, including messenger RNA (mRNA), transfer RNA (tRNA), and ribosomal RNA (rRNA). Each type has specific functions in the process of protein synthesis.



Deoxyribonucleic acid (DNA) **Ribonucleic acid (RNA)**

The sequence of nucleotides in DNA contains the instructions for the synthesis of proteins. Nucleic acids, particularly DNA, are responsible for the inheritance of traits from one generation to the next. DNA contains the genetic information that is passed on from parents to offspring. Through processes like DNA replication and cell division, the genetic material is faithfully transmitted during reproduction.

Nucleic acids also participate in various cellular processes, including gene regulation, enzyme activity, energy transfer (e.g., adenosine triphosphate or ATP), and cell signaling pathways. RNA molecules, such as tRNA and rRNA, have crucial roles in protein synthesis and the functioning of ribosomes.

Understanding nucleic acids and their structure-function relationships is essential to gain insights into the fundamental mechanisms of life and also in fields

such as genetic engineering, molecular biology, biotechnology, medicine, DNA fingerprinting, gene therapy, and DNA sequencing.

PROPERTIES AND FUNCTIONS OF PROTEINS

Proteins are large, complex molecules that play crucial roles in almost all biological processes. They are composed of amino acids linked together by peptide bonds to form polypeptide chains.

Proteins are synthesized through a process called translation. It occurs in the ribosomes, where the genetic information encoded in mRNA is used to assemble amino acids into a polypeptide chain. The order of amino acids in the protein is decided by the information in genetic code.

Proteins have a hierarchical structure that determines their function. The primary structure refers to the linear sequence of amino acids in the polypeptide chain. The secondary structure describes local folding patterns, such as alpha helices and beta sheets, stabilized by hydrogen bonds. The tertiary structure refers to the overall three-dimensional arrangement of the protein, including interactions between amino acid side chains. Some proteins may have a quaternary structure, involving the association of multiple polypeptide chains.

Proteins are fundamental to life and contribute to the structure, function, and regulation of living organisms. They have diverse functions in living organisms. They can act as enzymes, catalyzing biochemical reactions, and also as structural components, providing support and organization to cells and tissues. Proteins are also involved in cell signaling, transport of molecules (such as hemoglobin), immune responses (such as antibodies), gene regulation, and many other biological processes.

They play a significant role in maintaining disease free healthy life.

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PROPERTIES AND FUNCTIONS OF LIPIDS

Lipids are a class of biomolecules composed of carbon, hydrogen, and oxygen atoms. Unlike carbohydrates, lipids have a relatively lower proportion of oxygen. Hence, they are nonpolar in nature, insoluble in water and hydrophobic. Fats, oils, waxes, sterols, and related compounds come under the category of lipids.

There are many dietary sources for lipids, including animal-based sources such as meat, dairy products, and eggs. Plant-based sources include oils (olive oil, coconut oil, etc.), nuts, seeds, avocados, and some fruits (such as olives).

Lipids play essential roles in the human body and serve various functions. However, excessive consumption of certain types of lipids, such as saturated and trans fats, can have negative health effects. These fats are associated with an increased risk of cardiovascular diseases. It is important to consume a balanced diet that includes healthier lipid sources like unsaturated fats (found in nuts, seeds, fish, and vegetable oils) and to moderate the intake of saturated and trans fats.

There are several types of Lipids. Important ones are given below:

1. **Triglycerides:** Triglycerides are main components of fats and oils. They are esters of glycerol and three fatty acids. They are the most common type of lipid and serve as a major energy storage molecule in the body.
2. **Phospholipids:** Phospholipids have a similar structure to triglycerides, but one fatty acid is replaced by a phosphate group. They are a major component of cell membranes and play a crucial role in maintaining their structure and function.
3. **Sterols:** Sterols, such as cholesterol, are a type of lipid that have a specific multi-ring structure. They are essential for the production of hormones, bile acids, and the structure of cell membranes.
4. **Waxes:** Waxes are esters of long-chain fatty acids and long-chain alcohols. They have protective functions in plants and animals, providing a waterproof coating.

The main functions of Lipids are:

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1. **Energy Storage:** Triglycerides serve as a concentrated energy source in the body. They store and release energy as and when needed. They act as fuels for cellular processes.
2. **Structural Component:** Phospholipids are a fundamental component of cell membranes. They form a lipid bilayer that helps maintain the integrity and selective permeability of cells.
3. **Insulation and Protection:** Adipose tissue, which is composed of fat cells, acts as an insulating layer, helping to regulate body temperature. Lipids also provide padding and protection to vital organs.
4. **Hormone Production:** Steroids, a class of lipids derived from cholesterol, serve as precursors for the production of various hormones, including sex hormones and adrenal hormones.
5. **Absorption of Fat-Soluble Vitamins:** Lipids play a crucial role in the absorption and transport of fat-soluble vitamins (A, D, E, and K) in the body.
6. **Cellular Signaling:** Lipids, particularly certain types of phospholipids, act as signaling molecules and are involved in cell communication and signal transduction pathways.

IMPORTANCE OF SPECIAL BIOMOLECULES

ENZYMES

Enzymes are bio molecules, which act as catalysts in biochemical reactions. They speed up biochemical reactions, by factors of millions or more, without being consumed in the process. They are highly specific and selective in their activity. Enzymes are made of globular proteins with a complex three-dimensional structure.The precise arrangement of amino acids in the enzyme's structure determines its catalytic activity and specificity. Enzymes have an active site to which the substrate binds and the catalytic reaction occurs. Enzymes exhibit high substrate specificity, meaning they recognize and bind to specific substrates (reactant molecules) only. It is based on the shape, charge, and chemical properties of substrate. Due to this specificity, enzymes catalyze

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only specific reactions while remaining unreactive toward other molecules. Enzymes lower the activation energy of a chemical reaction, thereby accelerate the reaction rate. Enzyme catalysed reactions occur at normal temperature but with higher rate. Few examples of enzymes are proteases (catalyzing protein breakdown), amylases (breaking down starch), lipases (breaking down lipids), and DNA polymerases (catalyzing DNA synthesis).

Enzymes are essential for maintaining life processes and have significant practical applications in various fields. Few important applications of enzymes are:

1. Enzymes play several important roles in cellular metabolism.
2. Enzymes are employed as catalysts in industrial processes such as food production, pharmaceuticals, detergents, and biofuels.
3. Enzymes are used as cleaning agents.
4. Enzymes are used in biosensors like glucose sensor.

CLASSIFICATION OF ENZYMES

Enzymes are classified based on the type of reaction they catalyze. Classification **of enzymes into six major classes** with an example for each class is given below:

1. Oxidoreductases: They catalyze oxidation-reduction reactions by transferring electrons, protons (H^+), or oxygen atoms.

Example: Lactate dehydrogenase (LDH): It converts lactate to pyruvate by transferring electrons to NAD^+. It is a key enzyme in anaerobic respiration.

2. Transferases: They transfer a functional group (phosphate, methyl, amino) from one molecule to another.

Example: Hexokinase: It transfers a phosphate from ATP to glucose in glycolysis. It is the first step in glucose metabolism.

3. Hydrolases: They break chemical bonds using water (hydrolysis reaction).

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Example: Amylase: It breaks down starch into maltose by hydrolyzing glycosidic bonds. It is essential for carbohydrate digestion.

4. Lyases: They catalyse breaking of C-C, C-O, C-N bonds without hydrolysis or oxidation.

Example: Aldolase: It cleaves fructose-1,6-bisphosphate into two smaller molecules in glycolysis. It is essential for energy production in cells.

5. Isomerases: They catalyse conversion of one isomer of a molecule into another.

Example: Phosphoglucose isomerase: It converts glucose-6-phosphate into fructose-6-phosphate in glycolysis. It helps in regulating sugar metabolism.

6. Ligases: They catalyze the joining of two molecules using ATP.

Example: DNA Ligase: It joins DNA fragments by forming phosphodiester bond. It is Essential for DNA replication and repair.

PROPERTIES AND FUNCTIONS OF ENZYMES

Enzymes are biological catalysts. They catalyse vast array of chemical reactions necessary for growth, reproduction, and maintenance of living organisms. Below are few important properties and functions of enzymes:

Properties of Enzymes

1. Enzymes catalyse vast array of chemical reactions and significantly increase the rate of reactions by factors of millions or more. They lower the activation energy required for a reaction to proceed.

2. Enzymes exhibit specificity. They are highly specific to their substrates (the molecules they act upon).

3. Most enzyme-catalyzed reactions are reversible, meaning they can catalyze both the forward and reverse reactions.

4. Enzymes are temperature sensitive. They function most efficiently at an optimal temperature. Excessive heat can denature enzymes, destroying their catalytic activity.

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5. Enzymes are pH sensitive. Enzymes have an optimal pH range for activity. Deviations from this range can alter the enzyme's structure and reduce its activity.

6. Enzymes are not consumed in reactions and can be used repeatedly.

Functions of Enzymes

1. Main function of enzymes is catalysis of Metabolic Reactions. They facilitate the breakdown, synthesis, and transformation of molecules in metabolic pathways.

Examples: Digestive enzymes: Break down food into absorbable nutrients (e.g., amylase, lipase, protease).

2. Enzymes control the rate of metabolic pathways, ensuring that cellular processes occur at the right time and in the right amounts.

3.Enzymes are essential in energy-producing pathways like glycolysis, the Krebs cycle, and oxidative phosphorylation.

4. Enzymes like DNA polymerase and ligase are involved in DNA replication and repair.

5. Enzymes like kinases and phosphatases regulate cellular signaling pathways by modifying proteins.

6. Enzymes like cytochrome P450 help detoxify harmful substances in the liver.

7. Enzymes like lysozyme break down bacterial cell walls, aiding in immune defense.

8. Enzymes regulate processes like mitosis and meiosis, ensuring proper cell division and growth.

PROPERTIES AND FUNCTIONS OF VITAMINS

Vitamins are organic compounds that are essential for normal growth, development, and overall health of human body. They play a crucial role in maintaining health and preventing deficiency-related diseases. They are required by body in small amounts. A balanced diet rich in fruits, vegetables, whole grains, and lean proteins typically provides all the necessary vitamins for optimal health.

Vitamins are of two types: Water-Soluble and Fat-Soluble Vitamins:

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a) Water-soluble vitamins are easily soluble in water. They are not stored in the body; excess is excreted in urine. They are required to be taken through regular dietary intake.

e.g., B-complex vitamins, vitamin C

b) Fat-soluble vitamins are easily soluble in fats and oils. They are stored in the liver and fatty tissues. They can accumulate to toxic levels if consumed in excess.

e.g., Vitamins A, D, E, K.

Few important properties and functions of vitamins are given below:

Properties of Vitamins

1. Vitamins are essential nutrients. They are vital for maintaining health and must be obtained from food or supplements.
2. Vitamins are required in very small amounts compared to macronutrients (carbohydrates, proteins, fats).
3. Many vitamins act as coenzymes (non-protein molecules that assist enzymes in catalyzing reactions) or precursors to coenzymes.
4. Some vitamins are sensitive to heat, light, or air, which can degrade them during food processing or storage.
5. Vitamins often work together with other nutrients (e.g., minerals) to perform their functions effectively.

Functions of Vitamins

Vitamins play diverse roles in the body, including supporting metabolism, immune function, growth, and cellular repair. Few of the general functions of vitamins are given below:

General Functions of Vitamins:

1. B-complex vitamins play a critical role in energy production through conversion of food into energy (ATP).

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2. Vitamins C and E play the role of antioxidants and they protect cells from oxidative damage caused by free radicals.

3. Vitamins A, C, D, and E play key roles in maintaining immune function.

4. Vitamins D and K are required to maintain proper bone health. They are essential for calcium metabolism and bone formation.

5. Vitamin K is necessary for the synthesis of clotting factors and is mainly responsible for blood clotting.

6. Vitamins like folate and B12 are crucial for DNA synthesis and cell division and they are required for growth and repair of cells.

7. Vitamin A is essential for maintaining healthy vision.

Few specific functions of each vitamin group along with the problems caused due to their deficiency are given below:

Water-Soluble Vitamins

1. Vitamin B1 (Thiamine): Functions as Coenzyme in energy metabolism (carbohydrate breakdown). Deficiency leads to Beriberi, fatigue, nerve damage.

2. Vitamin B2 (Riboflavin): Involved in energy production and cellular function. Deficiency results in cracked lips, inflammation of the tongue, skin disorders.

3. Vitamin B3 (Niacin): It supports metabolism, DNA repair, and cell signaling. Deficiency leads to Pellagra (diarrhea, dermatitis, dementia).

4. Vitamin B5 (Pantothenic Acid): It is essential for synthesizing coenzyme A (CoA), involved in fatty acid metabolism. Deficiency results in fatigue and numbness.

5. Vitamin B6 (Pyridoxine): Involved in amino acid metabolism, neurotransmitter synthesis, and hemoglobin production. Its deficiency results in anemia, depression, weakened immune function.

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6. Vitamin B7 (Biotin): It functions as Coenzyme in fatty acid synthesis and amino acid metabolism. Its deficiency results in Hair loss, skin rashes, neurological symptoms.

7. Vitamin B9 (Folate/Folic Acid): It is essential for DNA synthesis, cell division, and red blood cell formation. Its deficiency results in Megaloblastic anaemia, neural tube defects in foetuses.

8. Vitamin B12 (Cobalamin): It is involved in red blood cell formation, DNA synthesis, and nerve function. Its deficiency results in pernicious anaemia, neurological disorders.

9. Vitamin C (Ascorbic Acid): It is an Antioxidant. It is essential in collagen synthesis, immune support, iron absorption. Its deficiency results in Scurvy (fatigue, gum disease, skin issues).

Fat-Soluble Vitamins

1. Vitamin A (Retinol, Beta-Carotene): It functions in vision (rhodopsin synthesis), immune function, skin health, and cell growth. Its deficiency results in night blindness, dry skin, weakened immunity. Excess intake of Vitamin A results in Hypervitaminosis A (toxicity).

2. Vitamin D (Calciferol): It regulates calcium and phosphorus absorption, bone health, immune function. Its deficiency results in Rickets (in children), osteomalacia (in adults). Excess intake of Vitamin D results in Hypercalcemia (high calcium levels).

3. Vitamin E (Tocopherol): It is an Antioxidant. It protects cell membranes, supports immune function. Its deficiency may cause nerve damage or muscle weakness. Excess intake of Vitamin E results in Bleeding disorders.

4. Vitamin K (Phylloquinone, Menaquinone): It is essential for blood clotting and bone metabolism. Its deficiency results in Bleeding disorders, weakened bones. Excess intake of Vitamin K may interfere with blood thinners.

Sources of Vitamins

Vitamin A: Carrots, sweet potatoes, spinach, liver.

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Vitamin D: Sunlight, fatty fish, fortified dairy products.

Vitamin E: Nuts, seeds, vegetable oils.

Vitamin K: Leafy greens, broccoli, soybeans.

B Vitamins: Whole grains, meat, eggs, dairy, legumes.

Vitamin C: Citrus fruits, strawberries, bell peppers, broccoli.

PROPERTIES AND FUNCTIONS OF HORMONES

Hormones are chemical substances produced by endocrine glands that regulate various physiological processes in the body. Few key properties of hormones are given below:

Properties of Hormones

1. Hormones are produced by ductless endocrine glands and released into the bloodstream. Through bloodstream they travel to different target organs or tissues, where they exert their effects.
2. Hormones are highly effective even in very low concentrations. They produce significant physiological effects even when present in very low amounts.
3. Some hormones act quickly (e.g., adrenaline), while others have long-lasting effects (e.g., growth hormone).
4. Hormones act on specific target cells or tissues that have receptors for them. This specificity is due to the presence of hormone-specific receptors on or in the target cells.
5. The production and release of hormones are controlled by negative or positive feedback mechanisms.

Functions of Hormones

Hormones regulate a wide range of physiological processes. Few of them are given below:

1. They are essential for Growth and Development. Growth Hormone (GH) stimulates growth of bones and tissues. Thyroid Hormones (T3, T4) are essential for normal growth and brain development.
2. They are essential for Metabolism. Insulin lowers blood glucose levels by promoting glucose uptake and storage. Glucagon raises blood glucose levels by stimulating glycogen breakdown. Thyroid Hormones regulate basal metabolic rate (BMR).

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3. They are crucial for reproduction. Estrogen and Progesterone regulate the menstrual cycle and maintain pregnancy. Testosterone promotes male secondary sexual characteristics and sperm production.
4. Hormones are responsible for stress response. Cortisol helps the body respond to stress by increasing blood glucose and suppressing the immune system. Adrenaline prepares the body for "fight or flight" by increasing heart rate and blood flow to muscles.
5. Hormones are responsible for Fluid and Electrolyte Balance. Antidiuretic Hormone promotes water reabsorption in the kidneys. Aldosterone regulates sodium and potassium balance.
6. Hormones are responsible for maintaining calcium levels in the body. Parathyroid Hormone (PTH) increases blood calcium levels. Calcitonin lowers blood calcium levels.
7. Hormones play an important role in the immune function of the body. Cortisol suppresses inflammation and immune responses. Thymosin promotes T-cell development.
8. Hormones are responsible for Behaviour and Mood. Serotonin and Dopamine regulate mood, appetite, and sleep. Oxytocin promotes bonding and social behaviour.

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MODULE- II

VTU SYLLABUS CONTENTS

BIOMOLECULES AND THEIR APPLICATIONS (QUALITATIVE):

BIOMOLECULES AND THEIR APPLICATIONS (QUALITATIVE)

Carbohydrates in cellulose-based water filters production, PHA and PLA in bioplastics production, Nucleic acids in vaccines and diagnosis (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting), , Proteins in food production (Proteins as food – whey protein and meat analogs, Plant based proteins), , lipids in biodiesel and cleaning agents/detergents production, Enzymes in biosensors fabrication (glucose-oxidase in biosensors) Enzymes in textile processing (lignolytic enzyme in bio-bleaching). Enzymes in food processing, and detergent formulation.

CARBOHYDRATES

CELLULOSE-BASED WATER FILTERS

Cellulose is a natural polymer found in the cell walls of plants, and it is used in water filtration due to its excellent filtration properties. Cellulose-based water filters are filtration systems that utilize cellulose as the primary filtering material.

The working principle of cellulose-based water filters is given below:

Cellulose fibers are processed and formed into a filter medium. Depending on the specific filtration needs, it is used as paper, membranes, or cartridges. Cellulose filters remove impurities from water by means of mechanical filtration. The cellulose fibers form a porous matrix that traps solid particles, sediment, and larger contaminants as water passes through the filter. This mechanism is effective in removing visible particles and improving water clarity.

Cellulose filters can be designed with micro and nanoporous structures that have specific pore sizes and can be used in microfiltration and ultrafiltration. These pores

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can be engineered to target different contaminants based on their size. For example, some cellulose filters use membranes with a pore size less than 0.5 µm to physically remove bacteria from water, such as Escherichia coli which is approximately 1–2 µm in length. But, the main drawback of this is the impair of the water flow by small pores, so that electrical pumps are often required to obtain a sufficient flow through the filter. Also, the small pores in the membranes can easily become clogged by organic matter in water.

The effectiveness of cellulose filters can be enhanced by incorporating additional filtration layers. For instance, activated carbon may be integrated with cellulose filters to increase the removal of organic compounds, odor, and taste.

The main advantages of Cellulose-based water filters are:

1. Biodegradable and Renewable: Cellulose is a natural and sustainable material and it is environmentally friendly compared to synthetic alternatives.
2. Cost-effective: Cellulose filters are generally more affordable than some other advanced filtration technologies, making them accessible to a wider range of users.
3. Easy to Use and Maintain: Cellulose filters are easy to install, replace, and maintain. They are available in various forms and sizes, suitable for different water filtration systems.
4. Effective Particle Removal: Cellulose filters effectively remove sediment, rust, and other larger particles, improving the overall clarity of water.

Cellulose filters may not effectively remove certain contaminants, such as viruses and extremely small particles. Therefore, for comprehensive water treatment, it's advisable to combine cellulose filters with other purification methods, such as UV disinfection or reverse osmosis, depending on the specific water quality requirements.

Similar to other water filtration systems, in cellulose water filters also, regular maintenance, including filter replacement and proper cleaning, is crucial to ensure optimal performance and prevent bacterial growth within the filter.

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BIOPLASTICS

Polymers are large molecules made up of repeating subunits called monomers. The strong covalent bonds between monomers in polymers make them resistant to degradation by biological agents like enzymes or microorganisms. Hence, synthetic polymers, including plastics, are generally non-biodegradable. They do not easily break down through natural biological processes. This is one of the major drawback of polymer materials. Non-biodegradable polymers can persist in the environment for an extended period, ranging from decades to centuries or even longer. This persistence leads to the accumulation of plastic waste in landfills, water bodies, and natural ecosystems.

Due to the environmental problems associated with non-biodegradable polymers, efforts have been made to develop and promote biodegradable alternatives. Biodegradable polymers, also called as bioplastics, are the polymers which are degraded easily by biological agents like enzymes or microorganisms. They are derived from renewable resources; such as plants or microorganisms. Using of bioplastics can address the environmental problems of plastic waste.

Two of the important bioplastics are given below:

1. POLYHYDROXYALKANOATES (PHAs)

Polyhydroxyalkanoates (PHAs) are biodegradable aliphatic polyesters. They are synthesized by natural biochemical method using certain bacteria. Cupriavidusnecator and Pseudomonas putida are examples of bacteria capable of producing PHAs. These bacteria consume natural carbon resources and convert them into PHAs inside their cells. The selected bacteria are cultivated in a bioreactor under controlled conditions. Various carbon sources, like sugars, fatty acids, and other essential nutrients (nitrogen, phosphorus, etc.) are added continuously to promote bacterial growth. Once the growth phase of bacteria is completed, addition of the carbon source is depleted. This triggers a metabolic shift in the bacteria. In this phase, the bacteria start accumulating PHA granules within their cells as intracellular storage. The accumulated PHAs act as a carbon and energy reserve for the bacteria.

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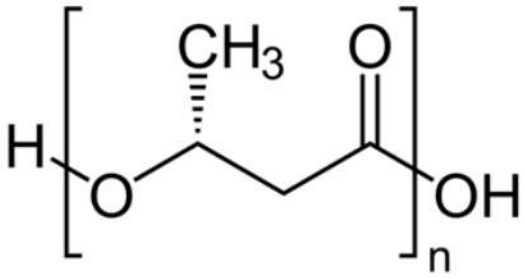
After sufficient PHA accumulation, the bacteria are harvested by various methods like centrifugation, filtration, and solvent extraction to extract PHA polymer. Depending on the microorganisms and the cultivation conditions, homo- or copolyesters with different hydroxylalkanoic acids like 3-, 4-, 5- and 6-hydroxyalkanoic acids can be obtained. Important commercially available PHAs are poly-3-hydroxybutyrate (PHB), poly-3-hydroxybutyrate-co-valerate (PHBV), and polyhydroxybutyrate-co-hexanoate (PHBH).



Structure of Poly-3-hydroxybutyrate (PHB)

Main advantages of PHAs are:

1. PHAs are biodegradable bioplastics. They can be broken down by natural processes into harmless substances like water, carbon dioxide, and biomass. Hence, PHA is an environmentally friendly alternative to conventional plastics that persist in the environment for hundreds of years.
2. PHAs are versatile bioplastics. They can be produced in a range of chemical compositions exhibiting varieties of chemical properties. They can be tailored to have different levels of flexibility, durability, and thermal stability, making it suitable for diverse applications.
3. PHAs are biocompatible and non-toxic, and hence suitable for certain medical and biomedical applications.

Applications:

PHAs are used in medical devices, drug delivery systems, and tissue engineering. PHAs are also used in various industries, including packaging, agriculture, textiles, and personal care products. They can be used to produce films, coatings, fibers, and molded products.

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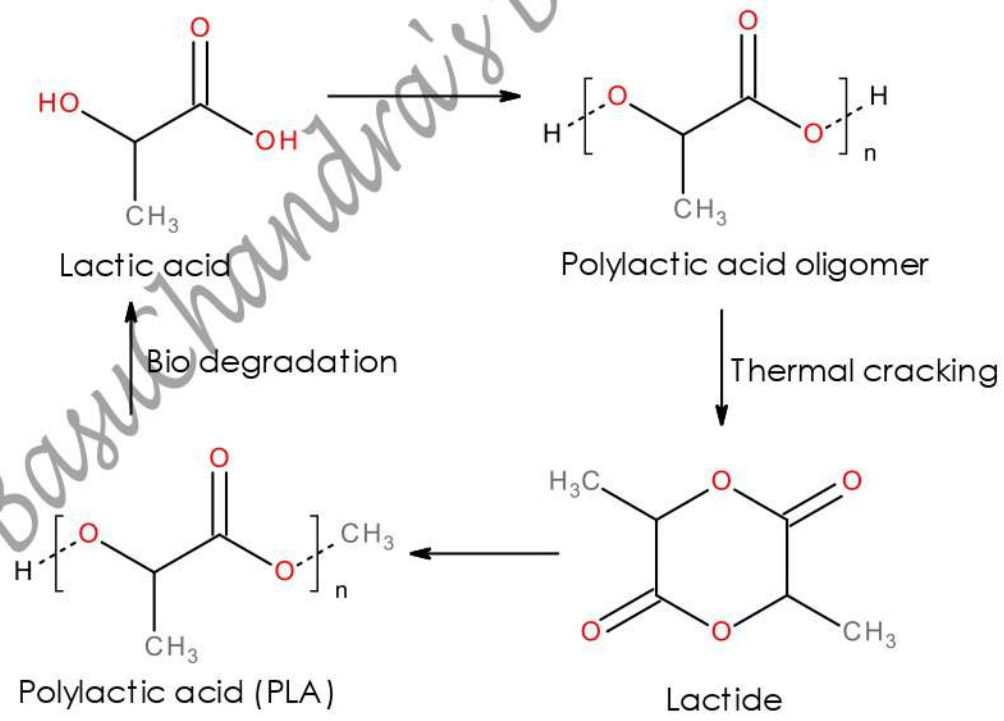
However, PHA production cost is currently higher compared to conventional plastics. Research and development efforts are ongoing to improve the production efficiency and reduce production costs, which could further promote the use of PHA as a sustainable bioplastic.

2. POLYLACTIC ACID (PLA)

PLA is an environmental friendly bioplastic obtained from polymerization of lactic acid. Lactic acid is obtained from fermentation of renewable carbon resources, such as corn starch, sugarcane, or other plant-based feedstocks using lactic acid bacteria. The process is known as lactic acid fermentation.

PLA belongs to the polyester family and it is obtained from polymerization of Lactic acid monomer. Steps involved in synthesis and degradation of PLA are given below. In the first step, Lactic acid solution is concentrated and subjected to condensation resulting in polylactic acid oligomers. At elevated temperature in the presence of a catalyst, oligomers undergo cyclization to give lactide. This undergoes ring opening polymerization resulting in PLA of higher molecular weight.



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Properties:

1. PLA is biodegradable and it can be broken down by natural processes. PLA can be easily degraded into carbon dioxide, water, and organic matter. This biodegradability reduces the environmental impact by accumulation of plastic waste.
2. PLA is derived from renewable resources, such as agricultural crops, which makes it a sustainable alternative to conventional petroleum-based plastics.
3. PLA is a versatile in terms of its physical and mechanical properties. It can be produced in various forms, including films, fibers, foams, and injection-moldedproducts. PLA can be transparent or opaque, rigid or flexible, and it can be modified with additives to achieve specific properties.

Applications:

PLA finds application in a wide range of industries and products. It is commonly used for packaging materials, disposable cutlery, food containers, disposable cups, and single-use bags. PLA is also utilized in textiles, 3D printing, and biomedical applications, such as sutures, drug delivery systems, and tissue engineering scaffolds.

VACCINES

Vaccines are biological substances that stimulate the immune system to recognize and defend against specific infectious diseases. Vaccines have been proven to be highly effective in preventing infectious diseases. They have a significant impact on public health, saving millions of lives worldwide.

NUCLEIC ACIDS IN VACCINES AND DIAGNOSIS

DNA VACCINE FOR RABIES

Rabies is a viral disease that affects the central nervous system of mammals, including humans. It is transmitted through the bite of an infected animal, usually a rabid dog, bat, raccoon, or other wildlife. Rabies is fatal once symptoms appear. It can

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be prevented by vaccination prior or post- viral transmission. Presently, vaccines made from killed rabies virus or recombinant proteins derived from the virus called as Inactivated Rabies Vaccines are used. They are administered through intramuscular injections. Recently, a new class of vaccines called as DNA vaccines are showing several potential advantages for rabies immunization.

A DNA vaccine for rabies is designed from a specific genetic material extracted from the rabies virus. When this is injected into host cells, an immune response is triggered and provides protection against the virus.

The design and mechanism of action of DNA vaccine is described below:

1. A plasmid DNA that carries genetic information is extracted from the rabies virus.
2. The plasmid is introduced into host cells, such as muscle cells.
3. Inside host cells, the plasmid DNA is transcribed and translated, leading to the production of viral proteins called as rabies glycoprotein (G protein).
4. The G protein so produced is recognized by the host's immune system as foreign antigens. This induces an immune response, generating protective antibodies against rabies.
5. These antibodies bind to the viral proteins, neutralizing the virus and preventing it from infecting cells.

Recently, DNA vaccines are showing several potential advantages for rabies immunization. They can be produced relatively quickly, are easy to modify and optimize, and have shown good safety profiles in preclinical and clinical studies. Although DNA vaccines have shown promise in preclinical studies and some early-stage clinical trials, their widespread use as licensed vaccines for rabies is still under development. Further research is needed to optimize vaccine design, enhance immunogenicity, and assess their effectiveness in larger human trials.

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RNA VACCINES FOR COVID-19

RNA vaccines have played a crucial role in fighting the COVID-19 pandemic globally. These vaccines were developed and deployed with remarkable speed and effectiveness to fight COVID-19. SARS-CoV-2 virus is the one that causes COVID-19.

RNA vaccines are specifically messenger RNA (mRNA) vaccines. The two most widely known mRNA vaccines for COVID-19 are the Pfizer-BioNTech (BNT162b2) and Moderna (mRNA-1273) vaccines.

The design and mechanism of action of mRNA vaccine is described below:

1. mRNA vaccines have a small piece of genetic material called messenger RNA (mRNA).
2. mRNA contains instructions for cells to produce a viral protein, which is similar to the spike protein found on the surface of the SARS-CoV-2 virus.
3. When mRNA is injected into the body's cells. It instructs the cells to produce the spike protein. This spike protein is harmless and does not cause the full-blown disease.

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Spike protein found on the surface of the SARS-CoV-2 virus

(Image by pearson0612 from Pixabay)

4. The spike protein produced by the cells triggers an immune response. The immune system recognizes the spike protein as foreign and results in an immune response against it. Immune response is initiated by production of antibodies or by the activation of immune cells.
5. The immune response generated by mRNA vaccines provides protection against COVID-19. If a vaccinated person is later exposed to the SARS-CoV-2 virus, their immune system recognizes the spike protein and mounts a rapid and robust response, preventing or minimizing the infection's severity.

mRNA vaccines have demonstrated high efficacy in preventing symptomatic COVID-19. Clinical trials have shown efficacy rates above 90% for both the Pfizer-BioNTech and Moderna vaccines, indicating strong protection against COVID-19. mRNA vaccines have undergone extensive testing for safety. They have been rigorously evaluated in large clinical trials and authorized for emergency use by regulatory authorities, such as the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA). Common side effects are generally mild and temporary, such as injection site pain, fatigue, or fever.

FORENSICS -DNA FINGERPRINTING

Each individual person exhibits a unique DNA profile and it can be considered as their genetic "fingerprint." This specific pattern of genetic information can be used to distinguish one person from another. DNA fingerprinting is a forensic technique, used to identify individuals and establish biological relationships based on their unique DNA profiles. This technique is also known as DNA profiling or genetic fingerprinting. It has revolutionized forensic science and become an invaluable tool in criminal investigations.

The mechanism of DNA fingerprinting technique used in forensics is described below:

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1. DNA (deoxyribonucleic acid) is a double-stranded helical molecule that carries the genetic information in almost all living organisms. DNA is composed of repeating units called nucleotides, and the sequence of these nucleotides varies among individuals, except for identical twins.
2. In DNA fingerprinting, specific regions of DNA in which short sequences of nucleotides are repeated multiple times are analyzed. Most commonly the regions called as short tandem repeats (STRs) are analyzed and it is called as STR analysis.These repetitive sequences are 2 to 6 base pairs in length, and are found in various regions of the human genome.
3. In STR analysis, the number of repeats in specific STR loci between DNA samples are identified and compared.
4. STR analysis is done using techniques such as polymerase chain reaction (PCR) and capillary electrophoresis.
5. The number of repeats in a Short Tandem Repeat (STR) loci, varies significantly from person to person. individuals.
6. The chances of two individuals having the same STR profile is very less.
7. Hence, it is used to discriminate one person from other person.

The procedure of DNA fingerprinting technique used in forensics is described below:

1. DNA samples can be collected from various sources, including blood, semen, saliva, hair roots, skin cells, and other biological materials left at crime scenes.
2. Once the DNA sample is collected, the DNA needs to be extracted and purified. Polymerase chain reaction (PCR) is then used to amplify the targeted STR regions. PCR makes millions of copies of the DNA, making it easier to analyze.
3. The amplified DNA is analyzed using methods such as capillary electrophoresis which separate the DNA fragments based on their size.
4. By comparing the patterns of DNA fragments from the crime scene sample and reference samples (such as suspects or victims), forensic scientists can determine if the DNA profiles match.

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5. The likelihood of a match between DNA profiles is calculated using statistical methods. This helps in assessing the strength of the evidence and determining the significance of the match.

DNA fingerprinting is used in various forensic scenarios like:

a) Identifying suspects and victims.
b) Excluding innocent individuals,
c) Linking suspects to crime scenes,
d) Resolving paternity and familial relationships.
e) In solving cold cases, exonerating wrongfully convicted individuals, and providing crucial evidence in court.

DNA fingerprinting has revolutionized forensic investigations. With high degree of accuracy and discriminatory power it is having a significant impact on criminal justice systems worldwide.

PROTEINS

PROTEINS AS FOOD

Proteins are one of the three macronutrients, alongside carbohydrates and fats. They are vital for the growth, repair, and maintenance of body tissues, as well as for the synthesis of enzymes, hormones, and antibodies. Proteins are an essential component of the human diet and are commonly consumed as a source of nutrition. Proteins are made of amino acids, which are the building blocks of protein molecules.

The recommended dietary allowance (RDA) for protein intake varies based on factors such as age, sex, weight, and activity level. Generally, it is recommended that protein intake should contribute around 10-35% of total daily calorie intake. Adequate protein intake is important for meeting nutritional needs and supporting overall health.

Protein sources are of two types:

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1. **Complete proteins:** These contain all the essential amino acids that the body cannot produce on its own. These are of high-quality and are highly digestible.

Examples of complete protein sources include animal-based proteins such as meat, fish, poultry, eggs, and dairy products like milk, cheese, whey and yogurt.

2. **Incomplete proteins:** These are protein sources which may lack one or more essential amino acids.

Examples of incomplete proteins are plant-based protein sources such as legumes (beans, lentils, chickpeas), soy products (tofu, tempeh), whole grains, nuts, and seeds. However, combining varieties of different incomplete protein sources in diet, one can get a complete amino acid profile from plant sources also.

WHEY PROTEIN

The liquid portion of milk that separates from it during cheese production is called as whey. Protein derived from whey is called as whey protein. Whey protein is a complete protein, meaning it contains all the essential amino acids required by the body. It is rich in branched-chain amino acids (BCAAs), which play a role in muscle protein synthesis and recovery. Whey protein is highly concentrated in protein, containing 70-90% protein by weight. Thus, it is a popular dietary supplement, used by athletes and bodybuilders to support muscle recovery and enhance performance. It is also commonly used in the food industry as an ingredient in various products.

The following steps are involved in whey protein production process:

1. The milk is pasteurized to eliminate harmful bacteria.
2. The collected milk is then used to produce cheese. During the cheese-making process, enzymes or acid are added to curdle the milk.
3. The liquid whey, which contains water, lactose (milk sugar), minerals, and whey proteins, is separated from the solid curds. This can be done through various methods such as centrifugation or filtration.
4. The whey is further filtered using ultrafiltration membranes to remove any remaining impurities.

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5. After filtration, the whey is subjected to additional concentration through processes like evaporation or reverse osmosis to increase the protein content. Here, the water is removed from the whey, resulting in a more concentrated protein solution.
6. The concentrated whey is subjected to processes such as ion exchange chromatography to isolate whey. This technique separates the whey protein from other components present in the whey, such as lactose and minerals.
7. After isolated, the whey protein is dried to create a powdered form. Spray drying is a commonly used method, where the protein solution is sprayed into a heated chamber, causing the liquid to evaporate and leaving behind the powdered whey protein.

MEAT ANALOGUES AS PROTEIN SOURCES

Meat analogues, are the products designed to resemble the taste, texture, and appearance of meat but are made entirely from plant-based ingredients. Meat analogues are made from plant proteins, such as soy, wheat gluten, peas, or mushrooms. These proteins are combined with other ingredients like oils, starches, and flavorings to mimic the taste, texture, and appearance of meat.

Meat analogues come in various forms, including burgers, sausages, nuggets, strips, and ground meat alternatives. They can be made from a single type of plant protein or a combination of different protein sources.

The main advantages of meat analogues are:

1. They are generally lower in fat and cholesterol compared to meat with higher dietary fiber.
2. Some meat analogues can also be fortified with additional nutrients, such as vitamins and minerals.
3. Consuming meat analogues will have lower environmental impact compared to conventional meat production. Plant-based meats require fewer resources,

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produce fewer greenhouse gas emissions, and contribute to reduced land and water use compared to animal agriculture.

4. Meat analogues can be grilled and baked and hence can be used in a wide range of recipes just like classical meat dishes.
5. Meat analogues can be a suitable option for individuals following vegetarian diets, as well as those looking to reduce their meat intake.

PLANT-BASED PROTEIN SOURCES

Plant-based protein sources are excellent alternatives for individuals to fulfill their protein requirement without relying on animal products. These protein sources are considered as incomplete proteins, because each source may lack one or more essential amino acids. However, combining varieties of different incomplete protein sources in diet, one can get a complete amino acid profile from plant sources also.

Few of the commonly available plant-based protein sources are:

1. **Legumes:** Legumes, including beans, lentils, chickpeas, and peas, are rich sources of protein.
2. **Soy Products:** Soybeans and soy-based products, such as tofu, tempeh, and edamame, are complete protein sources and offer a wide range of essential amino acids.
3. **Quinoa:** Quinoa is a seed that is often referred to as a pseudo-cereal due to its nutritional profile and culinary uses. It is a complete protein source.
4. **Nuts and Seeds:** Almonds, walnuts, peanuts, chia seeds, flaxseeds, and hemp seeds are examples of plant-based protein sources rich in healthy fats, fiber, and other nutrients.
5. **Whole Grains:** Some whole grains, including oats, quinoa, brown rice, and whole wheat, contain moderate amounts of protein.
6. **Seitan:** Seitan, also known as wheat gluten, is a high-protein food made from gluten, the main protein in wheat. It has a meat-like texture and is often used as a meat substitute in dishes.

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7. **Spirulina:** Spirulina is a type of blue-green algae that is rich in protein and contains all essential amino acids. It is available in powdered form and can be used as a supplement.
8. **Vegetables:** Vegetables are not high in protein content, but still contribute to overall protein intake. Some vegetables with relatively higher protein content include spinach, broccoli, brussels sprouts, and peas.
9. **Plant-Based Meat Substitutes:** Meat alternatives made from ingredients like soy, peas, or mushrooms are available in the market in the form of burgers, sausages, and nuggets. These products are designed to mimic the taste, texture, and appearance of traditional meat.

Plant based protein sources: Nuts,legumes and Spirulina.

LIPIDS

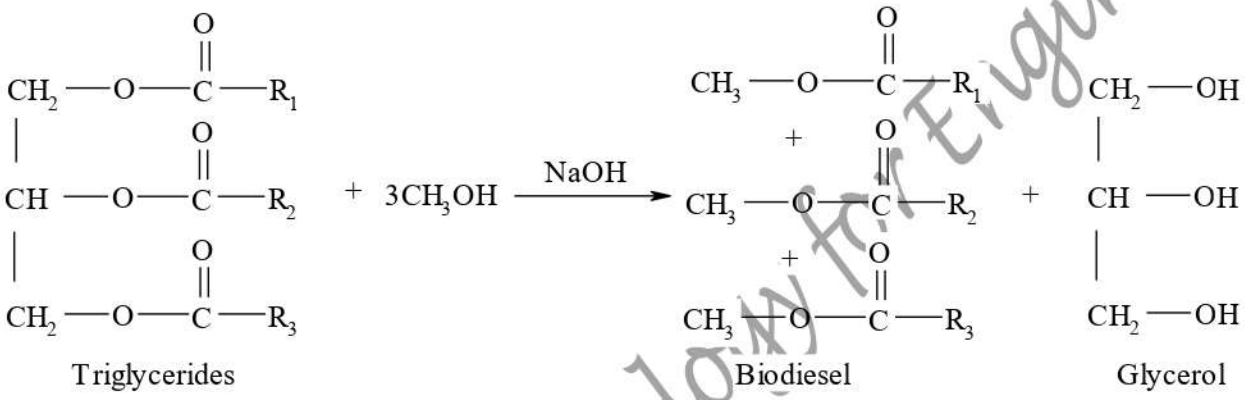
BIODIESEL

Biodiesel can be used as an alternative fuel to petroleum based diesel. It is produced from renewable sources, like vegetable oils or animal fats.Vegetable oils commonly used for biodiesel production include jatropa oil, pongamia oil, soybean oil, canola oil, palm oil, and sunflower oil. These oils are typically obtained from oilseed crops. Animal fats,derived from animal by-products, such as waste from meat processing industries can also be used.

Vegetable oils or animal fats mainly contain triglycerides (lipids). They are esters of glycerol and three fatty acids. Biodiesel is synthesized by transesterification of

triglycerides (lipids) with methyl alcohol using NaOH as catalyst. Transesterification is an organic reaction in which an ester is converted into other ester by reaction with an alcohol. During reaction, one bigger triglyceride molecule is converted in to three smaller molecules of methyl esters of fatty acid. Structure of these molecules is similar to structure of hydrocarbons present in petroleum diesel. Hence, it is called as biodiesel.

A typical transesterification reaction for synthesis of biodiesel is given below:



Where R1, R2 and R3 are the long chain fatty acids in the oil. Glycerol is soluble in water and hence can be separated from reaction mixture.

Main advantages of biodiesel are:

1. It does not contain sulfur. It is also free from aromatics.
2. It contains more oxygen & hence undergoes complete combustion compared to diesel and hence emits lesser amount of CO and hydrocarbons.
3. It is biodegradable and renewable.
4. It is cheaper than diesel.
5. It has higher cetane number compared to diesel.
6. It can be used in engine with no major modification in the engine hardware.
7. Recycled or used cooking oil, such as waste oil from restaurants or households, can be processed into biodiesel.

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CLEANING AGENTS/DETERGENTS

Detergents are cleaning agents used to remove dirt, grease, stains, and other unwanted materials from surfaces. They are chemically composed of substances called as surfactants. Surfactants are mainly responsible for the cleaning properties of detergents. Surfactants have both hydrophilic (water-loving) head group and a hydrophobic (water-repelling) tail. The head group is typically an ionized carboxylate or sulfonate group. The tail group is a hydrocarbon molecule. One common example is alkylbenzene sulfonate. Polar group interacts with water, whereas non-polar group interacts with oil stains and dirt thus facilitating the removal of oily stains and dirt from surfaces. Surfactants reduce the surface tension of water, allowing it to spread and penetrate more effectively.

Surfactants are synthetic chemicals which are similar to natural lipids in terms of amphiphilic nature that is having both polar and non-polar parts in same molecule.

In addition to surfactants, detergents may contain other additives, such as enzymes, builders, chelating agents, bleaching agents, and fragrances. These additional components enhance the cleaning efficiency, water softening, stain removal, and overall sensory experience of the detergent. Other additives used in detergents along with surfactants are:

1. **Enzymes:** Enzymes are biological molecules that can break down specific types of stains or dirt. They are commonly used in laundry detergents to remove protein-based stains like blood, grass, or food residues. Enzymes work by catalyzing chemical reactions that break down complex substances into simpler forms that can be easily washed away.
2. **Builders:** Builders, such as sodium tripolyphosphate or zeolites, are used to enhance the cleaning efficiency of detergents. They help to soften water by binding to mineral ions, which can interfere with the effectiveness of surfactants.
3. **Chelating Agents:** Chelating agents are compounds that bind to metal ions, such as calcium and magnesium, which are commonly found in hard water. By

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binding to these ions, chelating agents prevent them from interfering with the cleaning process.

4. **Bleaching Agents:** Bleaching agents, such as hydrogen peroxide or sodium hypochlorite, are used to remove stains and brighten fabrics. They work by oxidizing colored compounds, making them less visible.

5. **Fragrances:** Fragrances are added to cleaning agents to provide a pleasant scent after cleaning. They help mask any unpleasant odors associated with dirt.

ENZYMES

GLUCOSE-OXIDASE (GOx) IN BIOSENSORS

Biosensors are analytical devices that combine a biological component, such as an enzyme, with a transducer to convert a biological signal into a measurable output. Glucose oxidase (GOx) is an enzyme commonly used in biosensors for the detection and measurement of blood sugar levels in diabetic patients. The glucose bio sensor is one of the most widely used biosensor and account for 85% of the current world market for biosensors.

In portable glucose sensor, a disposable single-use strip is used. In this strip, electrodes are screen printed on the paper strip. Glucose oxidase (GOx) enzyme is immobilized on the electrode to measure the glucose level in blood. The method of testing the blood glucose involves pricking the finger with a lancet (a small, sharp needle), putting a drop of blood on a test strip and then placing the strip into a meter that displays the blood glucose level.

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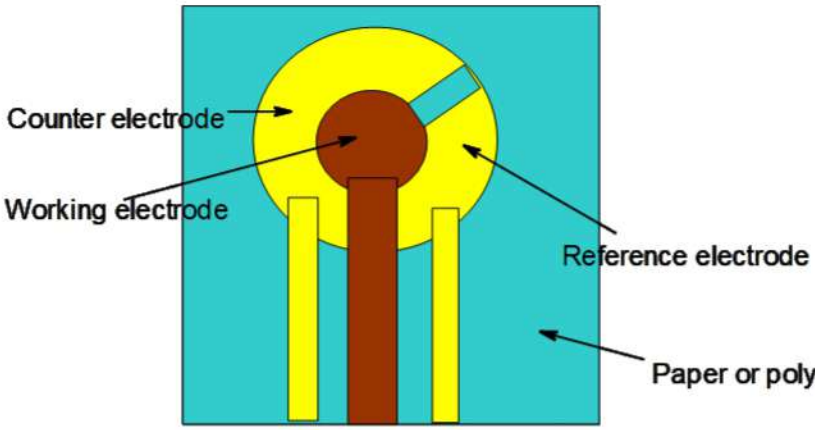

Screen printed paper strip disposable sensor

Glucose sensor

(Courtesy: Image by Gary VandenBergh from Pixabay)

Working principle of a Glucose sensor

Glucose biosensor is an electrochemical sensor. Clark oxygen electrode is the cell used in glucose biosensor. It is a three electrode cell, composed of a reference electrode, typically Ag/AgCl electrode, a working electrode, which is usually made of a noble metal or carbon and a counter electrode. Pt working electrode is immobilized with

Glucose Oxidase (GOx). GOx is an FAD (flavin adenine dinucleotide) enzyme, which very specifically oxidises ß-D-glucose. It oxidizes Glucose to Glucolactone and accepts electrons and gets reduced.

$$Glucose + GO_x(FAD) \rightarrow Glucolactone + GO_x(FADH2)$$

Reduced, deactivated enzyme consumes oxygen from the electrolyte, transfers electrons to it and reduces it to hydrogen peroxide. Thus, the amount of oxygen consumed is proportional to concentration of glucose. The concentration of glucose can be measured from decrease in oxygen concentration.

$$GO_x(FADH2) + O_2 \rightarrow GO_x(FAD) + H_2O_2$$

Glucose concentration can also be measured from the amount of hydrogen peroxide formed. Hydrogen peroxide is oxidized amperometrically at fixed voltage of 0.7 V and current generated is measured. Measured current is proportional to concentration of glucose.

$$H_2O_2 \rightarrow O_2 + 2e^- + 2H^+$$

There are several advantages of GOx in Biosensors. It has high specificity for glucose, making it suitable for accurate glucose detection in complex biological samples. GOx enzyme also exhibits good stability and a long shelf life.

ENZYMES IN TEXTILE PROCESSING

LIGNOLYTIC ENZYME IN BIO-BLEACHING

Pulp fibers are fibrous materials derived from wood, and they are used as a raw material in the production of paper and paper products. They provide the structural backbone and strength to paper.

Bio-bleaching refers to the use of biological agents, such as enzymes, to remove lignin from pulp fibers. Lignin is a complex polymer that provides rigidity and strength to plant cell walls. Its removal is necessary for producing high-quality paper products.

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Bio-bleaching results in bleached pulp which is brighter and more environmentally friendly.

Enzymes used to remove lignin from pulp fibers are known as Lignolytic enzymes. They are a group of enzymes produced by certain fungi and bacteria. The primary enzymes involved in lignin degradation are lignin peroxidase (LiP), manganese peroxidase (MnP), and versatile peroxidase (VP). These enzymes work by catalyzing oxidative cleavage reactions that lead to the cleavage of lignin's chemical bonds resulting in its degradation and solubilization. Lignolytic enzymes generate reactive oxygen species, such as hydrogen peroxide and free radicals. These attack the lignin polymer, breaking it into smaller and more soluble fragments. This process helps to de-lignify the pulp fibers and improves the brightness and cleanliness of the resulting paper product. Lignolytic enzymes have a high degree of specificity for lignin and can selectively degrade lignin without affecting cellulose which are responsible for paper's strength and durability.

Bio-bleaching is an environmental friendly process than traditional bleaching process where harsh chlorine-based chemicals, such as chlorine or chlorine dioxide, which can generate toxic by-products are used. Overall, lignolytic enzymes play a crucial role in the production of high-quality, environmentally friendly paper products.

ENZYMES IN FOOD PROCESSING

Enzymes are widely used in food processing. They are used to enhance flavor, texture, nutritional value, and shelf life of various food products. They act as biological catalysts in various chemical reactions that occur during processing of food materials without being consumed in the process. They are widely used in dairy, baking, brewing, meat processing, fruit juice production, and starch modification.

Some common applications of enzymes in food processing are given below:

1. **Baking Industry:** Amylases, Proteases and Lipases are commonly used in baking industry. Amylases break down starch into sugars, hence they are used to improve dough fermentation, texture, and crust color in bread making. Proteases

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are used to modify gluten proteins to improve dough elasticity and texture. Lipases are used to enhance dough stability and crumb structure of bread and cake.

2. **Dairy Industry:** Chymosin Lactase and Lipase are the enzymes commonly used in dairy industry. Chymosin is used in cheese production to coagulate milk proteins, forming curds. Lactase breaks down lactose into glucose and galactose and it is used in making lactose-free milk and dairy products. Lipases break down fats into free fatty acids and they are used to enhance flavor development in cheese by breaking down fats into free fatty acids.
3. **Fruit Juice and Beverage Industry:** Pectinases, Amylases and Cellulases are the enzymes commonly used in **Fruit Juice and Beverage Industries.** Pectinases clarify fruit juices by breaking down pectin, improving juice yield and clarity. Amylases convert starches into fermentable sugars in beer and alcohol production. Cellulases break down cellulose in plant materials to improve juice extraction.
4. **Meat Processing Industry:** Protease and Transglutaminase are commonly used in meat processing. Proteases tenderize meat by breaking down muscle proteins and connective tissues. Transglutaminase improves texture and binding properties in processed meat products.
5. **Starch and Sweetener Production Industry:** Amylase, Glucose Isomerase and Invertase are the enzymes used in Starch processing and Sweetener Production industries. Amylases convert starch into maltodextrins and sugars for syrups and sweeteners. Glucose Isomerase converts glucose to fructose, producing high-fructose corn syrup. Invertase produces high-fructose syrups.
6. **Brewing Industry:** Amylase and Protease are used in brewing industries. Amylases convert starches into fermentable sugars. Proteases improve beer clarity and foam stability.
7. **Flavor Enhancement:** Lipases and Proteases generate free fatty acids and amino acids. Hence, they are used to enhance flavors in cheese, sauces, and fermented foods.

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8. **Nutritional Improvement:** Phytases break down phytic acid in grains and legumes, improving the bioavailability of minerals like phosphorus, calcium, and iron.
9. **Shelf-Life Extension:** Glucose Oxidase reduces oxygen in packaged foods. Thus, it prevents spoilage of food and extends its shelf life.
10. **Natural preservative:** Lysozyme acts as a natural preservative in cheese and wine by inhibiting bacterial growth.

ENZYMES IN DETERGENT FORMULATION

Enzymes are capable of catalysing chemical reactions involved in the process of breaking down of tough stains and organic residues. Hence, they are widely used in **modern detergent formulations.** They provide effective, **environmentally** sustainable, and versatile cleaning solutions due to the following advantages:

- Enzymes work effectively in cold water, reducing energy consumption and saving costs.
- Enzymes are biodegradable and reduce the need for harsh chemicals, making detergents more environmentally sustainable.
- Enzymes target specific types of stains, improving overall cleaning performance along with maintaining fabric quality by preventing damage and discoloration.
- Synthetic surfactants and phosphates used in detergents are dangerous to the environment. By using enzymes, concentrations of surfactants and phosphates used in synthetic detergents can be minimized thus reducing their environmental impact.

An overview of the main enzymes used in detergent formulations and their role are given below:

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1. Proteases are enzymes that can catalyze the breaking down of protein-based stains such as blood, egg, grass, and sweat. They can selectively remove these stubborn stains preventing fabric discoloration. They are used in laundry detergents and dishwashing products.
2. Amylases are enzymes that can degrade starch-based stains like pasta, potato, gravy, and chocolate. They effectively remove food residues and prevents starch re-deposition on fabrics. They are used in laundry and dishwashing detergents.
3. Lipases catalyse the breaking down of fat and oil stains such as grease, butter, and salad oils. They are effective at low temperatures and prevent greasy residues on clothes. They are commonly used in laundry detergents.
4. Cellulases break down cellulose fibers on fabric surfaces, removing microfibrils and dirt particles. They soften fabrics, prevents pilling, and restores color brightness. They are used in laundry detergents, especially for cotton and other cellulose-based fabrics.
5. Mannanases degrade mannan-based stains, such as guar gum and food thickeners. They are effective against sticky and gummy stains. They are used in laundry detergents.
6. Pectinases break down pectin, a component of plant-based stains like fruit residues. They help to remove fruit stains and improve cleaning efficiency. They are used in laundry and dishwashing detergents.

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MODULE- III

ADAPTATION OF ANATOMICAL PRINCIPLES OF HUMAN ORGAN SYSTEMS FOR BIOENGINEERING DESIGN (QUALITATIVE)

Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for Parkinson's disease). Eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye). Heart as a pump system (architecture, electrical signaling - ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pace makers, defibrillators). Lungs as purification system (architecture, gas exchange mechanisms, spirometry, abnormal lung physiology - COPD, Ventilators, Heart-lung machine). Kidney as a filtration system (architecture, mechanism of filtration, CKD, dialysis systems).

HUMAN ORGAN SYSTEMS

The human body can be considered as a biological machine. Like a machine, the human body is composed of various interconnected parts that work together to perform specific functions. All these organs together are called as human organ systems. There are several organ systems in the human body, each with its own set of organs and functions.

The major organ systems with their brief functions is given below:

1. **Circulatory System:** The circulatory system is responsible for the transportation of oxygen, nutrients, hormones, and waste products throughout the body. The main organs involved are heart, blood vessels (arteries, veins, and capillaries), and blood.
2. **Respiratory System:** The respiratory system is involved in the exchange of oxygen and carbon dioxide between the body and the environment. It consists of the lungs, trachea, bronchi, and diaphragm.

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3. **Digestive System:** The digestive system processes food and absorbs nutrients to provide energy for the body. It includes organs such as the mouth, esophagus, stomach, small intestine, large intestine, liver, gallbladder, and pancreas.
4. **Nervous System:** The nervous system controls and coordinates the activities of the body. It includes the brain, spinal cord, and peripheral nerves.
5. **Muscular System:** The muscular system allows movement and provides support and stability to the body. It includes skeletal muscles, smooth muscles, and cardiac muscles.
6. **Skeletal System:** The skeletal system provides structure, support, and protection to the body. It consists of bones, joints, and cartilage.
7. **Endocrine System:** The endocrine system regulates body functions through the production and release of hormones. It includes glands such as the pituitary gland, thyroid gland, adrenal glands, and reproductive glands.
8. **Reproductive System:** The reproductive system is involved in the production of offspring. It differs between males and females and includes organs such as the ovaries, uterus, testes, and penis.
9. **Urinary System:** The urinary system is responsible for filtering and eliminating waste products from the body through the production and excretion of urine. It consists of the kidneys, bladder, ureters, and urethra.
10. **Integumentary System:** The integumentary system is the body's protective covering. It includes the skin, hair, nails, and sweat glands.

These organ systems work together to maintain the overall health and functioning of the human body.

Thus the human body can be considered as a machine with interconnected parts. There are some key similarities between **the human body and a machine** as given below:

1. **Organization and Structure:** Both machines and the human body have a defined structure and organization. Machines are built with specific components arranged in a particular manner to achieve their intended purpose. Similarly, the human body has organs, tissues, cells, and organelles working together to maintain overall function.

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2. **Input and Output:** Machines require input in the form of energy or materials to perform their tasks, and they produce output in the form of a desired product or result. Similarly, the human body takes in nutrients, oxygen, and sensory information as input, and it produces outputs such as movement, waste products, and sensory perceptions.
3. **Energy Conversion:** Both machines and the human body require energy to function. Machines typically convert energy from one form to another, such as electrical energy into mechanical energy. In the human body, energy is obtained from food through processes like digestion and metabolism, and it is converted into chemical energy (ATP) to power cellular processes.
4. **Control Systems:** Machines have control systems that regulate their operations, monitor performance, and make adjustments as needed. Similarly, the human body has complex control systems, such as the nervous system and endocrine system, which regulate and coordinate various bodily functions.
5. **Maintenance and Repair:** Machines require regular maintenance and occasional repairs to ensure optimal performance and longevity. Similarly, the human body has mechanisms for self-maintenance and repair, such as cell renewal, wound healing, and immune responses.

Despite these similarities, it is important to note that the human body is far more complex and sophisticated than any man-made machine. It possesses the ability to adapt, learn, and exhibit consciousness, which are not seen in conventional machines.

BIO ENGINEERING DESIGNS

Using the principles from biology and nature in the engineering field of design to create innovative solutions for various applications is called as bio design. In bio design inspiration and ideas are drawn from natural systems, natural processes, and natural structures. These are then used to design products, technologies, and systems that are more sustainable, efficient, and compatible with the environment.

Few examples of bio designs are given below:

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1. **Biomimicry:** Biomimicry is a design approach that imitates and applies biological solutions to human problems.

For instance, studying the structure and properties of lotus leaves has inspired the development of self-cleaning surfaces and coatings that repel water and dirt, mimicking the lotus effect.

Lotus leaf (Courtesy: Image by Zhu Bing from Pixabay)

2. **Bioinspired Materials:** Designers and scientists are exploring the development of new materials inspired by natural substances.

For example, Gecko feet are known for their remarkable adhesive properties, allowing geckos to cling to surfaces and even walk upside down on smooth vertical or overhead surfaces. The adhesive capabilities of gecko feet have been a source of inspiration for scientists to produce bio-inspired adhesives. Scientists have developed synthetic adhesives inspired by gecko feet, known as gecko-inspired adhesives or gecko tape. These adhesives can adhere to various surfaces and are reusable, leaving no residue upon detachment. Gecko-inspired adhesives have potential applications in technology and engineering like robotics, climbing equipment, and industrial manufacturing.

Studying natural gecko feet and understanding the mechanisms behind their adhesive capabilities has provided valuable insights for designing innovative adhesive materials and technologies.

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Gecko Feet (Courtesy:Image by Nel Botha from Pixabay)

3. **Bioluminescent Lighting:** Taking inspiration from bioluminescent organisms like fireflies and jellyfish, designers have created lighting solutions that emit light without the need for electricity. These bio-inspired lighting systems could be used for energy-efficient and sustainable illumination.

Firefly **Jellyfish**

4. **Sustainable Textiles:** Designers are exploring alternative materials and processes for creating textiles that are more sustainable and environmentally friendly.
   For example, using bio-based fibers derived from agricultural waste and developing textile dyes from natural sources to reduce pollution.
5. **Green Infrastructure:** Introducing nature-inspired designing such as green roofs, vertical gardens, and rainwater harvesting systems in urban spaces. This can enhance environmental sustainability in cities.

These examples show, how bio design can integrate biological concepts, processes, and materials into various design disciplines which can address challenges and create more sustainable and innovative solutions.

BRAIN

The brain is the center of the nervous system in humans. It is located within the skull. It is a complex and vital organ and is responsible for controlling and coordinating various bodily functions. The brain plays central role in processing information from the senses, enabling perception, thinking, and consciousness.

The brain is composed of billions of specialized cells called neurons. These neurons form intricate networks and pathways throughout the brain and the rest of the body. This network allows the transmission of information and communication with each other through electrical and chemical signals.

THE ARCHITECTURE OF THE BRAIN

The brain is a highly complex organ. It consists of several distinct regions, each with its own specialized functions. The structural organization and the arrangement of various components of brain is referred to as architecture of the brain. An overview of the major structural components of the brain and their functions is given briefly below:

1. **Cerebrum:** The cerebrum is the largest and most prominent part of the brain. It is divided into two hemispheres, called as the left and right cerebrums. These two are connected by a bundle of nerve fibers. The cerebrum is responsible for higher cognitive functions, including conscious thought, reasoning, memory, language processing, perception, and voluntary movements.
2. **Cerebellum:** The cerebellum is located at the back of the brain, below the cerebrum. The main functions of this part are motor control, coordination, balance, and posture. The cerebellum receives sensory information from the

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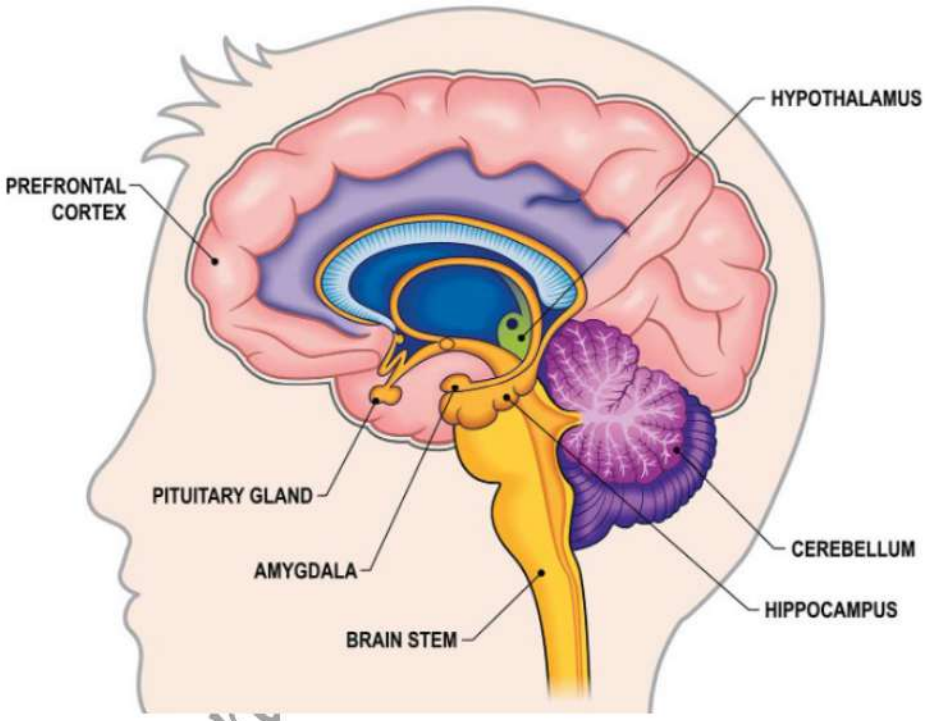
muscles and joints and helps in fine-tuning movements and maintaining equilibrium.

3. **Brainstem:** The brainstem is located at the base of the brain and connects the brain to the spinal cord. The brainstem is responsible for essential functions like regulating breathing, heart rate, blood pressure, and basic involuntary actions.
4. **Thalamus:** The thalamus is a structure located deep within the brain. It acts as a relay station. It receives sensory information from various senses and directs it to the appropriate regions of the cerebral cortex for processing.



Major structural components of the brain

5. **Hypothalamus:** The hypothalamus is a small area located below the thalamus. It plays a crucial role in maintaining various bodily functions, including body temperature, hunger, thirst, sleep, and hormone production through its control of the pituitary gland.
6. **Limbic System:** The limbic system is a group of interconnected brain structures involved in emotions, memory formation, and motivation. The main structures in limbic system are hippocampus (important for memory), and amygdala (associated with emotions and fear response).

7. **Cortex:** The cerebral cortex is the outermost layer of the cerebrum and is responsible for higher cognitive functions. It has four lobes and each lobe has distinct functions, such as motor control, sensory processing, language, and visual perception.

These are some of the major components of the brain's architecture. The brain's complexity arises from the intricate connections and interactions among its different regions. The integration of information and the execution of a wide range of cognitive and physiological processes are due to this complex network.

THE CENTRAL NERVOUS SYSTEM (CNS)

The central nervous system (CNS) is a major part of the nervous system. It includes the brain and spinal cord. It serves as the command center for the entire body. It receives and processes sensory information, initiates motor responses, and is responsible for coordinating and controlling of numerous physiological and cognitive functions.

Some key features and functions of the components of central nervous system are given below:

1. **Brain:** The brain is the central processing unit (CPU) of the CNS. It is responsible for higher cognitive functions, sensory processing, motor control, memory, emotions, and consciousness. It is protected by the skull. It is divided into different regions like cerebrum, cerebellum, and brainstem etc and each part plays different specific functions.
2. **Spinal Cord:** The spinal cord is a long, cylindrical bundle of nerves. It extends from the base of the brain down to the vertebral column. It serves as a conduction medium for nerve signals traveling between the brain and the rest of the body.
3. **Protection and Support:** The components of CNS are protected by various structures to ensure its safety. The brain is encased in the skull, and the spinal

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cord is surrounded by the bony vertebral column. Additionally, the CNS is further protected by membranes called meninges and cerebrospinal fluid, which act as a cushion.

The main functions of CNS are:

1. **Integration and Processing:** The CNS integrates and processes sensory information received from various parts of the body. It coordinates motor responses, initiating voluntary movements and controlling involuntary functions.
2. **Higher Cognitive Functions:** The CNS is responsible for complex cognitive processes, including learning, memory, language, problem-solving, decision-making, and conscious thought.
3. **Homeostasis:** The CNS regulates and maintains internal balance within the body through a process called homeostasis. It controls bodily functions such as heart rate, blood pressure, body temperature, respiration, and hormone production to ensure optimal physiological conditions.

The central nervous system works in conjunction with the peripheral nervous system.

PERIPHERAL NERVOUS SYSTEM (PNS)

The peripheral nervous system (PNS) consists of nerves that extends from the CNS to the rest of the body. It is the connection between CNS and rest of the body. It consists of nerves, ganglia (clusters of nerve cell bodies), and sensory receptors. The PNS plays a vital role in transmitting information between the CNS and the various organs, tissues, and cells throughout the body.

The main components of peripheral nervous system are:

1. **Nerves:** The PNS is composed of two types of nerves: cranial nerves and spinal nerves. Cranial nerves arise from the brain, while spinal nerves emerge from the spinal cord. These nerves contain bundles of nerve fibers called axons which transmit electrical signals between the CNS and the body.
2. **Sensory Receptors:** The sensory receptors of PNS, are located in the skin, muscles, joints, organs, and other tissues. They gather information from the

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external environment and internal body conditions. These receptors senses touch, temperature, pain, pressure, and position. Then transmit sensory signals to the CNS for processing and interpretation.

3. **Motor Neurons:** The motor neurons of PNS, carries motor signals from the CNS to the muscles and glands. They are responsible for voluntary and involuntary movements and controlling bodily functions.
4. **Autonomic Nervous System (ANS):** The ANS is a subdivision of the PNS. It is responsible for regulating involuntary bodily functions like heart rate, blood pressure, digestion, respiration, and glandular activity.
5. **Reflexes:** The PNS is involved in reflex actions, which are rapid, involuntary responses to stimuli. Reflexes often bypass the brain and involve a direct connection between sensory neurons and motor neurons in the spinal cord. This is required for quick responses to harmful or dangerous stimuli.

SIGNAL TRANSMISSION IN BRAIN

A neuron or nerve cell is the fundamental unit of the nervous system. Neurons are responsible for receiving, processing, and transmitting electrical and chemical signals within the nervous system. They communicate and coordinate between different parts of the body and mediate various physiological and cognitive processes.

The following steps are involved in signal transmission in the brain:

1. One neuron receives signal from other neuron in the form of neurotransmitter molecules.
2. Neurotransmitter molecules are transported from neuron to neuron through dendrites. Dendrites are branch-like extensions in the neuron's cell body. They have special structures called dendritic spines with large surface area that contain receptors for neurotransmitters. They receive incoming signals in the form of neurotransmitter molecules released from neighbouring neurons and transmit them towards the neuron cell body in the form of electrical signal.

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3. The cell body is the main part of the neuron that contains the nucleus and other cellular organelles. It is responsible for maintaining the cell's metabolic functions. It integrates incoming signals and determines whether the signals are strong enough to generate an outgoing signal and then generates outgoing signals.
4. The outgoing electrical signals from the neuron's cell body is called as action potential. An action potential is a rapid and brief electrical impulse. The action potential is transmitted along the axon. Axon is a long, slender projection of the neuron. Axon is responsible for transmitting signals over long distances to other target neuron cells, such as muscles or glands. The axon is covered by myelin, a fatty insulating substance that helps speed up the conduction of the electrical impulse.
5. At the end of the axon, there are specialized structures called axon terminals. When the action potential reaches the axon terminals, it triggers the release of neurotransmitters into the synapse. Synapse is a junction between present neuron to the next neuron cell. This allows the communication between neurons.
6. Synapses are the points of communication between one neuron cell and next target cell. At the terminal of one neuron cell, neurotransmitters are released.
7. These neurotransmitters diffuse across the synapse. At the beginning of target cell, neurotransmitters are received by neurotransmitter receptors.
8. These neurotransmitters generate electric potential in the target cell.
9. This process of conversion of chemical signal from neurotransmitter to electrical signal and vice versa repeats. The action potential travels from cell to cell allowing the continuation of signal transmission throughout the neural circuit.

Overall, signal transmission in the brain involves the inter conversion of electrical signals and chemical signals at synapses and also reception, integration, and propagation of these signals. This intricate process enables the communication and coordination of information throughout the brain.

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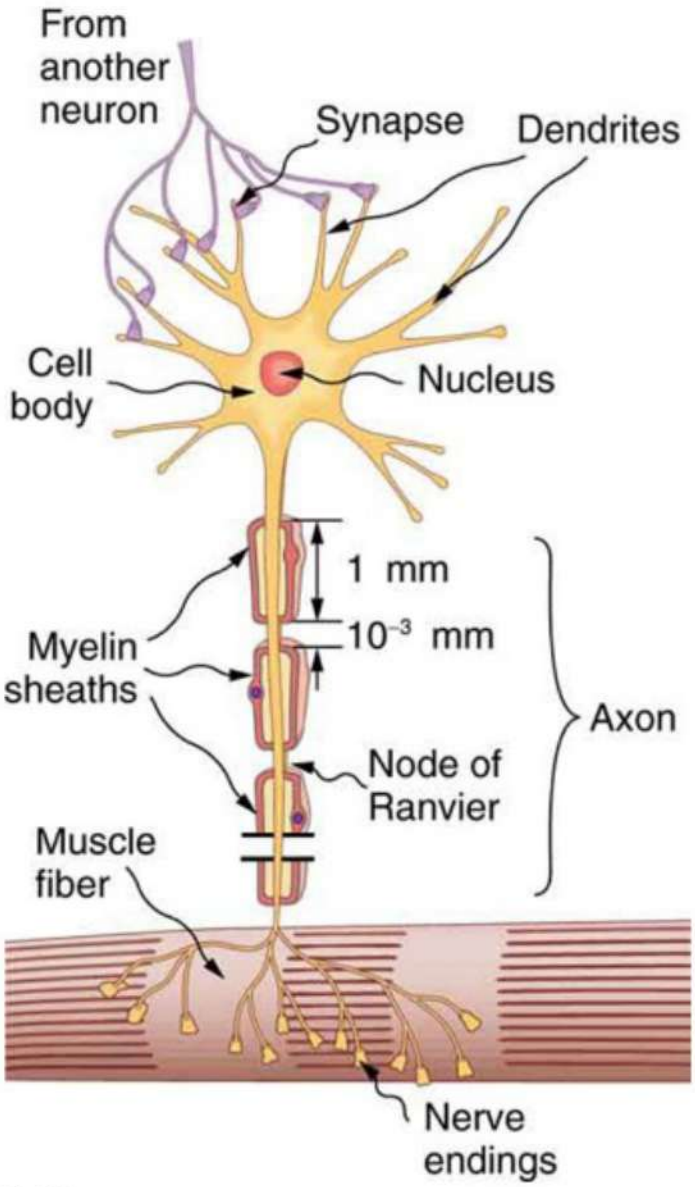

Neuron: The fundamental unit of the nervous system.

BRAIN AS A CPU SYSTEM

Brain can be compared with CPU (Central Processing Unit) of a computer because both process information. But they differ significantly in their structure, functioning, and capabilities. Human brain is more complex and sophisticated than any computer system we currently have.

Main similarities between the brain and a CPU system are:

Function	Brain	CPU
Information processing	The brain processes information from various sources, including sensory inputs and internal cognitive processes.	The CPU in a computer is responsible for executing instructions and processing data.
Parallel processing	The brain contains a vast network of interconnected neurons that work in parallel to process information and carry out various tasks simultaneously.	CPU has multiple cores that can perform computations simultaneously.
Task allocation	The brain has specialized regions responsible for different functions, such as vision, language processing, motor control, and memory.	In a CPU system, tasks are assigned to different cores based on their nature and computational requirements.
Speed and efficiency	The brain, with its intricate neural connections and optimized circuitry, is also capable of processing information at high speeds and adapting its functioning based on the demands of the task.	CPUs are designed to perform calculations quickly and efficiently.

Differences between the brain and a CPU system:

Function	Brain	CPU
Complexity	The brain is a complex organ with billions of neurons and trillions of	CPUs are engineered systems with a finite number of

	synapses. The brain's architecture, plasticity, and ability to learn and adapt far surpass the capabilities of any CPU system.	components and defined abilities.
Neural plasticity	The brain exhibits plasticity. It can rewire its connections and adapt its functioning based on experience and learning.	CPUs have fixed circuitry and instructions. Plasticity feature is not present in traditional CPU systems.
Energy efficiency	The brain is remarkably energy-efficient compared to computer CPUs. It can perform complex computations with relatively low energy consumption.	CPUs generate heat and consume large amount of energy.
Analog nature	The brain's neural activity is primarily analog in nature. Neurons communicate using electrical potentials and chemical signals.	CPUs operate using digital signals (1s and 0s).
Speed	The brain cannot function as fast as CPU in a specific defined task. But, it can process information in parallel and has cognitive capabilities. It can efficiently handle complex real-world tasks such as perception, learning, and creative problem-solving.	CPUs are faster than brain at executing specific tasks requiring precise calculations.

Thus, considering the brain as a CPU system can provide a simplified framework to understand certain aspects of its information processing capabilities. Computers excel in speed and executing specific tasks based on programmed instructions. However, the brain is an exceptionally intricate and dynamic organ. Brain surpasses

the limitations of current computer systems in terms of complexity, adaptability, and unique cognitive abilities and adaptability.

ELECTROENCEPHALOGRAM (EEG)

EEG is a neurophysiological technique used to measure and record the electrical activity of the brain. It is a non-invasive technique and does not require the insertion into the body through breaking of the skin or body tissues. EEG measures the electrical impulses generated by the neurons (brain cells) in the form of brainwaves. These electrical signals can provide valuable information about brain function and activity.

Electroencephalogram (EEG)

In EEG procedure, multiple electrodes are placed on the scalp. These electrodes are small metal discs or sensors. They detect and pick up the electrical signals

generated by the brain. The electrical activity recorded by EEG is displayed as waveforms known as brainwaves. Different states of brain activity, such as wakefulness, relaxation, sleep, or specific cognitive processes produces different brainwave patterns in EEG.

Few of the wave patterns, their frequency in Hz and related brain activity are given below:

1. Delta waves (0.5-4 Hz): Predominant during deep sleep.
2. Theta waves (4-8 Hz): Associated with drowsiness, daydreaming, and meditation.
3. Alpha waves (8-13 Hz): Present during relaxed states, closed eyes, and light meditation.
4. Beta waves (13-30 Hz): Common during wakefulness, active thinking, and concentration.
5. Gamma waves (above 30 Hz): Associated with higher cognitive processing, attention, and perception.

EEG technique is widely used in clinical, research, and diagnostic applications. Few of the applications of EEG are:

1. It is commonly used to assess and diagnose various neurological conditions, such as epilepsy (seizure disorder), sleep disorders, brain tumors, and cognitive disorders.
2. It is used to assess brain damage from head injury.
3. EEG can also be used to study brain activity during cognitive tasks.
4. EEG can be used to evaluate brain responses to stimuli,
5. EEG is used to monitor anesthesia during surgery.
6. It can also be used to confirm brain death in someone in a persistent coma.
7. It can be used to explore brain patterns in neuroscience research.

EEG is a valuable non-invasive tool in neuroscience and clinical practice. It provides insights into brain activity and can help diagnose and monitor various neurological conditions.

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ROBOTIC ARMS FOR PROSTHETICS

Prosthetics are artificial devices used to replace or augment missing or impaired body parts. They are designed to improve the mobility and functionality of individuals who have lost limbs or are born with limb differences. Prosthetic limbs can be used to replace arms, legs, hands, feet, or other body parts. There are various types of prosthetics available, depending on the specific needs of the individual.

Advancements in materials, engineering, and technology have resulted in more capable and comfortable prosthetic devices. For example, with 3D printing technology, it is possible to create customized prosthetics for individual needs at lower cost. With neural interface and robotics technology, mind-controlled or brain-controlled prosthetics have been designed. In these prosthetics, users can control their prosthetic limbs through their thoughts.

Robotic prosthetics arms are advanced mechanical devices designed to replace or augment the functionality of a missing or impaired human arm. These are also called as bionic arms. They are a type of upper limb prosthetics. These prosthetic arms are made of modern materials and use robotics, sensors for improved functionality, control and natural movement.

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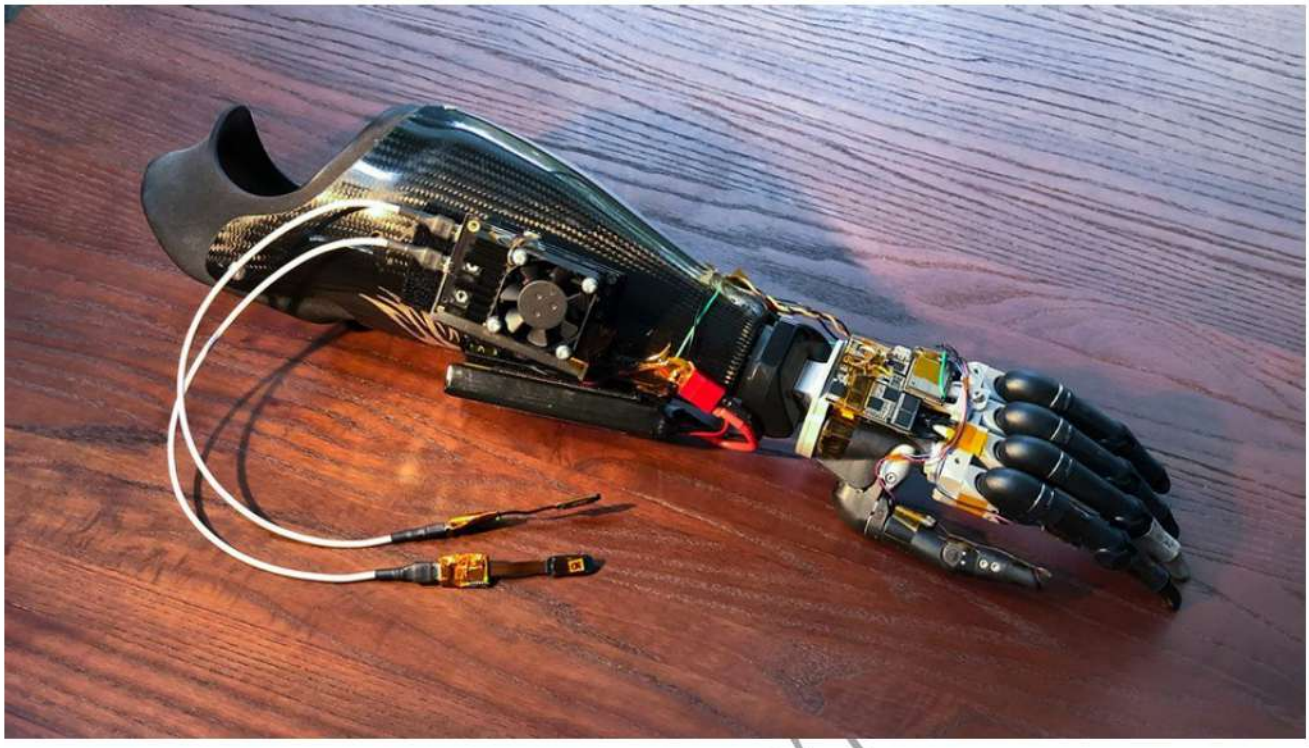

Robotic prosthetics arms

The main components of robotic prosthetics arms and their function are described below:

1. **Design and Construction:** Robotic arms for prosthetics are made from lightweight and durable materials such as carbon fiber, aluminum, or titanium. They are engineered to mimic the structure and movement capabilities of a natural human arm, including the elbow, wrist, and hand.
2. **Actuators and Motors:** Robotic arms use various actuators and motors to generate the necessary movements and force. Electric motors are used for precise control of joint angles and grip strength. Hydraulic actuators are used in some designs.
3. **Sensors:** Robotic arms are equipped with sensors that provide feedback and enable control of the arm's movements. Sensors can detect and interpret the electrical signals generated by the muscles in the residual limb.
4. **Control Systems:** Robotic arms use sophisticated control systems to interpret input signals from sensors and translate them into desired movements. These

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control systems can be pre-programmed with machine learning algorithms to adapt and respond to the user's intended actions.

5. **Gripping and Manipulation:** Prosthetic arms include various types of grippers that enable individuals to grasp and manipulate objects. These can range from simple hooks or claws to more complex hand-like designs with individual finger movement and tactile feedback.
6. **Interface and Control Options:** Robotic arms have different interface options to control the limb. These include switches, voice commands, and brain-computer interfaces that interpret brain signals.
7. **Customization and Adaptability:** Robotic arms can be customized and adjusted to fit the specific needs and preferences of the individual user. They can be calibrated to accommodate various tasks and activities, such as gripping delicate objects or performing more forceful actions.
8. **Power Source:** Robotic arms require a power source to operate. This can be a rechargeable battery pack that provides sufficient power for a day's use. They enable the user to perform daily activities without interruption.

Robotic arms for prosthetics aim to restore functional capabilities and enhance the quality of life for individuals with limb loss or limb impairment.

ENGINEERING SOLUTIONS FOR PARKINSON'S DISEASE

Parkinson's disease is a neurodegenerative disorder. It is characterized by the progressive degeneration and dysfunction of the nervous system, particularly the neurons (nerve cells) in the brain. It is named after James Parkinson, the British physician who first described the condition in 1817. Parkinson's disease is characterized by the progressive loss of dopamine-producing cells in the brain, particularly in a region called the substantia nigra. Dopamine is a neurotransmitter chemical involved in regulating movement. The deficiency of this neurotransmitter leads to the motor symptoms. Parkinson's disease mainly affects movement control.

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The most common symptoms of Parkinson's disease are related to movement and include tremors (typically starting in the hands), rigidity or stiffness of muscles, bradykinesia (slowness of movement), and postural instability (balance problems).

There is currently no cure for Parkinson's disease. Various treatments can help manage the symptoms and improve quality of life. Engineering technologies can help improve quality of life, manage symptoms, and assist with daily activities.

Some of the engineering solutions commonly employed for Parkinson's disease are:

1. **Deep Brain Stimulation (DBS):** DBS is a surgical procedure to implant electrodes in specific areas of the brain to deliver electrical impulses. These impulses help control abnormal brain activity associated with Parkinson's disease. They reduce motor symptoms such as tremors, rigidity, and bradykinesia. The DBS system uses a pacemaker-like device, which is programmed and controlled externally. DBS is an effective treatment option for some individuals with advanced Parkinson's disease.
2. **Wearable Devices:** Wearable devices like smart watches can be used to monitor tremors, sleep patterns, and medication schedules. They provide valuable data for managing symptoms and optimizing treatment plans.
3. **Assistive Devices:** Adaptive clothing with magnetic fasteners can make dressing easier for individuals with limited dexterity. Devices such as voice-controlled home automation systems can assist with controlling lights, appliances, and other household functions.
4. **Virtual Reality (VR) and Gaming:** Virtual reality and gaming platforms with motion-controlled interfaces can engage users in fun and interactive activities. These promote physical activity, coordination, and cognitive function.
5. **Medication Delivery Systems:** Engineering solutions can also help individuals manage their medication schedules and optimize drug delivery. Smart pillboxes and medication dispensers can provide reminders and dispense medication at specific times. Implantable devices can deliver medication in a controlled and continuous manner.

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Engineering solutions for Parkinson's disease continue to advance as technology progresses. These solutions aim to alleviate symptoms, enhance mobility and functionality, and improve the overall quality of life for individuals living with Parkinson's disease.

EYE

ARCHITECTURE OF EYE

The human eye is a complex organ which helps us to perceive the visual world. The structure of eye has various components which work together to facilitate vision. Main components of eye and their role are given below:

1. **Cornea:** The cornea is a clear, dome-shaped structure located at the front of the eye. It helps to focus incoming light onto the lens and plays a crucial role in the eye's overall optical power.

2. **Iris:** The iris is the coloured part of the eye. The central opening in the iris is called as pupil. The amount of light entering the eye depends on size of pupil. Iris has muscles that can control the size of the pupil and regulates the amount of light entering the eye by adjusting the size of the pupil.

3. **Lens:** Lens is located behind the iris and pupil. The lens is a transparent, flexible structure that focuses light onto the retina. It changes shape to allow the eye to focus on objects at different distances, a process known as accommodation.

4. **Retina:** The retina is a thin, light-sensitive layer located at the back of the eye. It contains specialized cells called photoreceptors. Rods and cones are photoreceptor cells present in retina. Photoreceptors present in retina convert light into electrical signals. The retina also contains other cells that process and transmit these signals to the brain through the optic nerve.

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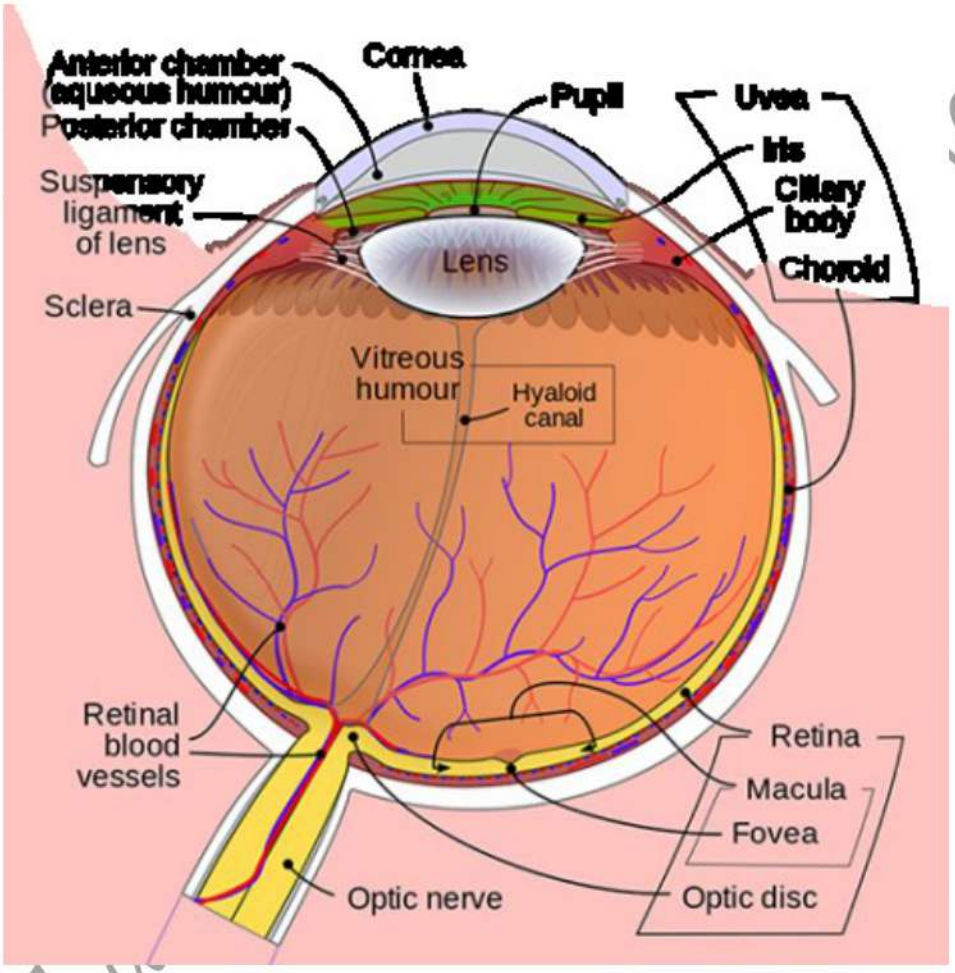
5. **Optic Nerve:** The optic nerve is a bundle of nerve fibers that carries visual information from the retina to the brain. They carry electrical signals from the back of the eye and transmit to the visual processing centres in the brain.



Architecture of eye

ARCHITECTURE OF ROD AND CONE CELLS

Rod and cone cells are photoreceptor cells located in the retina of the eye. They play a crucial role in the process of vision by converting light into electrical signals.

These signals are then transferred to the brain through the optic nerve. Both rod and cone cells are responsible for visual perception, but, they differ in their structure and function. An overview of the architecture of rod and cone cells is given below:

Rod Cells:

Rod cells are photoreceptor cells in the retina of the eye. They are long and cylindrical in shape. They are more in number than cone cells. On average, there are approximately 92 million rod cells in the human retina compared to 6 million cone cells. Rod cells are highly sensitive to light and are specialized for vision in dim or low-light conditions. They do not provide colour vision or fine detail perception. They are found concentrated at the outer edges of the retina and are responsible for peripheral vision. They can detect motion, and provide vision in low-light conditions. Rod cells contain a photopigment called rhodopsin. Rod cells transmit the electrical signals from the rods to the ganglion cells in the retina. This pathway is responsible for transmitting visual information to the brain.

Cone Cells:

Cone cells are cone-shaped and have a shorter and wider structure compared to rod cells. They are mainly concentrated in the central region of the retina, known as the fovea. Cone cells contain three types of photo-pigments. These pigments are responsible for colour vision and are most sensitive to different wavelengths of light: short (blue), medium (green), and long (red) wavelengths. They provide the ability to distinguish colours and perceive fine details. Cone cells have a lower sensitivity to light compared to rod cells and require brighter light conditions (photopic vision) for activation. They are responsible for bright light vision. Cone cells also transmit the electrical signals to the ganglion cells for further processing and transmission to the brain.

The rod and cone cells together in the retina are responsible for visual sensitivity across the visual field. The high density of cones in the fovea helps in high-resolution

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and colour vision in the central visual field. The rods present in the peripheral retina helps in better sensitivity in low-light conditions and detection of motion.

EYE AS A CAMERA SYSTEM

The eye can be considered as a complex camera system due to its similarities in capturing and processing of visual information. The simple analogy between eye and camera is given below:

Different Parts	Eye	Camera
Cornea and Lens	In the eye, the cornea and lens focus incoming light onto the retina, which acts as the eye's "image sensor."	In a camera, the lens focuses light onto the image sensor or film.
Iris and Pupil: Aperture	The iris in the eye controls the size of the pupil which regulates the amount light that enters eye.	Camera's aperture and shutter speed regulates the amount of light that enters the camera.
Retina : Image sensor	The retina in the eye contains specialized cells called photoreceptors, mainly rods and cones cells. They capture light and convert it into electrical signals.	The image sensor or film in a camera captures image.
Rods and Cones: Pixels	Rods are highly sensitive to light and are responsible for vision in low-light conditions, while cones are responsible for colour vision and high-resolution details.	Pixels in a camera's image sensor play the role of rods and cones cells in the eye.
Optic Nerve: Cables or connectors	The optic nerve carries visual information from the retina to the brain for further processing and interpretation.	In camera, the cables or connectors transmit electrical signals from the camera's image sensor to the processing

		unit.
Image Processing	In the eye, the information is processed in the retina and transmitted to the brain for interpretation.	In a camera, the image sensor's signals are processed by the camera's electronics and can be further modified or enhanced digitally.
Focus and Accommodation	The eye has ability to adjust focus and accommodation, allowing for clear vision at different distances.	Cameras have autofocus and manual focus mechanisms.
Eyelids and Tears	The eyelids in the eye act as a protective cover. Tears lubricate and cleanse the eye's surface, maintaining optimal vision and protecting against dryness.	The lens cap or camera body cover is used to shield the camera's lens and sensor. Lens cleaning solutions are used to keep camera lenses clean.

The eye-camera analogy helps in understanding the basic functions of the eye. But, the eye is a highly complex and sophisticated biological system which is more capable than even the most advanced cameras.

CATARACT

Cataract is a common eye problem in which natural lens inside the eye is clouded, leading to blurred or hazy vision. It is usually associated with aging but can also occur due to other factors such as eye injuries, certain medications, or due to other medical problems.

The main symptoms of cataract are:

1. Blurred or cloudy vision.

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2. Sensitivity to glare, especially from bright lights or sunlight.

3. Difficulty seeing clearly at night.

4. Colors appearing faded or yellowed.

5. Frequent changes in eyeglass prescription.

Cataract surgery and Intraocular Lenses: Cataract surgery is a common surgical procedure performed to remove a cataract, which is the clouding of the natural lens inside the eye. Cataract surgery aims to restore clear vision for individuals affected by cataracts and improve quality of life. In this surgery, cataract is removed first by surgical method. After removal of cataracts, cloudy, dysfunctional lens is replaced by implanting a clear, artificial lens. Intraocular Lenses (IOLs) are the artificial lenses surgically implanted inside the eye to replace the natural lens. The IOL is placed inside the remaining lens capsule, providing focus and clarity. IOLs can correct refractive errors, and eliminate the need for glasses or contact lenses.

In the early stages of cataracts, some lifestyle modifications and visual aids may help to improve vision temporarily, such as:

- Wearing eyeglasses or contact lenses with a new prescription.
- Using anti-glare sunglasses to reduce glare and improve vision in bright light.
- Increasing lighting for reading or other close-up activities.
- Magnifying lenses or reading glasses for near vision tasks.

The cataracts cannot be prevented or reversed with medications, eye drops, or other non-surgical treatments. Once a cataract starts affecting a person's quality of life or daily activities, surgical intervention is the main solution.

OPTICAL CORRECTIONS

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When the eye is unable to properly focus light onto the retina, it results in blurred or distorted vision. This vision problem is called as refractive errors. Optical corrections are the techniques and devices used to improve or correct refractive errors. These corrections aim to enhance visual acuity and clarity. Optical corrections enhance the eye's ability to focus light properly onto the retina.

The most common types of optical corrections are:

1. **Eyeglasses:** Eyeglasses (spectacles) are the most widely used form of optical correction. They consist of lenses which help to refract light properly and focus correctly on the retina, resulting in clearer vision. A specific lens is used to address specific refractive errors, such as near-sightedness (myopia), far-sightedness (hyperopia), blurred or distorted vision (astigmatism), and age related vision problems (presbyopia).

2. **Contact Lenses:** Contact lenses are thin, curved lenses that are placed directly on the surface of the eye. They work similar to eyeglasses but provide a more natural field of view and avoid the visual obstruction caused by frames.

3. **Cataract surgery and Intraocular Lenses:** Cataract surgery is a common surgical procedure performed to remove a cataract, which is the clouding of the natural lens inside the eye. Cataracts can cause blurry or hazy vision, difficulty seeing in low light, glare sensitivity, and overall visual impairment. Cataract surgery aims to restore clear vision and improve quality of life for individuals affected by cataracts. After removal of cataracts, cloudy, dysfunctional lens is replaced by implanting a clear, artificial lens. Intraocular Lenses (IOLs) are the artificial lenses surgically implanted inside the eye to replace the natural lens. The IOL is placed inside the remaining lens capsule, providing focus and clarity. IOLs can correct refractive errors, and eliminate the need for glasses or contact lenses.
4. **Refractive Surgery:** Refractive surgery, such as LASIK (Laser-Assisted in Situ Keratomileusis) and PRK (Photorefractive Keratectomy), aims to permanently reshape the cornea to correct refractive errors. These procedures use laser

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technology to precisely remove corneal tissue, altering its curvature to improve focusing ability. Refractive surgery is effective for long-term vision correction without relying on glasses or contact lenses.

The appropriate optical correction method varies based on individual factors such as the type and severity of refractive error, overall eye health, lifestyle preferences, and personal suitability.

LENS MATERIALS

Different types of lens materials are used in eye wears. The choice of lens material depends on factors such as prescription strength, desired thickness, durability, and visual clarity. Some common lens materials used in eyeglasses are:

1. **Glass:** Glass lenses were widely used in the past but have become less common now. They offer excellent optical clarity but are heavier and more fragile than other materials.
2. **Plastic (CR-39):** CR-39 is a lightweight plastic material commonly used for eyeglass lenses. It is less prone to shattering than glass and is relatively cheaper. However, it has lower impact-resistance than other materials and also it is required in thicker lenses for higher prescriptions.
3. **Polycarbonate:** Polycarbonate lenses are lightweight, impact-resistant, and thinner than CR-39 plastic lenses. They provide excellent durability and are a popular choice, especially for children's eyewear, safety glasses, and sports eyewear. Polycarbonate lenses also provide built-in UV protection.
4. **High-Index Plastic:** High-index plastic lenses are thinner and lighter than traditional plastic lenses. They are particularly beneficial for individuals with stronger prescriptions. They can reduce lens thickness and improve cosmetic appearance.

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5. **Photochromic:** Photochromic lenses have the ability to darken when exposed to UV light and return to a clear state in the indoors. They provide the convenience of both regular eyeglasses and sunglasses in one lens.

BIONIC EYE

A bionic eye, is a visual implant designed to restore partial vision to individuals with severe vision loss or blindness. It is also known as a retinal prosthesis or artificial retina. Here, the technology is used to bypass non-functioning parts of the eye. Then, the cells in remaining partially functioning parts like the retina or optic nerve are directly stimulated to create visual perceptions. Bionic eye implantation is useless if optic nerve or visual cortex or retina are completely damaged.

Main components of bionic eye are:

1. **External Components:** A bionic eye system has an external camera or sensor that captures visual information from the environment. This device can be worn on glasses or as a head-mounted unit.
2. **Processing Unit:** The captured visual information is processed by an external unit based on specific algorithms. This converts the visual signals into electrical signals and transfers the signal to implanted device.
3. **Implant:** The implant is surgically placed inside the eye or connected to the optic nerve. It consists of an array of electrodes that stimulate the remaining retinal cells or the optic nerve to transmit electrical signals to the brain.
4. **Visual Perception:** The brain interprets the electrical signals received from the bionic eye as visual sensations. The resulting visual perceptions may be limited and vary among individuals. They may appear as patterns of light or dark areas or the perception of simple shapes or objects.

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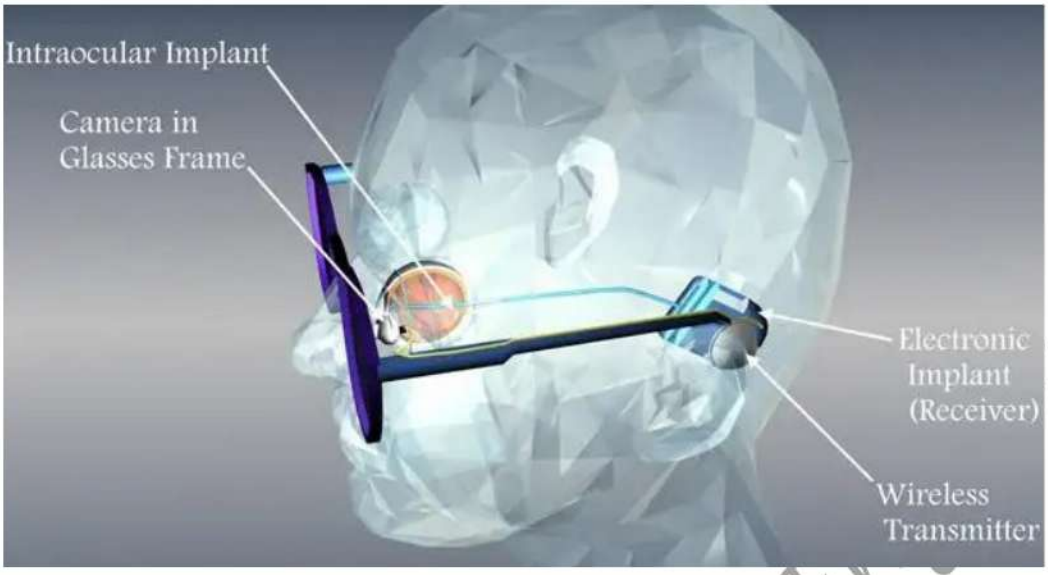

Bionic eye technology

Bionic eye technology is still evolving, and different research groups and companies worldwide are actively working on developing and improving the technology. Bionic eyes hold promise for restoring vision, but the current technology is not able to fully replicate the complexity and resolution of natural vision. The level of visual improvement varies among individuals. The technology is currently more effective for individuals with severe vision loss rather than complete blindness.

HEART

HEART AS A PUMP SYSTEM: ARCHITECTURE OF HEART

The heart functions as a powerful pumping system responsible for circulating blood throughout the body. It works continuously to ensure the delivery of oxygen, nutrients, hormones, and other vital substances to the tissues and organs.

The heart pumps blood in a coordinated and cyclical manner through a series of contractions and relaxations. The architecture of the human heart and its function as pumping system is described below:

1. The heart has four chambers: two upper chambers called atrium (left and right atrium) and two lower chambers called ventricles (Left and right ventricles).

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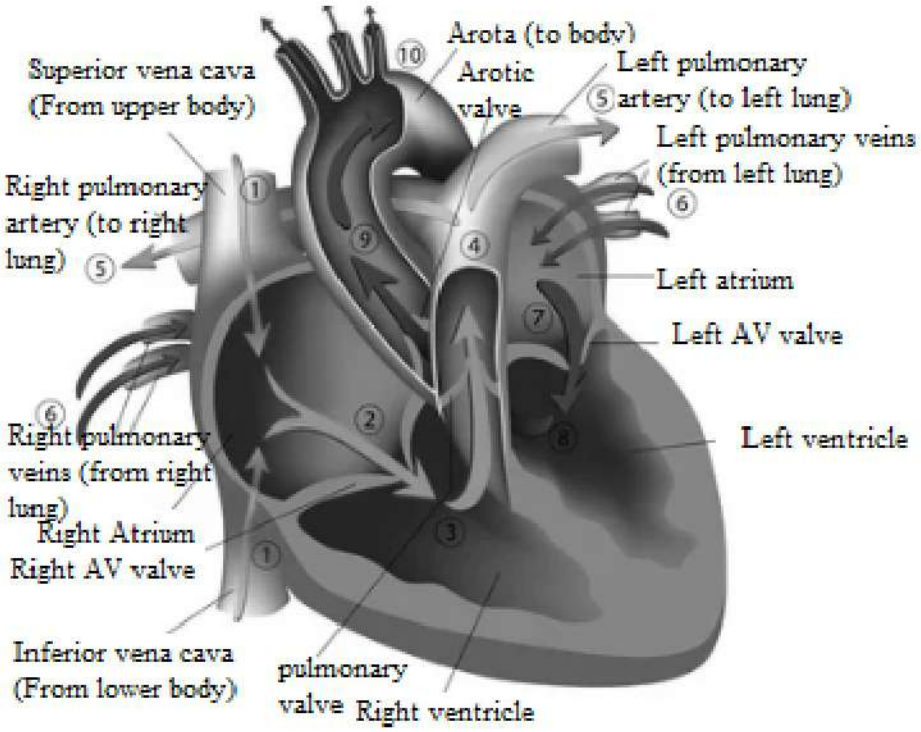
2. Deoxygenated blood from the body's tissues, enters the heart through two large veins called the superior vena cava (from the upper body) and inferior vena cava (from the lower body). This blood enters the right atrium.
3. When the right atrium contracts, it pushes the deoxygenated blood through a valve called the tricuspid valve into the right ventricle.



Blood Flow pathway in heart

4. The right ventricle contracts, pumping the deoxygenated blood through another valve called the pulmonary valve into the pulmonary artery.
5. The pulmonary artery carries the deoxygenated blood to the lungs. In the lungs, the blood picks up oxygen and releases carbon dioxide, a waste product.
6. Oxygenated blood from the lungs enters the heart through four pulmonary veins and flows into the left atrium.
7. When the left atrium contracts, it pushes the oxygenated blood through a valve called the mitral valve into the left ventricle.

8. The left ventricle, which is the strongest chamber of the heart, contracts and pumps the oxygenated blood through another valve called the aortic valve into the aorta.
9. The aorta is the largest artery in the body and carries the oxygenated blood to all the organs and tissues, supplying them with oxygen and nutrients.
10. The organs and tissues extract oxygen and nutrients from the blood and release waste products, such as carbon dioxide, into the bloodstream.
11. Deoxygenated blood, carrying waste products, returns to the heart through the superior vena cava and inferior vena cava, and the whole process starts again.

So, in simple terms, deoxygenated blood from various organs enters the heart, gets pumped to the lungs to pick up oxygen. Oxygenated blood returns to the heart, and then gets pumped out to the body's organs and tissues. This continuous circulation of blood is vital for delivering oxygen and nutrients to the body and removing waste products. The coordinated contractions and relaxations of the heart chambers, along with the opening and closing of valves, ensure efficient blood flow throughout the cardiovascular system.

ELECTRICAL CONDUCTION SYSTEM IN HEART

The electrical signal system in the heart is responsible for rhythmic contractions of the heart muscle. It ensures that the chambers of the heart contract in a synchronized manner, allowing efficient pumping of blood. The simple working of electrical signal system in the heart is given below:

1. **Sinoatrial (SA) Node:** The electrical signal originates in a specialized group of cells called the SA node, located in the upper part of the right atrium. The SA node acts as the natural pacemaker of the heart. It generates electrical impulses that initiate each heartbeat.

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2. **Atrial Conduction:** The electrical impulses generated by the SA node spread across the atria and causes contraction of atria. This contraction pushes blood from the atria into the ventricles.
3. **Atrioventricular (AV) Node:** The AV node is located at the junction between the atria and ventricles. The AV node acts as a gatekeeper. It delays the electrical impulses briefly at the atria and allows it to fully contract. This results in complete transfer of blood from atria to ventricle and thus, efficient filling of the ventricles.
4. **Bundle of His and Bundle Branches:** After passing through the AV node, the electrical signals travel down a pathway called the bundle of His. This pathway splits into two branches, known as the left and right bundle branches, which extend into the ventricles.: It carries the electrical signal from the AV node to the ventricles.
5. **Purkinje Fibers:** The electrical signals then spread rapidly through specialized fibers called Purkinje fibers. These fibers distribute the electrical impulses throughout the ventricles, causing them to contract in a coordinated manner.
6. **Ventricular Contraction:** The synchronized contraction of the ventricles pumps blood out of the heart. The electrical signals initiate the contraction of the ventricles from the bottom upwards, ensuring an efficient squeezing action.
7. **Relaxation and Recharging:** After contraction, the electrical signals cease, and the heart muscles relax. This relaxation phase allows the chambers to refill with blood in preparation for the next heartbeat.

The electrical signal system in the heart ensures a regular and coordinated rhythm of contractions. This allows the heart to efficiently pump blood throughout the body. Any disruptions or abnormalities in the electrical system can lead to heart rhythm disorders, such as arrhythmias.

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HEART-RELATED ISSUES

There are various conditions and diseases that can affect the normal functioning of the heart. These can range from minor problems to life-threatening conditions. Some common heart-related issues are briefly described below:

1. **Coronary Artery Disease (CAD):** CAD occurs when the coronary arteries, which supply blood to the heart muscle, become narrowed or blocked due to the build-up of plaque. It can lead to chest pain (angina) or a heart attack if the blood flow to the heart is significantly reduced or blocked.
2. **Heart Failure:** Heart failure is a condition in which the heart is unable to pump enough blood to meet the body's needs. It can occur due to various reasons, including weakened heart muscles (cardiomyopathy), high blood pressure, heart valve problems, or previous heart attacks.
3. **Arrhythmias:** Arrhythmias are abnormal heart rhythms. It can be too fast heartbeat (tachycardia), too slow heart beat (bradycardia), or irregular heart beat. They can lead to symptoms like palpitations, dizziness, or fainting.
4. **Heart Valve Disorders:** Heart valve disorders involve problems with the valves that regulate blood flow in the heart. Conditions such as valve stenosis (narrowing) or valve regurgitation (leakage) can cause the heart to work harder. This leads to symptoms like fatigue, shortness of breath, or chest discomfort.
5. **Congenital Heart Defects:** Congenital heart defects are structural abnormalities present at birth that affect the heart's normal functioning. These defects can range from simple issues that may not cause significant problems to complex conditions requiring surgical intervention.
6. **Cardiac Arrest:** Cardiac arrest is a sudden and abrupt loss of heart function. It causes heart to stop beating effectively. It results in the cessation of blood flow to the body's vital organs. It is a medical emergency that requires immediate intervention, such as CPR and defibrillation, to restore normal heart rhythm.
7. **Cardiomyopathy:** Cardiomyopathy refers to diseases that affect the heart muscle, causing it to become enlarged, thickened, or stiff. This can impair the heart's ability to pump blood effectively.

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8. **Heart Infections:** Infections of the heart, such as endocarditis or myocarditis, can damage the heart valves, muscle, or lining and interfere with normal heart function.

ECG (ELECTROCARDIOGRAM)

ECG (electrocardiogram) is a valuable diagnostic tool which is used to assess and monitor various heart problems. It records the electrical activity of the heart. It provides important information about the heart's rhythm, rate, and electrical conduction. It is a commonly used tool in cardiology to evaluate heart health, detect abnormalities, and monitor the heart's function.

During an ECG, small, adhesive electrodes are placed on specific locations on the chest, arms, and legs. These electrodes detect the electrical signals generated by the heart as it contracts and relaxes. The electrodes are connected to an ECG machine. The electrical signals received from heart are amplified and recorded as a series of waveforms on graph paper.

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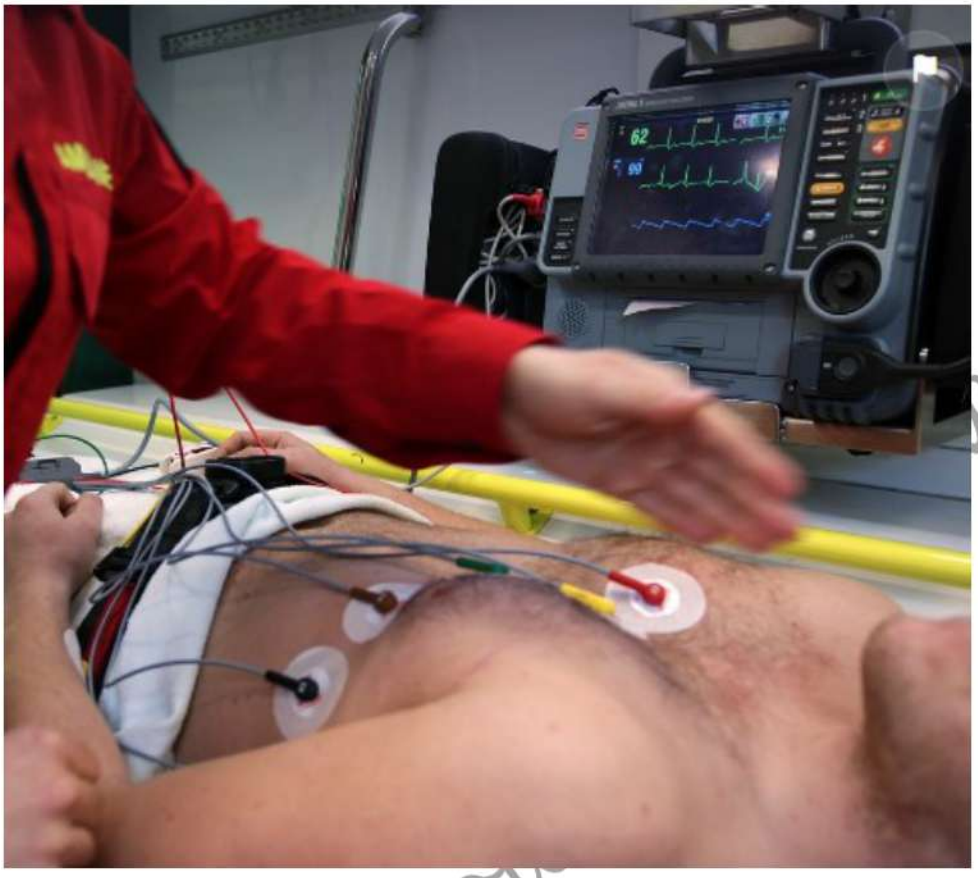

ECG (electrocardiogram)

The ECG provides valuable information about various aspects of heart function, including:

a) Heart rate,
b) Abnormalities in the heart's rhythm, such as atrial fibrillation, ventricular tachycardia, bradycardia, or heart block.
c) Abnormalities in the electrical conduction of heart.
d) Insufficient blood flow (ischemia) to certain areas of the heart called as Ischemic Heart Disease.
e) Past damage to heart from a heart attack (myocardial infarction).
f) Structural abnormalities in the heart, such as hypertrophy (enlargement) of the heart chambers.
g) To monitor and evaluate the functioning of Pacemaker and Implantable Cardioverter Defibrillator (ICD) device.

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ECG monitoring is a non-invasive and relatively simple procedure. It provides valuable information about the heart's electrical activity and diagnosing and managing of various heart problems.

REASONS FOR BLOCKAGES OF BLOOD VESSELS

Blockages of blood vessels is known as atherosclerosis. It occurs due to the build-up of fatty deposits called plaque on the inner walls of the arteries. Several factors contribute to the development of these blockages. Some common reasons for blockages of blood vessels are given below:

1. **High Cholesterol:** High levels of low-density lipoprotein (LDL) cholesterol and lower levels of high-density lipoprotein (HDL) cholesterol in the blood can lead to the accumulation of plaque in the arteries. LDL cholesterol can deposit in the arterial walls, promoting inflammation and the formation of plaque.
2. **Smoking:** Smoking damages the inner lining of blood vessels, making them more susceptible to plaque build-up. Smoking also lowers levels of high-density lipoprotein (HDL) cholesterol, which is called as good cholesterol.
3. **High Blood Pressure:** High blood pressure causes stress and damage to the arterial walls, making them more prone to the formation of plaque.
4. **Diabetes:** People with diabetes often have high blood sugar levels, which can damage the blood vessels and promote the development of atherosclerosis.
5. **Obesity:** Excess body weight, especially around the waist, is associated with increased inflammation and the release of certain substances that contribute to the formation of plaque.
6. **Sedentary Lifestyle:** Lack of regular physical activity can contribute to the development of blockages. Exercise helps maintain healthy blood vessels, promotes better cholesterol levels, and improves overall cardiovascular health.
7. **Unhealthy Diet:** A diet high in saturated fats, trans fats, cholesterol, and refined carbohydrates can increase the risk of atherosclerosis. These types of foods contribute to elevated LDL cholesterol levels and inflammation.

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8. **Genetics:** Some individuals may have a genetic predisposition to developing atherosclerosis. Certain gene variations can affect how the body metabolizes cholesterol and processes fats, increasing the risk of plaque formation.
9. **Age and Gender:** The risk of developing blockages increases with age. Men are generally at higher risk than premenopausal women, although the risk evens out after menopause.

These factors can interact with each other, compounding the risk of developing atherosclerosis. Making healthy lifestyle choices, managing risk factors, and seeking appropriate medical care can help reduce the risk of blockages and related cardiovascular conditions.

DESIGN OF STENTS

Stents are medical devices used to treat blockages or narrowings in blood vessels, particularly in coronary arteries. The design of stents has evolved over time to improve their effectiveness and safety.

The presently used stents are made of metal alloys, such as stainless steel, cobalt-chromium, or platinum-chromium. The stent consists of a tubular mesh-like structure with small openings called struts. The struts allow the stent to expand and provide support to the blood vessel. Balloon-expandable stents are the most common type. They are mounted on a deflated balloon catheter and inserted into the narrowed artery. When the balloon is inflated, the stent expands, pressing against the arterial wall and restoring blood flow. Once expanded, the balloon is deflated and removed, leaving the stent in place.

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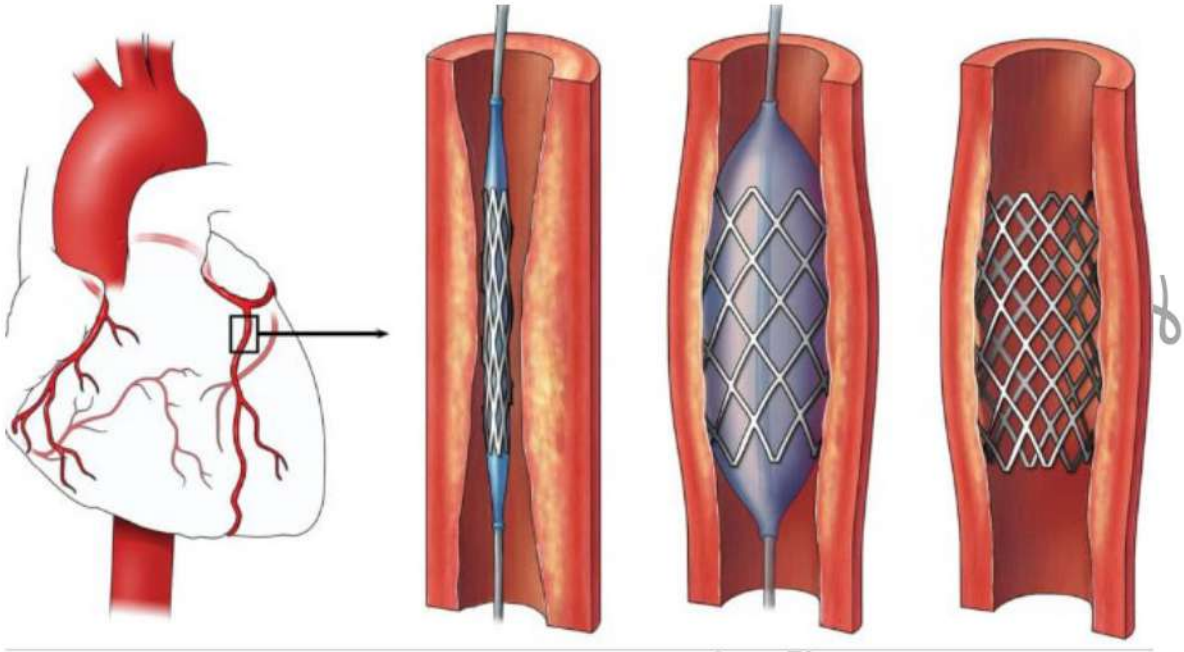

Design of stents

There is a continuous advancement in the design of stents. Stents with improved deliverability, enhanced scaffolding properties, better long-term outcomes, and reduced risk of complications are being introduced.

Few examples of modern stent materials are:

1. Self-Expanding stents made of materials with shape memory properties, such as nitinol. These stents have a pre-set shape and can expand on their own when deployed.
2. Bio-resorbable stents which can gradually dissolve over time, leaving behind a restored artery. These stents are made from materials that are naturally absorbed by the body, such as polylactic acid (PLA) or polyglycolic acid (PGA). As the stent dissolves, the artery can regain its natural function.
3. Stents with thinner struts which exhbits improved deliverability, reduced risk of vessel injury, and potentially better long-term outcomes.
4. Biocompatible stent materials which are not harmful or toxic to the body and do not trigger an immune response. This ensures that the stent can be safely implanted without causing adverse reactions or complications.

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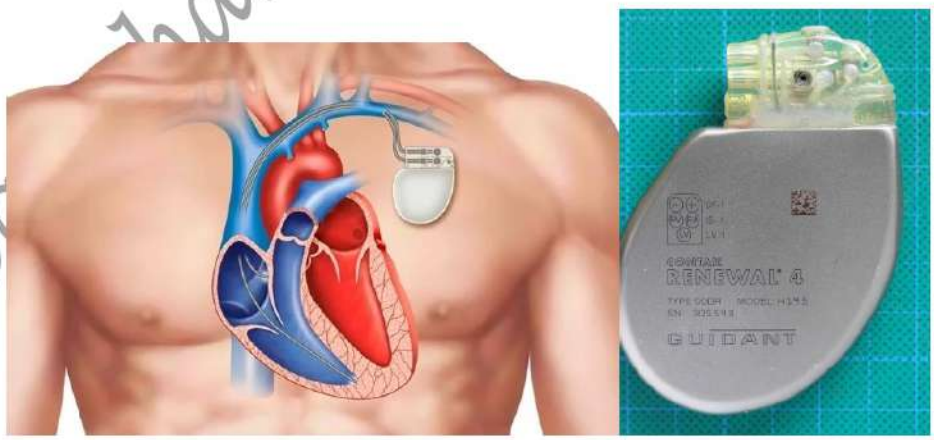
5. Drug-Eluting Stents, which are coated with medications that help prevent the re-narrowing of the treated artery, a condition called restenosis. The drug coating is typically a polymer that slowly releases an anti-proliferative drug, inhibiting the growth of cells that could cause re-narrowing.

Stent selection depends on various factors, including the patient's condition, the location of the blockage, and the specific characteristics of the stent.

PACEMAKERS

Pacemakers are small electronic medical devices which are implanted in the body to assist heart functioning. Pacemakers monitor the heart's electrical activity and deliver electrical signals to the heart. These electrical impulses stimulate the heart muscles. They ensure that the heart beats at a regular pace. They help to regulate abnormal heart rhythms, specifically those that are too slow (bradycardia). They help prevent symptoms such as fatigue, dizziness, and fainting which result from a slow heart rhythm.

Pacemakers are implanted through a surgical procedure under local anesthesia. A small incision is made near the collarbone, and the pacemaker leads (thin wires with electrodes) are threaded through a vein into the heart chambers. The leads are then attached to the pacemaker device, which is placed under the skin.



Pacemaker

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Pacemakers are powered by batteries which can last for several years. The battery life depends on various factors such as the type of pacemaker, the programming settings, and the patient's heart rhythm. When the battery runs low, the pacemaker needs to be replaced or the battery can be changed during a simple outpatient procedure. After pacemaker implantation, regular check-ups with a cardiologist are necessary to ensure proper functioning of the device. The pacemaker can be programmed and adjusted during to optimize its performance based on the patient's heart condition.

DEFIBRILLATORS

Cardiac arrest is condition in which the heart suddenly stops beating effectively. This leads to cessation of blood flow to the body's vital organs. It is a sudden and life-threatening medical emergency condition. It requires immediate intervention to prevent death and minimize potential complications.

During cardiac arrest, the electrical signals that control the heart's pumping action become disrupted, leading to an abnormal heart rhythm. This is called as ventricular fibrillation. A normal heart rhythm can be restored in certain cases of cardiac arrest by delivering an electric shock.

Defibrillators are medical devices that are used to deliver an electric shock to the heart when a person is experiencing a cardiac arrest. The purpose of a defibrillator is to restore a normal heart rhythm by stopping abnormal electrical activity in the heart. It helps the natural pacemaker of the heart to resume control. Accessing and using of a defibrillator within the first few minutes of cardiac arrest can significantly improve the chances of survival.

There are two main types of defibrillators:

1. **Automated External Defibrillators (AEDs):** AEDs are portable devices commonly found in public places such as airports, schools, and sports arenas.

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They can be used by a non-medical personnel without any medical training. In the process, adhesive electrode pads of defibrillator are placed on the chest of the person experiencing cardiac arrest. AEDs analyze the heart rhythm and, if necessary, deliver a shock to restore a normal rhythm. The shock temporarily stops the heart's electrical activity, allowing the heart's natural pacemaker to regain control and restore a normal rhythm. These devices are crucial in the early stages of a cardiac arrest before professional medical help arrives.

2. **Implantable Cardioverter Defibrillators (ICDs):** ICDs are surgically implanted in individuals who are at high risk of developing life-threatening arrhythmias. They continuously monitor the heart's rhythm and, if an abnormal rhythm is detected, can deliver a shock to restore a normal rhythm. ICDs are typically recommended for people with a history of serious arrhythmias or certain heart conditions.

Defibrillation is just one component of the overall chain of survival in cardiac arrest situations. The main critical factors which can improve a person's chances of survival are early recognition, immediate cardiopulmonary resuscitation (CPR), early access to emergency medical services, and timely defibrillation.

LUNGS

ARCHITECTURE OF LUNGS

The lung is an important organ in the human body. It is part of the respiratory system. It is mainly involved in the process of breathing. The primary function of lungs is to facilitate the exchange of gases between the air we breathe and the bloodstream.

The architecture of lungs is complex and well organized. Knowledge of the architecture of lung is essential for understanding the relationship between morphology -functions of the lung component. This helps to understand the mechanism of gas exchange process.

An overview of the main structures and components of the lungs along with their working mechanism is described below:

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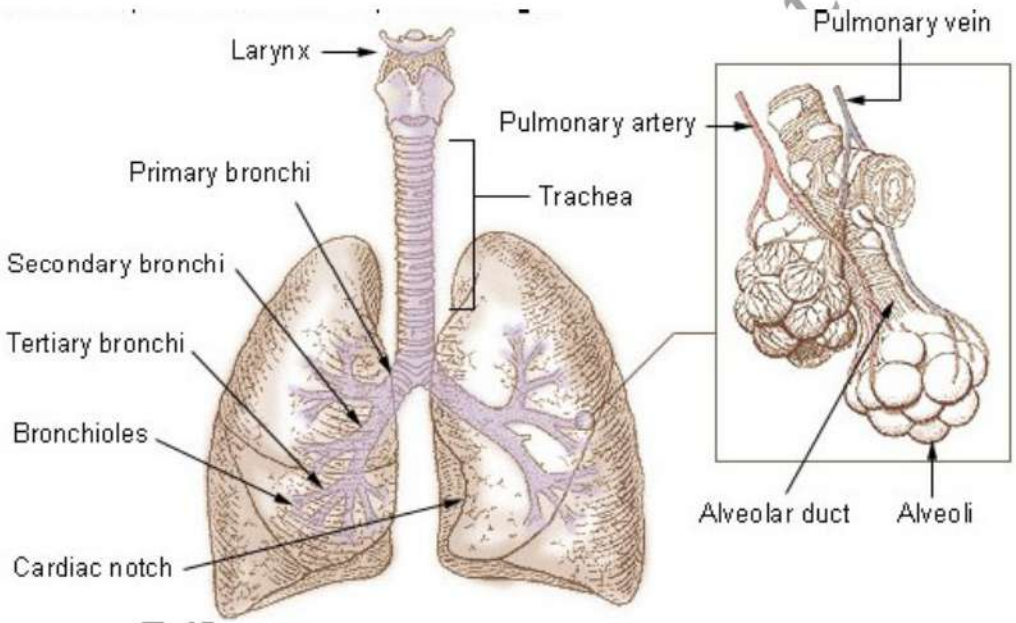
1. **Trachea:** When we breathe, air enters in to the nose and passes through nasal passages to wind pipe. This wind pipe is called as the trachea. It is a rigid tube made of cartilage rings that connects the larynx (voice box) to the bronchi. It serves as the main passageway for air to enter and exit the lungs.
2. **Bronchi:** At the entry point of lungs, the trachea divides into two main branches one leading to each lung called as bronchi.
3. **Bronchioles:** The bronchi further divide into small, narrow branches known as bronchioles. This branching network extends throughout the lungs. Bronchioles contain smooth muscle which can contract or expand easily to regulate airflow.



4. **Alveoli:** Alveoli are tiny, grape-like sacs at the ends of the bronchioles. They are the primary sites of gas exchange in the lungs. Each lung contains millions of alveoli. They provide a large surface area for efficient oxygen and carbon dioxide exchange.

5. **Pulmonary Capillaries:** Alveolus are surrounded by networks of tiny blood vessels called as pulmonary capillaries. These capillaries have thin walls

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which allow the exchange of gases between the air in the alveoli and the bloodstream.

6. **Diaphragm:** The diaphragm is a dome-shaped muscle located below the lungs. It plays a crucial role in respiration. It contracts and flattens during inhalation and hence increases the volume of the chest cavity. This allows the lungs to expand during inhalation.

Thus, the architecture of the lungs is designed to maximize the surface area available for gas exchange.

GAS EXCHANGE MECHANISMS IN LUNGS

The lung is the main organ involved in the process of breathing. Breathing is inhaling oxygen-rich air and exhaling carbon dioxide rich air.

Mechanism of gas exchange during inhaling and exhaling is briefly described here:

1. The first step is contraction of diaphragm. When diaphragm contracts, the muscles in the rib cage expand the chest cavity. This causes the expansion of lungs also. This expansion lowers the pressure inside the lungs. Due to difference in pressure, air is drawn in to the respiratory system.

2. Air enters in to the nose and passes through nasal passages to trachea.

3. Then, air passes through bronchi, bronchioles and diffuses across the thin walls of the alveoli (tiny air sacs in the lungs) into the surrounding blood capillaries.

4. The concentration of oxygen in the alveoli of the lungs is higher than in the blood. Hence, oxygen diffuses into the bloodstream. There, oxygen binds to iron of hemoglobin molecules within red blood cells, forming oxyhemoglobin.

5. This oxygenated blood is then transported to the heart, which pumps it to various organs and tissues in the body.

6. Concentration of oxygen within cell is less than in oxygenated blood, thus at cell walls, the oxygen molecules dissociate from the oxyhemoglobin and diffuse across the cells.

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7. In this way oxygen is supplied to each and every cell where it is utilized for cellular respiration, energy production, and metabolism.

8. At the same time, carbon dioxide, a waste product generated by cellular metabolism binds to iron of hemoglobin molecules within red blood cells, forming carboxy-hemoglobin.

9. This carboxy-hemoglobin in blood is transported to the heart, which pumps it back into the alveoli of lungs and exhaled out.

Thus, gas exchange in the lungs occurs primarily through a process of diffusion. It involves the movement of gases (oxygen and carbon dioxide) across the thin walls of the alveoli (air sacs) and the surrounding pulmonary capillaries.

LUNGS AS PURIFICATION SYSTEM

The lungs have certain built-in mechanisms to filter out larger particles and foreign substances present in the inhaled air. This mechanism helps to filter the air and protect the respiratory system from harmful substances. The role of lungs in purification mechanism is briefly described here:

1. When we breathe, the nose and nasal passages act as the first line of defense. They trap larger particles such as dust, pollen, and pollutants.
2. The mucus lining of the respiratory tract also helps to trap and remove particles, preventing them from reaching deeper into the lungs.
3. The cilia, tiny hair-like structures lining the airways, then work in coordination with the mucus to move trapped particles upward, where they can be expelled by coughing or swallowing.
4. The respiratory system, including the lungs, has an immune defense system that helps protect against pathogens such as bacteria, viruses, and fungi.
5. The mucus in the respiratory tract contains antibodies and enzymes that can neutralize or destroy pathogens.
6. The lungs plays main role in removing carbon dioxide from blood. Carbon dioxide can act as an acid when dissolved and can reduce pH of blood.

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7. The lungs help regulate the pH balance of the blood by controlling the levels of carbon dioxide. Carbon dioxide can act as an acid when dissolved in water. Lungs remove carbon dioxide from blood and helps in maintaining optimal pH level of blood.

Thus, the lungs are responsible for breathing, gas exchange, oxygenation of blood, removal of carbon dioxide, acid-base balance regulation, and defense against foreign particles. Hence, the proper functioning of lungs is crucial for maintaining the body's overall health and well-being.

ABNORMAL LUNG PHYSIOLOGY

The main role of the lungs is to facilitate the exchange of oxygen and carbon dioxide between the body and the environment. Any deviation from the normal functioning of the lungs is referred to as abnormal lung physiology. Deviation from normal lung functioning can occur due to various factors like exposing to dangerous chemicals in the environment, lifestyle choices like smoking and various infections. These can result in various abnormalities of lungs. Few of them are given below:

1. **Chronic obstructive pulmonary disease (COPD):** It is characterized by airway obstruction that makes it difficult for air to flow in and out of the lungs. Obstruction can be caused by inflammation, narrowing of the airways, excessive mucus production, or a combination of these factors. This leads to symptoms like shortness of breath, wheezing, and decreased airflow.
2. **Asthma:** Asthma is a chronic condition characterized by inflammation and narrowing of the airways. It leads to recurrent episodes of wheezing, breathlessness, chest tightness, and coughing. Asthma symptoms are triggered by allergens, weather conditions and respiratory infections.
3. **Respiratory Infections:** Pneumonia is an infection of lungs caused by bacteria, viruses, fungi, or other microorganisms. It results in inflammation in the air sacs of the lungs. Symptoms associated with pneumonia are fever, cough, chest pain, and difficulty breathing. Bronchitis is another common infection of

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lungs. These infections can lead to fluid accumulation in the lungs, impaired gas exchange, and decreased lung function.

4. **Pulmonary Fibrosis:** Pulmonary fibrosis results in the scarring and thickening of lung tissue. This leads to progressive loss of lung function and impaired oxygen transfer. It can be caused by exposure to environmental toxins, or certain medications.
5. **Lung Cancer:** Lung cancer can affect lung physiology by causing the growth of abnormal cells within the lungs. Tumors can obstruct the airways, impair lung function, and interfere with gas exchange. Lung cancer can also spread to other parts of the body, further impacting lung function.

These are few examples of conditions that can result in abnormal lung physiology. Each condition can affect lung function in a different way. Diagnosis and treatment of abnormal lung physiology require a thorough evaluation by a healthcare professional, typically a pulmonologist or respiratory therapist, who can conduct tests, interpret results, and develop an appropriate management plan.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Chronic Obstructive Pulmonary Disease (COPD) is a chronic lung disease that affects the airways and lungs. It results in breathing difficulty due to the obstruction or narrowing of the airways. This obstruction is caused by chronic inflammation and damage to the lung tissue. The airways lose their elasticity, and the walls of the air sacs (alveoli) become inflamed and thickened, leading to reduced airflow. COPD is a chronic disease. A chronic disease refers to a long-lasting medical condition that generally progresses over time and often has no known cure.

The main causes of COPD are:

1. COPD is usually caused by long-term exposure to irritants, mainly cigarette smoke.

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2. It can also be caused by exposure to occupational dust and chemicals, indoor air pollution.
3. It can also be caused due to genetic factors.

The main symptoms of COPD are:

1. The most common symptoms of COPD include chronic cough, sputum production, shortness of breath, and wheezing. These symptoms often develop gradually and worsen over time.
2. Individuals with COPD may experience frequent respiratory infections, fatigue, and exercise intolerance.

COPD can be diagnosed based on a combination of symptoms, medical history, lung function tests, and imaging studies. Spirometry is a key test used to measure the reduced lung function and limited airflow in case of COPD patient.

There is no cure for COPD. The disease should be effectively managed to reduce symptoms, slow down disease progression, and improve quality of life. The main treatment options for COPD are:

1. Lifestyle changes like leaving smoking, reducing exposure to air pollutants and regular exercises.
2. Regular medications like bronchodilator inhalers which reduce swelling in the airways and relax the airways to keep them open.
3. Pulmonary rehabilitation programs and oxygen therapy.
4. Vaccinations against respiratory infections.

SPIROMETRY

Spirometry is a breathing test used to assess lung function. This test is performed using a device called as spirometer. During a spirometry test, a person is asked to breathe in and out through a mouthpiece connected to the spirometer. The spirometer records the volume and flow rate of the inhaled and exhaled air. Results are used to evaluate lung capacity and lung function.

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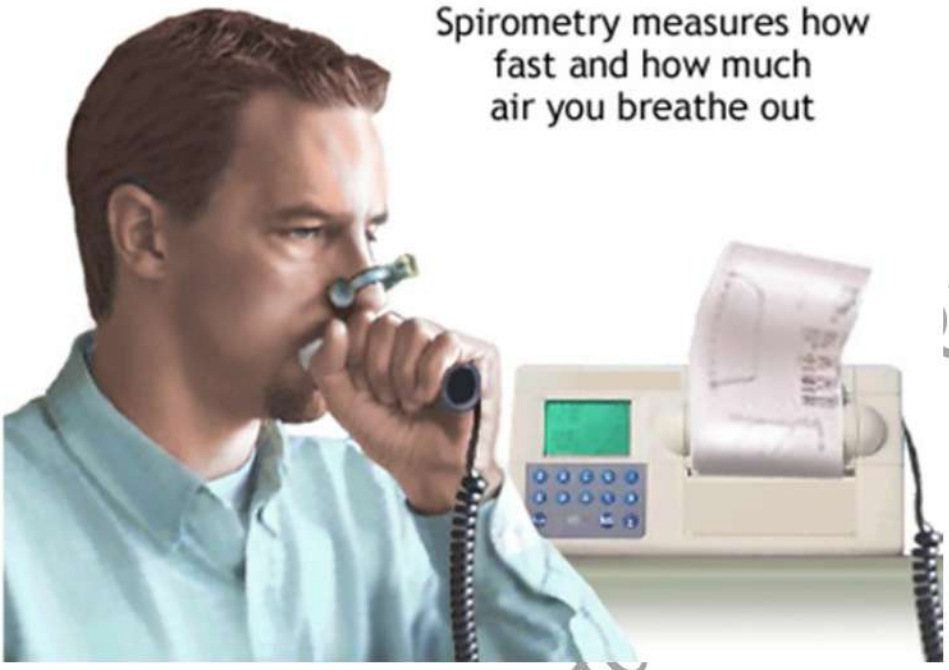

Spirometry

Spirometry is used to assess lung function:

1. When a person is having wheezing, shortness of breath, or a cough.
2. To diagnose problems like asthma, COPD, lung cancer, or pulmonary fibrosis.
3. To determine if the disease is improving or worsening and whether medications or inhalers are working properly.

Few of the key parameters measured in spirometry are:

1. Forced Vital Capacity (FVC): FVC is the maximum amount of air a person can forcefully exhale after taking a deep breath. It represents the total volume of air that can be expelled from the lungs.
2. Forced Expiratory Volume in 1 second (FEV1): FEV1 is the volume of air forcefully exhaled during the first second of the FVC operation. It is an important parameter used to assess airflow limitation.

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3. Peak Expiratory Flow (PEF): PEF is the maximum flow rate achieved during a forced expiration. It reflects the maximum speed at which a person can exhale air and is useful in monitoring conditions such as asthma.
4. Forced Expiratory Flow (FEF): FEF measures the flow rate during the middle portion of the forced expiration. Different FEF values can provide information about airflow at different points during exhalation.

VENTILATOR

When a patient condition is serious, and unable to breathe adequately on their own. Then, breathing support to patient is given using a medical device called as a ventilator. It is also known as a mechanical ventilator or a respirator. Ventilators are life-saving devices. With ventilators, healthcare providers can control and monitor breathing parameters, adjust oxygen levels, and support patients until their condition improves.

Ventilators are generally used in the following cases:

1. Ventilators are used in intensive care units (ICUs) to manage critically ill patients who require advanced respiratory support. They provide support to patients until their condition stabilizes or improves.
2. Ventilators can provide temporary respiratory assistance to patients with acute respiratory distress syndrome (ARDS), pneumonia, chronic obstructive pulmonary disease (COPD), or other conditions. They can maintain vital functions and buy time for the underlying condition to improve.

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Ventilator

3. During certain surgical procedures, general anesthesia can cause the suppression of natural breathing. Ventilators are used to provide controlled ventilation to patients while they are under anesthesia.

During ventilation Process, a predetermined volume of air/oxygen mixture humidified and supplied into the patient's lungs. This is known as the tidal volume. The patient is allowed to exhale passively. In some cases, a positive pressure is applied to assist with exhalation and prevent alveolar collapse. Healthcare professionals closely monitor the patient's vital signs and respiratory parameters using the ventilator's displays and alarms. They make adjustments to ventilation settings as needed, based on the patient's condition, blood gas analysis, and clinical assessment.

The four main components of ventilator are a control panel, breath delivery system, sensors and alarms. A control panel is used to input parameters such as tidal volume, respiratory rate, and oxygen concentration. Breath delivery system delivers the mixture of air and oxygen to the patient. Sensors and alarms are used to sense and alert the patient's respiratory parameters, such as airway pressure, tidal volume, and oxygen saturation levels.

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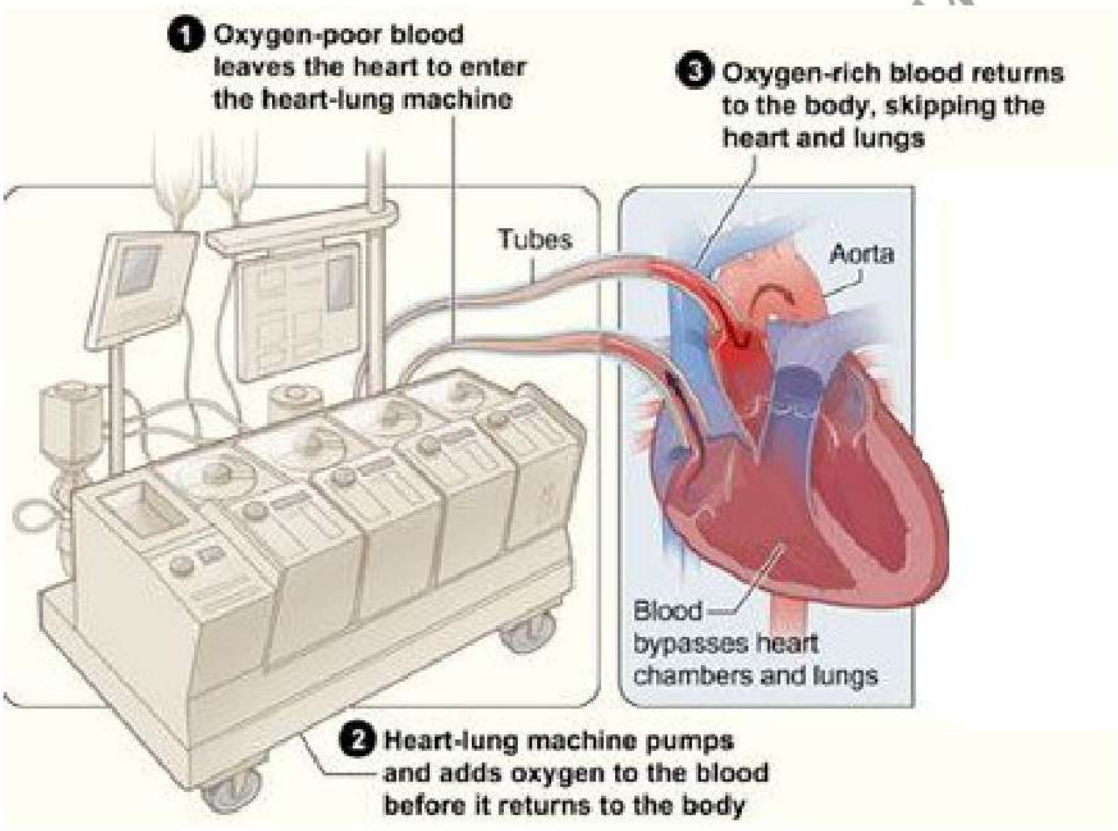
THE HEART-LUNG MACHINE

The heart-lung machine is a medical device which is used temporarily to perform the functions of the heart and lungs. It is used during complex cardiac surgeries. It acts like a bypass machine. It allows surgeons to temporarily stop the heart and create a still and bloodless surgical field. So that, they can perform intricate procedures on the heart and major blood vessels.



The heart-lung machine

The main components and functions of heart-lung machine are:

1. Pump: Pump serves the role of heart. It pumps oxygenated blood through the body and maintains blood flow to vital organs and tissues. It prevents damage of other organs like brain, kidney by supplying adequate amount of oxygen during the surgical procedure.

2. Oxygenator: This performs the function of the lungs. It adds oxygen to the blood outside the body and removes waste gases like carbon dioxide from it. It supplies oxygenated blood to pump so that it is circulated in the body.
3. Heat exchanger: It regulates the temperature of the blood when it passes through the machine and ensures that it is appropriate when returned to the patient's body.
4. Filters: They remove any air bubbles, clots, or other debris from the blood before it is circulated back into the patient's body.

The heart-lung machine has significantly contributed to the advancement of cardiac surgery and saved countless lives by allowing surgeons to perform complex procedures that were once considered impossible.

KIDNEYS

The kidneys are two bean-shaped organs. They are located just below the rib cage, one on each side of the spine. The primary function of the kidneys is to filter waste products from blood stream, while maintaining electrolyte balance. The filtered waste and excess fluids form urine. Healthy kidneys filter about a half cup of blood every minute, removing wastes and extra water to make urine. Blood circulates through kidneys many times a day and each time waste products are filtered out.

Apart from filtration of waste materials from blood stream, kidneys have other important functions as given below:

1. Blood Pressure Regulation: The kidneys produce a hormone called renin. Renin activates a series of reactions that help control blood pressure. And also, kidneys regulate the volume of blood plasma by adjusting the amount of water and sodium in blood.

2. Red Blood Cell Production: The kidneys produce a hormone called erythropoietin, which stimulates the bone marrow to produce red blood cells. Red blood cells are essential for oxygen transportation throughout the body.
3. Vitamin D Activation: The kidneys play a role in activating vitamin D, which is important for calcium absorption in the intestines. Vitamin D helps maintain proper bone health and plays a role in several other physiological processes.
4. Kidneys also remove excess acid produced by the cells and maintain acid-base balance in the body. Without this balance, nerves, muscles, and other tissues in your body may not work normally.
5. The kidneys maintain electrolyte balance in the body by filtration and reabsorption mechanism. This electrolyte and acid-base balance is highly required for proper functioning of nerves, muscles, and other tissues.

Thus, kidneys play a critical role in overall health and proper functioning of the body.

KIDNEY AS A FILTRATION SYSTEM

ARCHITECTURE OF KIDNEYS AND MECHANISM OF FILTRATION

The main role of kidney is to filter out waste products from blood stream. Hence, it is considered as a filtration system. The main components of kidney and their role in filtration mechanism are briefly described below:

1. **Renal artery:** The renal artery is a blood vessel through which oxygenated, unfiltered blood is supplied to the kidneys. Upon entering the kidney, the renal artery branches into smaller arteries, which further divide to form a complex network of blood vessels throughout the nephron. These branches ensure a rich blood supply to the nephrons for filtration.
2. **Nephron:** A nephron is the functional filtering unit of the kidney. It is responsible for filtering blood, reabsorbing essential substances, and producing urine. Each

kidney contains millions of nephrons. Each nephron contains two main parts, one is the glomerulus which performs filtration, and other a tubule. The tubule reabsorbs and returns needed substances to blood and removes wastes.

3. **Glomerulus:** The glomerulus is present in the initial part of a nephron. It is a network of specialized capillaries (tiny blood vessels) with small pores or openings in their walls. These small pores allow the passage of small molecules and fluid while preventing the passage of larger particles like blood cells and proteins. They are mainly responsible for filtration of blood.

As blood flows into each nephron, it enters in to the glomerulus with high pressure. Under high pressure, smaller molecules, wastes, and fluid are forced out through small pores and pass into the tubule. While, only larger molecules, such as proteins and blood cells, stay in the blood vessel. Thus, waste materials are filtered out of blood and moved to filtrate. Along with waste, other essential small molecules like electrolytes, and water are also removed and passed on to filtrate.

4. **Tubule:** The blood filtrate from glomerulus enters in to tubule. Tubule has highly coiled and convoluted structure with a large surface area. As the filtrate moves through the tubule, water and essential electrolytes, such as sodium, potassium, and calcium, are re-absorbed back. These essential small molecules are passed into the bloodstream through blood vessel which runs alongside the tubule. Thus, the tubule returns needed substances to blood and removes wastes. This process helps maintain proper hydration and electrolyte levels in the body. The remaining fluid and wastes in the tubule becomes urine.
5. **Collecting Ducts:** The collecting ducts receive urine from multiple nephrons.
6. **Ureter:** Urine is passed from the kidneys to the urinary bladder through two muscular tubes called ureters. Each ureter connects one kidney to the urinary bladder. Kidneys, ureters, and bladder are part of your urinary system.

Overall, the architecture of kidneys allows the efficient filtration of waste products and reabsorption of necessary electrolytes and fluid in to blood stream.

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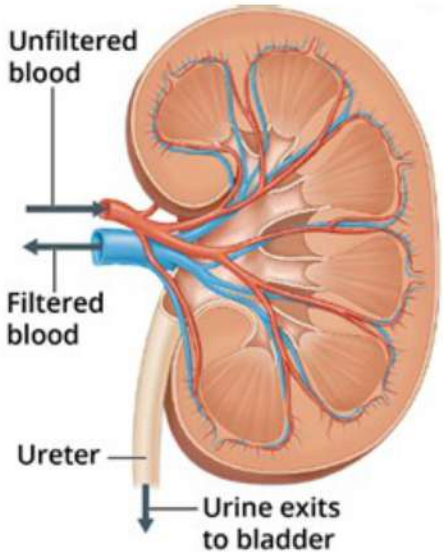

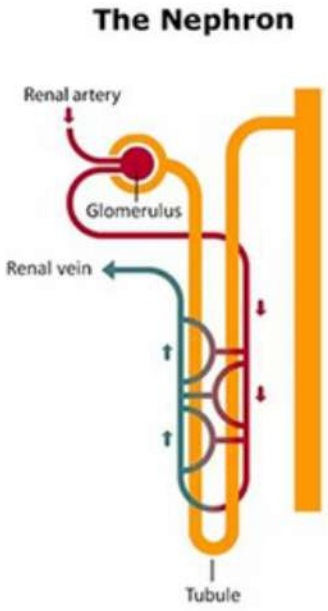

CHRONIC KIDNEY DISEASE (CKD)

The main role of kidneys is to filter waste products and excess fluids from the blood stream. Due to various reasons, kidneys may gradually and progressively loose their ability to filter wastes from blood. This condition is called as chronic kidney disease. It is a serious health issue that can lead to many other complications. It is referred to as "chronic" because it is a progressive disease that lasts for a long duration. In the early stages, CKD may not show noticeable symptoms. But, as the disease progresses, symptoms like fatigue, weakness, loss of appetite, weight loss, dry itchy skin, changes in urine output (increased or decreased), fluid retention, swelling (particularly in the legs and ankles), difficulty and increased or decreased urination at night are observed.

Chronic kidney disease can result in:

a) Building up of toxic wastes in blood.
b) Imbalance in the concentration of electrolytes.
c) Improper production of hormones that regulate blood pressure.
d) Improper production of hormones that produce red blood cells resulting in anemia.

e) CKD can result in a sudden loss of kidney function and may finally progress to end-stage renal disease (ESRD).

Main causes which may result in chronic kidney disease are:

a) Diabetes: High blood sugar levels over time can damage the small blood vessels in the kidneys.
b) Hypertension (high blood pressure): Uncontrolled high blood pressure can strain the kidneys, leading to damage.
c) Certain kidney conditions like glomerulonephritis and polycystic kidney disease.
d) Recurrent kidney infections, urinary tract obstructions, and
e) Prolonged use of certain medications.

Kidney's filtering ability is measured in terms of estimated glomerular filtration rate (eGFR). Based on the eGFR values, Chronic kidney disease can be classified into five stages. Stage 1 (mild kidney damage with normal eGFR) to Stage 5 (severe kidney damage with low eGFR). Stage 5 is the end-stage renal disease, where the kidneys are no longer functioning adequately to sustain life. This condition requires requiring dialysis or kidney transplantation.

The following are the treatment options for CKD:

a) Taking prescribed medications as directed.
b) Controlling underlying conditions like diabetes and hypertension.
c) Lifestyle modifications, such as a healthy diet, regular exercise, staying hydrated, avoiding excessive alcohol consumption and quitting smoking.
d) Avoiding substances that can further damage the kidneys (such as certain medications and excessive use of painkillers).evaluating kidney function through blood tests, urine tests, imaging studies, and kidney biopsy in some cases.

In advanced stages of CKD, when kidney function is severely impaired, main treatment options are dialysis or kidney transplantation. Kidney transplantation is the replacing of the damaged kidneys with a healthy kidney from a donor. Dialysis is using a machine to filter waste and excess fluid from the blood.

DIALYSIS SYSTEMS

Dialysis is the main treatment option for the people in advanced stages of chronic kidney disease (CKD) like end-stage renal disease (ESRD) or kidney failure. At this stage, kidneys can function at around 10% to 15% of their normal function. Even though, kidney transplantation is the major solution to these problems, dialysis serves as a life-sustaining treatment.

Dialysis system is used to remove waste products, excess fluids, and toxins from the blood when the kidneys are unable to perform these functions adequately.

Dialysis system

Dialysis systems help to maintain the balance of fluids and electrolytes in the body, control blood pressure, and manage other complications associated with kidney failure.

The main components and operation principle of dialysis unit is given below:

1. The core part of dialysis unit is the hemodialysis machine. This machine is responsible for filtering and purifying the blood during the dialysis process.
2. The patient is connected to hemodialysis machine using the appropriate vascular access site. The blood from patient's body is pumped to the machine for purification.
3. The main working part of hemodialysis machine is the dialyzer, also known as an artificial kidney. It contains a semi-permeable membrane.
4. Patient's blood is supplied to one side of the semi-permeable membrane and to the other side of the membrane a specially formulated solution called as dialysate is circulated.
5. The dialysate is a carefully formulated solution with lower concentration than blood sample.
6. Due to this difference in concentration, waste products and excess fluid from the patient's blood move across semipermeable membrane to the dialysate solution.
7. The dialysis unit has a system for preparing and delivering dialysate. The dialysate is mixed accurately and delivered at the appropriate temperature and flow rate.
8. The dialysis unit is equipped with various monitoring devices to track the patient's vital signs, such as blood pressure, heart rate, oxygen saturation, and temperature. And also, alarms and safety mechanisms are in place to detect and respond to any issues or emergencies during the dialysis treatment.

Thus, the dialyzer unit filters the blood, mimicking the natural functioning of the kidneys. It filters waste products, excess fluid, and toxins from the blood, while retaining essential substances. By removing these substances, dialysis prevents their harmful buildup in the body.

Dialysis can alleviate various symptoms associated with kidney failure, such as fatigue, nausea, vomiting, loss of appetite, itching, and difficulty breathing. By removing waste products and excess fluids, dialysis can improve the overall well-being and quality of life for individuals with kidney failure. Dialysis can be used as a

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temporary measure while awaiting a kidney transplant. It helps maintain the individual's health and stability until a suitable donor organ becomes available.

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MODULE- IV

NATURE-BIOINSPIRED MATERIALS AND MECHANISMS (QUALITATIVE)

Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swim suits), Kingfisher beak (Bullet train). Human Blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perflourocarbons (PFCs).

NATURE-BIOINSPIRED MATERIALS AND MECHANISMS

INTRODUCTION

Nature and biological things have been a significant source of inspiration for human discovery throughout history. Observing and studying the natural world have led to groundbreaking discoveries and innovations in various fields. Scientists and engineers havedesigned new materials, equipment's from the ideas generated from their understanding and observation of nature.

Some examples of bio-inspired materials and mechanisms are mentioned below:

1. Velcro is a classic example of a bio-inspired material inspired by the burrs of the burdock plant.
2. The beak of the kingfisher bird was the inspiration to the design of bullet trains with reduced noise.
3. Gecko-inspired adhesives were developed by mimicking the fine structures of gecko's feet.
4. The lotus effect has inspired the development of water-repellent and self-cleaning surfaces.

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5. Structure of bones, seashells, or plant fibers have inspired the development of composite materials.
6. Robots have been created by mimicking the movements and behaviors of humans and animals.
7. Echolocation of animals is the inspiration for ultrasonography and sonars.
8. Flying mechanism of birds was the inspiration for development of GPS technology and aircraft.
9. Photosynthesis is the inspiration for the development of photovoltaic cells, and bionic leaf.

These are just a few examples of how nature's designs have guided the development of bio-inspired materials and mechanisms. Nature is incredibly complex, and its design is the result of millions of years of evolution and adaptation. Understanding nature's design can be both fascinating and challenging. As our knowledge and technology advances, we are gradually unlocking the secrets of nature's design and continue to create innovative technologies that can solve complex challenges and improve various aspects of our lives.

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ECHO-LOCATION

Echo-location is a biological phenomenon used by some animals. Few animals are capable of sending (emitting) high-frequency sounds. These emitted sound waves hit objects in the surroundings, bounce off and return to the animal's ears or other sensory organs. The echoes that bounce back after hitting objects are interpreted by animals. This is used by certain animals for navigation and hunting by perceiving their environment. By analyzing the echoes, animals can determine the location, distance, shape, and even movement of objects.

Over a thousand animal and bird species mainly bats, dolphins, and whales, are well-known for their ability to use echo-location. These animals emit high-frequency sounds, and listen for the returning echoes. They have specialized anatomical

structures, for sending sounds and listening to echo sound. Fortunately, in many cases the frequency of emitted sound is so high that, human beings cannot hear it.

When the emitted sound waves encounter an object, they bounce off and return to the animal's ears or other sensory organs. By analyzing the time taken for the echoes to return and the changes in frequency or intensity, the animal can create a mental map of its environment. This allows them to navigate, find prey, avoid obstacles, and even communicate with other members of their species.

Animals that use echo-location have highly developed auditory systems and specialized brain regions dedicated to processing the information received. They use this remarkable ability to navigate in environments where vision is limited, such as in darkness or underwater.

Few examples of echo-location are given below:

1. Bats make echo-locating sounds in their larynxes and emit them through their mouths. Bats can scream at high frequency sounds up to 140 decibels, which is as loud as a jet engine 30 meter away. Fortunately, these sounds are too high-pitched for humans to hear. To avoid being deafened by its own sound, a bat turns off its middle ear just before making sound. But, restores its hearing a split second later to listen for echoes. Bats can detect an insect up to 5 meter away, work out its size and hardness. They can sense and avoid wires as fine as human hairs. They can fly at night as well as in dark caves and can locate night-flying insects that birds can't find.
2. Dolphins and whales make sound by squeezing air through nasal passages near the blowhole. These soundwaves then pass into the forehead, where a big blob of fat called the melon focuses soundwaves into a high-frequency beam. Echo-locating sounds sent are so loud that the ears of dolphins and whales are shielded to protect them. Dolphins and whales use this method to work out an object's distance, direction, speed, density and size. Using echolocation, they can detect small objects at longer distances which they cannot normally see. Dolphin

can move its head and aim the sound beam at different parts of a fish, and can differentiate between different species of fish. Dolphins and whales use echo-location technique to see in muddy waters and also in dark ocean depths.

Echo-location has also inspired the development of technologies for human use. For example, sonar systems used in submarines and ships employ similar principles to detect underwater objects and navigate through the water. Additionally, researchers have been exploring the use of echo-location inspired technology to assist people with visual impairments in navigating their surroundings.

ULTRASONOGRAPHY

Ultrasonography is a medical imaging technique in which high-frequency sound waves are used to create images of the internal structures of the body.

Ultrasonography is inspired by the natural phenomenon of echo-location. Itis similar to echo-location in terms of transmission and reception of sound waves. But, both are used in completely different contexts and for different purposes.

In this technique, an ultrasound transducer is used to send high-frequency sound waves into the body. Sound waves of high frequency in the range of 2 to 18 megahertz are transmitted through the skin and tissues. When these waves encounter boundaries between different tissues or organs, they reflect back as echoes. The transducer also acts as a receiver, picking up the echoes as they return. The echoes are then processed by a computer to create real-time images of the internal structures being examined. The resulting images can be interpreted by a trained healthcare professional.

Ultrasonography can provide detailed images of various organs and structures within the body such as abdomen, pelvis, heart, blood vessels, liver, kidneys, bladder, and reproductive organs. It is commonly used for diagnostic purposes to evaluate conditions such as pregnancy, gallstones, tumors, cysts, organ abnormalities, and vascular conditions. In this technique, images are generated instantaneously in real-time. It can be used for dynamic observations such as the movement of blood flow and the beating of the heart.

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It is a non-invasive and safe imaging method widely used in medical diagnostics. It does not use any dangerous ionizing radiation.

Ultrasonography is similar to echo-location in terms of transmission and reception of sound waves. But, both are used in completely different contexts and for different purposes. Echo-location is a natural biological mechanism used by animals for navigation and perception, whereas ultrasonography is a technology developed by human beings for medical diagnostic purposes.

SONAR

SONAR is an acronym for "Sound Navigation and Ranging". It is a technology in which sound waves are used to detect and locate objects underwater. It is inspired by echolocation and operates on the same principle as echolocation used by certain animals. Thus, both echolocation and sonar use of sound waves for perceiving and detecting objects in the environment, but they are used in different contexts.

Sonar systems emit sound waves of high-frequency into the water. These sound waves travel through the water and hit the objects such as submarines, ships, or underwater structures in their path and bounces off. The echoes produced by these sound waves are then detected by a receiver and analyzed to determine the distance, direction, and composition of the objects.

There are mainly two types of sonar systems:

1. **Active Sonar:** In this type of sonar, the sound waves are emitted by the sonar system itself. The system emits pulses of sound and listens for the returning echoes. This type of sonar provides real-time information and is commonly used for navigation, detecting underwater hazards, and locating underwater objects.
2. **Passive Sonar:** Passive sonar systems do not emit sound waves. They just listen for sounds generated by other sources, such as marine animals, machinery, or other vessels. Passive sonar is used for detecting and tracking

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underwater objects without emitting signals that might reveal the presence of the sonar system.

Sonar technology has various applications, including:

1. Military and Defense: Sonar systems are used in naval operations for detecting submarines, underwater mines, and other threats. They help in submarine navigation, target detection, and anti-submarine warfare.
2. Fisheries and Marine Research: Sonar is utilized to locate fish, assess fish population, and study marine ecosystems. It aids in fisheries management and scientific research.
3. Underwater Mapping and Exploration: Sonar systems are employed to map and explore the seafloor, search for underwater archaeological sites, locate sunken vessels, and study geological formations.
4. Offshore Industry: Sonar is used in offshore oil and gas operations for locating and inspecting underwater infrastructure, such as pipelines and platforms.

Hence, SONAR is a technology developed by humans for object detection and ranging of objects using sound waves. It was inspired by the natural phenomena of echo-location observed in certain animals. It plays a vital role in underwater operations like navigation, exploration, defense, and scientific research.

PHOTOSYNTHESIS

Photosynthesis is a biological process by which green plants and certain other organisms (algae, and some bacteria) transform energy form sun light into chemical energy. In this process, plants take up sunlight, water, and carbon dioxide to create oxygen and energy in the form of glucose as shown in the reaction.

Photosynthesis is fundamental for sustaining life on Earth. Photosynthesis is the primary source of energy for ecosystem. Glucose formed by photosynthesis is consumed by other organisms transferring energy in the food chain. Oxygen, released as a byproduct of photosynthesis is essential for the survival of aerobic organisms, including

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humans. It is required for cellular respiration, by which organisms break down glucose to release energy. In addition, photosynthesis plays an important role in reducing carbon dioxide (CO2) from the atmosphere. Carbon dioxide is a greenhouse gas that contributes to global warming and climate change. By taking up CO2 during photosynthesis, plants act as natural sink for CO_2. Thus, without photosynthesis, life would not be existing on earth.

Important steps involved in the process of photosynthesis are:

1. Light Absorption: Plants have specialized pigments that enable them to absorb light energy from the sun. The primary pigment involved in light absorption is chlorophyll, which is located in the chloroplasts of plant cells.
2. Carbon Dioxide Fixation: Carbon dioxide (CO_2) from the atmosphere enters the leaves of plants through small openings called stomata.

$$6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

3. Conversion of Light Energy: The energy absorbed by chlorophyll molecules is used to power series of chemical reactions that convert carbon dioxide (CO_2) and water (H_2O) into glucose ($C_6H_{12}O_6$) and oxygen (O_2).
4. Oxygen Release: Oxygen (O_2) is released into the atmosphere as a byproduct of photosynthesis. This oxygen is essential for supporting aerobic respiration in many organisms, including humans.

The glucose produced by plants during photosynthesis can be used for their growth and reproduction. It can also be stored as starch for later use. Additionally, the glucose produced during photosynthesis is the foundation of food chains and provides energy for organisms that cannot perform photosynthesis themselves.

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PHOTOVOLTAIC CELLS

Photovoltaic (PV) cells, are also known as solar cells. These are the devices that convert sunlight directly into electricity using the photovoltaic effect. Photovoltaic cells work similar to photosynthesis by absorbing solar energy. The first photovoltaic cells were made in the early 1900s, but they were not very efficient. In the 1950s, scientists began to study the structure of plant cells and how they convert sunlight into energy. This research led to the development of more efficient photovoltaic cells.In photosynthesis solar energy is converted to chemical energy in the form of glucose whereas in photovoltaic cells solar light energy is converted into electricity. The materials used in PV cells should be capable of ejecting electrons from their surface by absorbing solar energy. Even though many such materials are available, silicon semiconductor is the most widely used one.

The main steps involved in the energy conversion process in a PV cell are:

1. Light Absorption: The sunlight is composed of small pockets of energy called as photons. Energy carried by each photon depends on its frequency. When photon strikes the surface of photovoltaic material, energy of the photon is completely transferred to an electron present in the outermost orbital of the material.
2. Electron Excitation: Energy of the photon causes electron to break free from its atom, get excited, and move to a higher energy state.
3. Generation of Electron-Hole Pairs: The excited electron leaves behind a positively charged hole within the material. Thus, an electron-hole pair is created.
4. Electron Flow: The freed electrons are separated from holes and made to move through the external circuit.
5. Electrical Current: This movement of electrons through external circuit creates an electric current.
6. Electrical Output: The electric current generated by the flow of electrons can be harnessed and used to power electrical devices or stored in batteries for later use.

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7. Ultimately, the electrons from external circuit are made to come back and occupy the holes and get ready for next cycle of excitation. In this way, electric current can be produced from solar cells as long as they are exposed to sunlight.

The solar PV cells can play an important role in meeting our future energy requirements and also in solving environmental pollution problems. Few aspects of significance of solar PV cells are:

1. Renewable Energy: Electric current produced from PV cells using solar energy is a clean and renewable source of power. This reduces dependence on non-renewable fossil fuel energy sources and contributes to a sustainable energy future.
2. Environmental Impact: PV cells produce electricity without emitting greenhouse gases or other pollutants contributing to a greener and healthier planet.
3. Energy Independence: PV cells generate electricity on-site and closer to the point of consumption. This reduces transmission losses and dependence on centralized power grids.
4. Cost Reduction: With advancements in PV technology, the cost of solar cells and PV systems has significantly decreased over the years. This will further reduce in future, making solar energy more affordable and accessible.

Thus, solar PV cells can play a significant role due to their ability to generate clean, renewable energy, reduce environmental impact, enhance energy security, and contribute to sustainable development.

BIONIC LEAF

A bionic leaf is an artificial system inspired by natural photosynthesis. In the process of photosynthesis, chlorophyll present in plant leaf takes up sunlight, water, and carbon dioxide and converts them into usable chemical energy, such as glucose.

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A bionic leaf is designed to mimic the process of photosynthesis. It is constructed with the objective of converting sunlight, water, and carbon dioxide into usable energy, such as hydrogen or other chemicals.

The concept of bionic leaf was originated in the year 2009 at MIT, where the researchers first developed the "artificial leaf". It was a device made from silicon and an anode electro-catalyst for the oxidation of water. This was capable of splitting water into hydrogen and oxygen gases. In 2012, researchers from Harvard University joined MIT team and they together merged the concept of the artificial leaf with genetically engineered bacteria that feed on the hydrogen and convert CO_2 in the air into alcohol fuels or chemicals. The first version of the Bionic Leaf was created by them in 2015.

The construction and working of a typical Bionic Leaf is described below:

1. Bionic Leaf uses solar photovoltaic cells as a system to capture sunlight. The captured sunlight is utilized for water splitting, where water molecules (H_2O) are split into hydrogen (H_2) and oxygen (O_2).
2. The second main component of bionic leaf is Electro-catalysts required for splitting of water in to H_2 and O_2.

$$2\,H_2O \rightarrow 2\,H_2 + O_2$$

3. Generation of H_2 and O_2 is similar to the steps of natural photosynthesis. The hydrogen generated can be used as a clean fuel source, while the oxygen is released into the environment.
4. In the next step, a specialized bacterium is incorporated in to bionic leaf. These bacteria can take up generated hydrogen and carbon dioxide from the environment and convert them into useful chemicals. This process is known as microbial electro-synthesis. It is similar to formation of glucose in photosynthesis.

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5. Thus in "bionic leaf", advanced bio-hybrid systems consisting of specialized bacteria, with artificial components like solar cell, catalysts, and electrodes are used to convert solar energy into chemical energy.

The bionic leaf technology can play an important role in future for the applications such as renewable energy production, carbon dioxide capture, and the production of sustainable chemical fuels and other chemicals. It can contribute to create a carbon-free, clean energy system.

Bionic leaf is still considered as an emerging technology. Research and development works are in progress to make these systems practical and scalable for real-world applications.

BIRD FLYING

Bird flying is the ability of birds to take off and fly through the air using their wings. Birds are different from other animals due to this unique flying nature. Birds have developed this ability as a result of a combination of anatomical, physiological, and behavioral adaptations. These adaptations have evolved gradually over millions of years. The flying nature helps birds in feeding, breeding, avoiding predators, migrating and adapting to various environmental conditions.

Bird flight is a complex form of loco-motion. It involves different types of motions like taking off, sustain lift, control their flight path, and adapt to different flight conditions and landing. Through adaptations over millions of years, birds have developed specializations in their wings and other body parts, and acquired different forms of flight.

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BASIC MECHANICS OF BIRD FLIGHT

The basic mechanics of bird flight can be understood by considering combination of aerodynamic principles, wing movements and functions of different body parts. The basic mechanics of bird flight is briefly described below:

1. Bird flight begins with takeoff. Birds use their legs to push off the ground, generating an initial upward force. The primary muscles involved in takeoff are the leg muscles.
2. After taking off, they flap their wings rapidly to generate lift and thrust. The wings have a curved shape known as an airfoil. This curved surface is responsible for creating a pressure difference between the upper and lower wing surfaces. The upward (up-stroke) and downward (down-stroke) movements of wings generates lift.
3. During the down-stroke, the wings move downward and forward. As the wings move downward, air is pushed downward. This creates an equal and opposite upward force (lift) against the gravity. This upward force is responsible for generation of lift by birds.
4. As the bird completes the down-stroke, it starts the upstroke. During the upstroke, the wings move upward and backward. This motion reduces drag and allows the bird to reset its wings for the next down-stroke. The upstroke also contributes to lift and thrust by redirecting air currents smoothly over the wings.
5. During flight, birds can change the angle of their wings, creating different amounts of lift.
6. Thrust is produced by the flapping motion of the wings. This thrust is responsible for propelling the bird forward. Birds have different flapping frequencies depending on their size, and species. Smaller birds generally have higher wing flapping frequencies and move quickly. While larger birds have slower wing flapping frequencies that provide greater lift and stability.
7. The tail plays a crucial role in flight control. Birds use their tail feathers to adjust their balance, stability, and direction. By tilting or spreading their tail feathers, birds can make quick turns or maintain a straight flight path.

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8. Some birds, such as raptors and seabirds, can soar and glide for extended periods without flapping their wings constantly. Soaring involves flying without flapping their wings. Gliding refers to flying with extended wings. They make use of rising air currents. By adjusting their wings and body posture, they can effectively ride these air currents and maintain flight with minimal energy expenditure. Soaring and gliding allow birds to cover long distances, conserve energy during migration, and search for food efficiently.

Birds possess a few unique aerodynamic and anatomical features which are mainly responsible for their ability to fly. Few of them are mentioned below:

1. **Streamlined body shapes:** The streamlined body shape of birds reduces drag and enables efficient movement through the air.
2. **Specialized curved wings:** Birds have lightweight and curved shape wings, that allow their efficient movement through the air. The curvature of the upper surface of the wings, which is called as the airfoil is responsible for generating lift as air flows over them.
3. **Sleek feathers:** The feathers are located at the tip of the wings. They are lightweight, yet strong, and provide the necessary surface area for generating lift and moving in the air. They minimize air resistance and allow for smooth airflow over their bodies. They also provide insulation and protection.
4. **Lightweight skeletons:** Birds have a lightweight skeleton, consisting of thin and hollow bones. These hollow bones reduce the overall weight of the bird, making it easier for them to take flight.
5. **Powerful flight muscles:** The powerful primary flight muscles, such as the pectoralis major, and supracoracoideus muscles contract and help birds to flap their wings.
6. **Tail:** The tail plays a crucial role in flight control. Birds use their tail feathers to provide stability, and change direction. They can adjust the position and orientation of their tail to maintain balance and move effectively.

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7. **Energy and Metabolism:** Flying requires a considerable amount of energy. Birds have a high metabolic rate and efficient respiratory and circulatory systems to provide oxygen and nutrients to the flight muscles.
8. **Respiratory system:**Birds have a unique respiratory system for a continuous and efficient supply of oxygen during flight. Air flows unidirectional through a system of air sacs in lungs and hence more amount of oxygen is constantly supplied from the lungs to flight muscles.
9. **Circulatory system:**Birds have a highly efficient circulatory system to deliver oxygenated blood to the flight muscles. The heart of bird has a four-chambered structure and it can beat rapidly to supply oxygenated blood. Birds also have high metabolic rate to supply the high energy needs of flying.

The ability of bird to fly is due to combination of aerodynamic principles, and different specialized body parts. A complex involvement of forces, muscle coordination, and adaptations allow the birds to navigate the skies with grace and precision.

BIRD FLYING AND AIRCRAFT

Birds have natural ability to fly and this has been evolved over millions of years through biological adaptations. On the other hand, flying of the aircraft is due to well-defined aerodynamic principles, materials, and technology. Birds rely on their muscular power and flapping motion to generate lift and thrust for flight. Birds control their flight by flapping their wings, moving their tails, and changing the shape of their wings. In contrast, aircraft use engines, such as jet engines or propellers, to provide the necessary propulsion. Birds have a wide variety of wing shapes, each adapted to their specific flying requirements, such as soaring, hovering, or high-speed flight. Aircraft wings are designed based on aerodynamic principles to optimize lift and minimize drag.Aircraft are controlled by the pilot using the rudder, ailerons, and elevators. Birds take rest and can fly in certain weather conditions while aircraft being a machine can fly in a wider range of conditions, for long periods of time without landing.

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Although, airplanes and birds' flight mechanisms are fundamentally different, the observation and study of bird flight has provided valuable insights and inspiration for improving aircraft performance and efficiency.Since humans first aspired to fly, they have observed birds' flight and sought to replicate their abilities in the development of aircraft. Several aspects of bird flight have influenced the design of airplanes and other flying machines.

Here are some specific examples of how aircraft design has been inspired by bird flying mechanisms:

1. Birds have wings with camber. Camber is a curve in the wing, which helps to generate lift. Camber similar to bird is used in the design of aircraft wings.
2. Birds have vortex generators on their wings which help to improve airflow over the wing. Similar vortex generators are used in the design of aircraft wings.
3. Ailerons are control surfaces that are located on the trailing edge of the wing. Ailerons are used in aircraft design to control their roll.
4. Birds move their tails to control their yaw. In aircrafts, rudders are used to control the yaw of the aircraft.

The design of aircraft is constantly evolving. Still there is much to learn from birds flying and navigation skills. As we learn more about how birds fly, navigate and communicate, we can develop even more efficient and aerodynamic aircraft.

NAVIGATION OF BIRDS

Birds exhibit remarkable navigational skills. They travel thousands of miles annually to find food, mates, and breeding grounds and find their way back to specific place. They do not use any technology like GPS to navigate. Instead, they use variety of biological cues to navigate. Some of the cues used by birds for navigation are:

1. Birds use the position of sun in the sky as a reference point to determine their direction of travel during the day time.
2. They use the position of stars as landmark to navigate during night.
3. Birds have specialized organs called "magneto-receptors". They can sense the Earth's magnetic field. This helps them determine their latitude and longitude and aids in long-distance navigation.
4. Birds can also use landmarks, such as coastlines, rivers, and mountains, to navigate.

The navigation techniques of birds are still not fully understood. Scientists could learn from birds' navigation abilities in order to improve the GPS technology which is being used presently for navigation. Birds are able to use the Earth's magnetic field to navigate even when there are no other cues available. This suggests that there is some potential for scientists to learn from birds in order to improve GPS technology.

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GLOBAL POSITIONING SYSTEM (GPS)

GPS stands for Global Positioning System. It is a satellite-based navigation system. It helps users to determine their precise location and time anywhere on the Earth or near the Earth's surface.

GPS technology was developed and maintained by the United States government, primarily for military use, but it is freely accessible to anyone with a GPS receiver. GPS receivers are the devices capable of picking up radio signals coming from multiple GPS satellites. Now a days, they are used everywhere in various devices such as smartphones, tablets, vehicles, ships and aircrafts.

The GPS system consists of a network of satellites orbiting the Earth at above 20350 KM above the surface of the Earth. There are 31 satellites in the GPS constellation and they are distributed in six equally spaced different orbits. Out of 31 satellites 24 are operational at any point of time. Each one of this satellite circles earth

twice in a day in any one of those six orbits. Together, they provide continuous worldwide coverage. GPS satellites continuously broadcast radio signals providing information about their location and the precise time of the transmission of signal. They are equipped with atomic clocks to provide the information on precise time.

Radio signal transmitted by GPS satellite travels with speed of light (c). If t_1 is time at which signal is emitted by GPS satellite and t_2 is the time at which radio signal is received by GPS receiver, then distance of GPS receiver from satellite is calculated from the formula,

$$Distance = Velocity \times time = c \times (t_2 - t_1)$$

In a similar way GPS receiver calculates its distance from different satellites. By knowing the distances to the satellites and the locations of the satellites, the receiver can calculate its own location.

GPS technology uses "trilateration" method to determine the location of a receiver on the Earth's surface. Trilateration is a method of finding the location of a point by measuring the distance of the point to three or more other known points.

To obtain the geolocation and time information, a GPS receiver should receive signals from at least four GPS satellites simultaneously. The accuracy of GPS location finding depends on the number of satellites used. The more satellites used, the more accurate the location finding. When distance of GPS receiver from four satellites is measured, the accuracy of the location is about 10 meters. With more satellites, the accuracy can be improved to a few centimetres.

The major drawback of GPS technology is that the radio signals are weak and can be affected by obstacles like tall buildings, mountains, or dense vegetation. This may result in reduced accuracy or temporary loss of signal. Therefore, to enhance accuracy and reliability of GPS technology, multiple satellite systems launched and operated by other countries like GLONASS (Russian), Galileo (European Union), and

BeiDou (Chinese) are used along with GPS of USA. Modern GPS receivers are capable of receiving signals from all these satellites. This allows for more satellites to be available for positioning, even in challenging environments where direct line-of-sight to GPS satellites may be obstructed. All these satellite systems collectively used for navigation purpose are referred to as GNSS (Global Navigation Satellite System).

APPLICATIONS OF GPS TECHNOLOGY

Main applications of GPS technology are:

1. **Navigation:** GPS is widely used for navigation in vehicles, ships, aircraft, and for pedestrian use. It helps people find their way to a particular destination and provides real-time location information.
2. **Mapping and Surveying:** GPS is used in mapping and surveying to create accurate maps and geospatial data.
3. **Timing and Synchronization:** GPS is utilized as a highly precise time reference for various applications requiring accurate timing, such as telecommunications and financial transactions.
4. **Location-based Services:** Many apps and services use GPS technology to provide location-specific information and services, such as finding nearby restaurants, gas stations, or points of interest.
5. **Tracking and Monitoring:** GPS is used to track the movement of vehicles, assets, wildlife, and even individuals for various tracking and monitoring purposes.
6. **Emergency Services:** GPS technology is essential for emergency services to locate and respond to distress calls accurately.

GPS TECHNOLOGY AND AIRCRAFT

GPS technology is widely used in modern aircraft navigation. It provides accurate and reliable information on position, speed, and time of aircraft. It is much

superior to earlier two-dimensional ground-based navigation system. It provides four-dimensional positioning and hence improves safety, and reduces accidents.

Some of the uses of GPS technology in aircraft are given below:

1. **Navigation:** The use of GPS technology in aircraft has completely revolutionized the navigation by improving safety, efficiency, and overall flight experience. GPS provides accurate position information. The GPS receiver in the aircraft calculates the aircraft's position based on the signals from the GPS satellites. Pilots can know the position, altitude, heading, and groundspeed of aircraft in real-time, which are essential for navigation. This information is displayed on cockpit navigation displays. This information is used to fly along a planned route, to approach and land at an airport, and to avoid obstacles.
2. **Approach and landing:** GPS is also used by pilots to approach and land their aircraft at airports. The GPS receiver in the aircraft calculates the aircraft's position and altitude relative to the airport, and this information is then used by the aircraft's autopilot to guide the aircraft to the runway. This enhances safety, especially in challenging weather conditions. GPS technology enables aircraft to follow precise, curved flight paths during approach and landing.
3. **Aircraft tracking:** GPS is used by air traffic controllers to monitor and manage air traffic more efficiently. They can track the position of an aircraft in real time. This information is used by air traffic controllers to ensure that aircraft are separated safely. It can also be used to locate an aircraft in the event of an emergency.
4. **Automatic Dependent Surveillance-Broadcast (ADS-B):**ADS-B is a technology that uses GPS to broadcast an aircraft's position and other information to ground air traffic control station and nearby aircraft. This information is used to track aircraft and to ensure that they are operating safely. This enhances situational awareness and contributes to a more effective and safer air traffic management system.

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Pilot using GPS technology for navigation in aircraft

5. **Flight Planning:** Pilots use GPS data during flight planning to calculate fuel requirements, establish the most efficient routes. Thus, more efficient, optimized, flexible, and user-preferred route structures can be developed. This results in significant savings from shortened flight times and reduced fuel consumption.

LOTUS LEAF EFFECT

The lotus leaves have unique rough and water-repellent surface. Water falling on its surface just rolls off taking dirt and other contaminants with it. Thus, the surface of lotus leaf is naturally cleaned without manual intervention or using any external cleaning agents. This self-cleaning, super-hydrophobic nature of lotus leaves has led to the development of new concept called as 'Lotus effect'. This effect has inspired the material scientists to develop the new materials and coatings which exhibit super hydrophobic (water-repellent) and self-cleaning surface properties.

The super-hydrophobic, self-cleaning nature of lotus leaf is attributed to the following structural features:

a) Lotus leaf has combination of micro- and nano-structures with optimized Geometry. It has tiny bumps and ridges that are about 10-20 micrometres in size. These bumps and ridges create a rough surface that traps air. This layer of air prevents the water droplet from fully wetting the leaf surface, and it also makes it difficult for the water droplet to adhere to the leaf surface.

b) The whole surface of lotus leaf is covered with short wax tubules. These wax tubules are made of non-polar, hydrophobic, hydrocarbon molecules. These are accumulated in clusters with high density on the surface.

Super-hydrophobic self-cleaning surface of lotus leaf – Lotus effect

c) Due to these structural features, lotus leaf exhibits very low surface energy. When water comes in contact with its surface, it forms high contact angle of 162°. The hydrophobicity of a surface can be measured by its contact angle. Higher the contact angle, higher is the hydrophobicity of a surface. Surfaces with a contact angle < 90° are referred to as hydrophilic and those with an angle >90° are referred to as hydrophobic. Lotus leaf forms contact angle of 162° and is considered as ultra-hydrophobic. With this contact angle, only 0.6% of the

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surface of a water droplet is in contact with surface of the leaf. Thus, the contact between surface of lotus leaf with water, dirt or any other contaminant is very less. Hence, when water falls on the surface of lotus leaf, it automatically rolls off the surface, taking dirt and contaminants with it.

SUPER HYDROPHOBIC AND SELF-CLEANING SURFACES

Lotus effect has inspired the material scientists to develop the new materials and coatings which exhibit super hydrophobic (water-repellent) and self-cleaning surface properties. Materials with these properties can find several applications in various industries like textiles, and architecture.

Few examples for the materials used in super hydrophobic, self-cleaning surfaces are mentioned below:

a) Fluorinated polymers such as perfluoroalkyl polymers or polytetrafluoroethylene (PTFE), are widely used as hydrophobic agents due to their excellent water-repellent properties.
b) Organo-silane compounds are another class of compounds used in these materials. They can be chemically modified to obtain specific surface characteristics.
c) Silicone based polymers are known for their water-repellent nature and are often used in self-cleaning coatings.
d) The surface of few nanostructured materials can be made rough and super-hydrophobic. For example, silica nanoparticles, Titanium dioxide nanoparticles, carbon nanotubes and graphene-based materials are used as hydrophobic, water repellent agents.
e) Acrylic resins, cat-ionic and an-ionic surfactants, solvents and binders are used to coat and spread the above mentioned materials evenly on any surface to convert surface into hydrophobic, self-cleaning.

Few of the applications of materials exhibiting lotus leaf effect are given below:

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a) Super hydrophobic self-cleaning coatings have been developed inspired by the lotus leaf effect. These coatings can be applied over various surfaces, such as glass, metal, and ceramics. These coatings create a super-hydrophobic surface that repels water, and prevent the adhesion of dust, dirt, and pollutants. This imparts self-cleaning properties to their surface. Hence, their surface need not be cleaned manually, which can save time and money.

b) Textiles and Fabrics can be coated with super-hydrophobic, self-cleaning materials to make their surface water-resistant and stain-resistant. These textiles are easier to clean and provide protection from moisture. They find applications in outdoor apparel, raincoats, sportswear etc. The Swiss companies HeiQ and SchoellerTextil have developed stain-resistant textiles under the brand names "HeiQ Eco Dry" and "nanosphere" respectively. Stains of tomato sauce, coffee and red wine can be easily washed away from these clothes.

c) These materials are used in architectural designs to make outer surface of the high rise buildings and structures self-cleaning. This can help keep buildings cleaner and reduce maintenance costs.

d) Super hydrophobic coatings have been applied to electronic devices to protect them from water damage and improve their reliability.

e) These materials are used to create surfaces that repel ice and snow in in cold weather climates. They can reduce the usage of de-icing agents.

f) These materials are used on the surfaces as water-repellents. They can resist water damage and increase durability of the material.

Thus, development of super hydrophobic, self-cleaning surfaces inspired from lotus leaf effect is a promising new technology with a wide range of potential applications. As technology and research progress, new and more environmentally friendly materials may be developed which can provide innovative solutions to many problems.

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PLANT BURRS

Seeds and fruits of certain plants have hooks or teeth on their surface and these are called as plant burrs. These hooks and teeth are naturally designed to attach themselves to fur, feathers, or fabric of passing animals. This is the natural mechanism used by these plants to spread and disperse their seeds to new locations. This aids in the plant's reproduction and distribution.

One of the most well-known examples of a plant burr is the burdock plant (Arctium spp.), which has round and spiky burrs. Other common plants with burrs are Devil's-claw (Proboscidealouisianica) and Stickseed (Hackeliavirginiana).

Plant burr of the burdock plant

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These burrs have inspired the invention of Velcro due to their ability to attach to fabrics and create a strong grip.

VELCRO

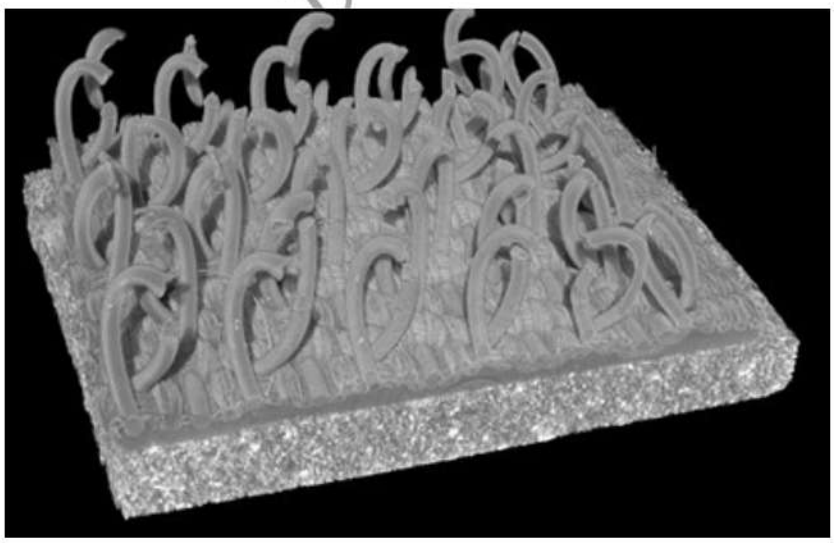
Velcro is a type of fastener system used to fasten two different pieces of material. It has two components present on the two different surfaces of material. They are:

a) Hook side consists of tiny hooks made of a sturdy material like nylon polymer. The hooks are arranged in a dense array on one surface of the strip.
b) Loop side is made of a softer material with small loops that can easily interlock with the hooks on the other side.

When hook strip and loop strip are pressed together, the hooks catch onto the loops, and fasten two parts creating a secure bond. Two sides can be separated by simply pulling the two strips apart. Closing and opening that can be repeated many times. Velcro is a strong and secure fastener which can be opened and closed quickly. It is highly durable and can be used repeatedly. Due to these advantages, it has become a convenient fastening solution for wide variety of applications. Velcro is used in a wide range of everyday consumer products and specialized industrial needs, including clothing, shoes, bags, medical devices, sporting goods, aerospace and automotive applications.

Discovery and design of Velcro is one of the older examples for biomimicry and bio-design. Velcro was invented by a Swiss engineer named George de Mestral in the year 1941. This invention was inspired by the burrs he found on himself and on his dog. Once when George de Mestral was hunting with his dog in rural Switzerland, he noticed small burrs from Burdock plant stuck to his pant legs and his dog's fur. He wondered how the tiny hooks of the burrs were sticking to him. He examined the burr under a microscope and observed an interlocking mechanism between the small hooks of the burr and loops of the fur/fabric.This sparked an idea to mimic the structure as a potential fastener.

Hooks found on the burr of the burdock plant



Hooks on the nylon Velcro

De Mestral spent several years developing the concept and experimenting with different materials and finally duplicated the hook and loop fastener inspired by the

Burdock plant. He settled on using nylon polymer as the basis for the hooks and loops due to its durability and flexibility. In 1948, de Mestral patented his invention, which he named "Velcro." The name is a combination of the French words "velours" (velvet) and "crochet" (hook). The result of his new invention was Velcro ® brand fasteners.

Velcro quickly gained popularity for its ease of use and versatility. It found numerous applications in various industries, from clothing and footwear to medical devices and aerospace. Over the years, Velcro has become an essential fastening solution in everyday life, making it one of the most successful and recognizable inventions to come from nature's inspiration.

SHARK SKIN

The skin of shark is naturally designed to maximize its swimming efficiency. The rough surface of shark skin helps reduce drag and increase thrust as it swims. The surface of the shark skin is covered with small, tooth-like scales referred to as "denticles." The denticles have a riblet-like structure, resembling tiny V-shaped ridges. These ridges are aligned in the direction from the shark's head to tail. The denticles overlap with each other, creating a "chainmail-like" armor on the shark's skin. The denticles tend to become smaller in size as they move from the head of the shark's body to its tail. The riblet structure found on shark skin is completely different from smooth scales found on many other fish.

The superior hydrodynamic performance of shark is due to these denticles and their arrangement in a specific pattern. The overlapping arrangement of denticles helps in reducing drag and minimizing turbulence as water flows over the skin. The denticles create a low-pressure zone, called a leading-edge vortex, as the water moves over the skin. This low-pressure area increases thrust and reduces drag.

The gradual reduction in size of denticles optimizes hydrodynamics and allows smoother transition of water flow over the shark's body. This allows sharks to move through the water with less resistance, conserving energy during long-distance swims. The reduction in drag enables sharks to attain higher speeds with less effort. Due to

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systematic arrangement of denticles, the body of the shark is more streamlined. Due to this, shark can make quick turns and changes in its directions. The presence of denticles also contributes to reduce noise generated by water flow over the skin and hence shark can swim silently. Apart from their hydrodynamic advantages, denticles also provide some level of protection to the shark's skin from injuries.

FRICTION REDUCING SWIM SUITS

Friction-reducing swim suits are primarily used in competitive swimming to enhance the performance of athletes. Scientists and engineers were inspired by the unique design of shark skin, developed the friction-reducing swim suits. Using the principle behind the shark skin design to develop friction reducing swim suits is another example of bio-design and biomimicry.

The denticles on shark skin are structured in a way that minimizes drag and enhances thrust as the shark moves through the water. In the early 2000s, researchers tried to replicate this structure in swim suits, hoping to reduce the frictional drag experienced by swimmers and to improve their speed and efficiency in the water.

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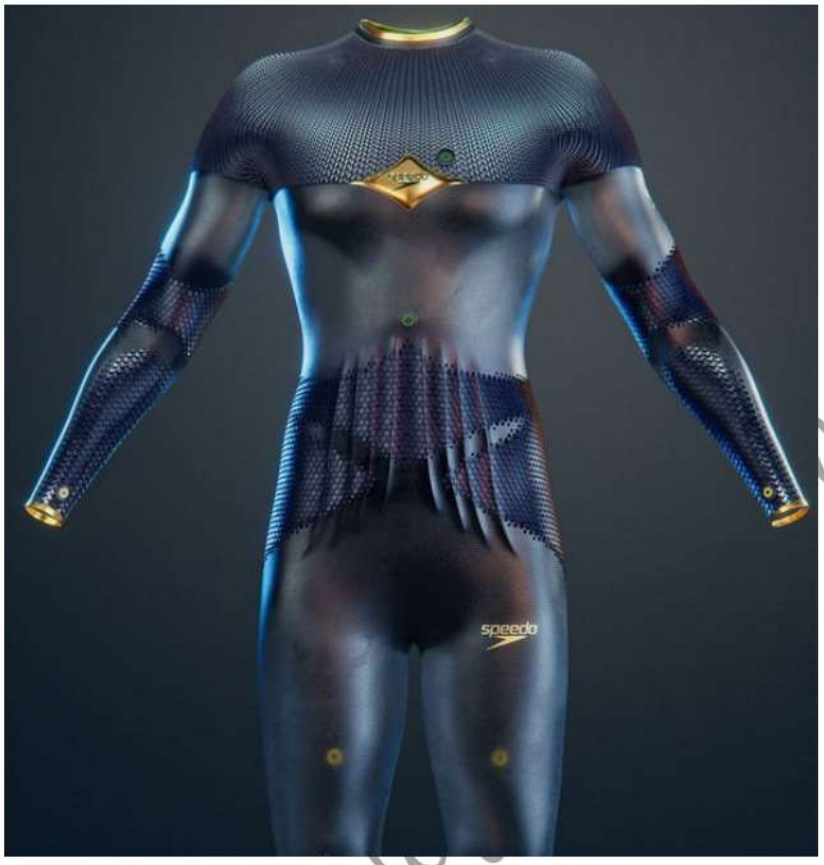

Recent shark skin swimsuit from Speedo

One of the first companies to market a shark skin swimsuit was Speedo. Their Fastskin II swimsuit was designed to mimic the texture of shark skin. The swimsuit was made of a fabric with tiny ridges that were arranged in the same way as shark denticles. Speedo's Fastskin II swimsuit was a huge success. It was worn by many of the top swimmers in the world, and it helped to set numerous world records. With the advancement of material science and 3D printing technology, high-tech swim suits with more realistic texture that mimics shark skin are being developed by the companies.

KINGFISHER BEAK

The kingfisher bird survives eating fish. As its name suggest, kingfisher bird is master at catching fish. Kingfisher birds are capable of slicing through the air and dive into the water to catch prey without barely making a splash.

Fish have a defensive strategy that is hard to overcome. They have specialized receptors known as a lateral line along their body. These can sense any waves generated from even slight movement of water and suddenly, in a flash change their position. This can be observed when we try to catch fish. As soon as our hand touches the water, fish detects it and suddenly escapes.

A kingfisher has special glare-reducing cells in its eyes so that it can detect fish in water from long distance. It approaches the water surface at a speed of over thirty-six feet (eleven meters) per second, and dives in to water without making a splash and manages to grab a fish in its beak before the fish detects it and flees.

This ability of kingfisher to approach fish without generating any detectable waves in water is attributed to the shape of the kingfisher beak and head.

The kingfisher has a long, pointed and cone-shaped beaks and heads. The diameter of the beak steadily increases from the tip to the head. The tip point of the conical beak has a little surface area and offers no resistance to the water upon entry. Thus, kingfisher enters in to water without making any noisy splash above water and also without creating a compression wave below the surface of the water. The evenly and gradually enlarging cross-section of the beak keeps fluid flowing smoothly around it as it penetrates further into the water column. This gives the kingfisher bird, crucial milliseconds to reach the fish before the fish can detect it and flee. The length and shape of the beak are the critical factors. The longer the beak with more gradually expansion of the angle of the wedge then the resistance produced will be lesser.

Birds with shorter, fatter, or rounder beak with small increase in the wedge angle cannot catch fish. This is because; they offer more resistance to flow of water around them resulting in a splashy noise and a compression wave. Fish can detect this easily and flees.

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Kingfisher beak and front shape of a bullet train

BULLET TRAIN

The Shinkansen bullet train is a network of high-speed trains in Japan. They travel throughout Japan at high-speeds of 240–320 km/hr (150–200 mph). They carry millions of passengers every year. These rails are known for comfort, safety, lower cost and energy efficiency.

When Shinkansen bullet trains were first introduced, engineers faced a unique problem. Due to travelling at high speeds of 240–320 km/hr (150–200 mph), they generated loud noise that could be heard up to 400 metres away. And also, at the exit of tunnel, they used to produce a loud "tunnel boom", disturbing nearby residents. This

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was due to build up atmospheric pressure waves in front of the train due to air resistance. Therefore, to function effectively without creating so much noise, the Shinkansen trains needed a structural redesign. They needed a new design that would prevent abrupt changes in air resistance around them.

To find the new design, engineers looked in to kingfisher's beak as a model. The beak of a kingfisher is a remarkable example of nature's design efficiency. The beak of kingfisher is long, pointed, and streamlined. The tip point of the conical shaped beak has a little surface area and offers no resistance to the water upon entry. Thus, kingfisher enters in to water without making any noisy splash above water and also without creating a compression wave below the surface of the water. The aerodynamic principles that make the kingfisher's beak efficient in minimizing water resistance have been adapted to the design of the front of high-speed bullet trains.

The geometry of the front part of the bullet train was designed similar to the shape of a kingfisher's beak. The redesigned trains became quieter and more efficient due to decreased air resistance. The problem of higher noise and tunnel boom were resolved. And also, the redesigned trains required 15% less electrical energy while traveling 10 % faster than before.

Design of front of bullet train taking kingfisher's beak as an inspiration serves as best example for bio-design and bio-mimicry.

HUMAN BLOOD SUBSTITUTES

Human blood substitutes are substances that are used as substitute for blood or blood components. They are also known as artificial blood. They are developed to use in situations where there is a shortage of blood or where the patient's blood is incompatible with donor's blood. Human blood substitutes are not a perfect replacement for blood. Their composition is different and they cannot have all the properties of the natural blood. They can sometimes cause severe side effects. However, they can be a life-saving treatment to address various medical and emergency situations where there is no other option.

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There are two main types of human blood substitutes:

1. **Fluid substitutes:** These are substances used to replace the fluid or plasma volume of blood. They are also called as volume expanders. Saline solution or a combination of saline solution with other substances, such as gelatin or starch or albumin is commonly used as fluid substitute.
2. **Oxygen carrier substitutes:** These are substances used to replace oxygen carrying component of the blood. Hemoglobin present in red blood cells is the protein that carries oxygen in blood. Iron present in hemoglobin molecule has a high affinity for oxygen. It readily picks up oxygen from lungs and transports it to different parts of the body. On the way back, it picks up carbon dioxide from different parts of the body and releases it back to lungs. The Oxygen carrier substitutes used in synthetic blood are designed to mimic the oxygen carrying ability of hemoglobin.

There are two main Oxygen carrier substitutes used in the human blood substitutes:

a) Hemoglobin-Based Oxygen Carriers (HBOCs)
b) Perflourocarbons (PFCs).

HEMOGLOBIN-BASED OXYGEN CARRIERS (HBOCs)

In case of Hemoglobin-Based Oxygen Carriers (HBOCs) natural Hemoglobin molecule is used as oxygen carrier. Hemoglobin used in these substitutes is isolated from human or animal sources (such as cow or pig blood) or can be genetically engineered in the laboratory. Hemoglobin used in HBOCs works through same mechanism as that of natural Hemoglobin present in the red blood cells. However, HBOCs have several advantages compared to natural blood as mentioned below:

a) In case of real blood, there are several blood groups. Sometimes, the required blood group may not be available or it can mismatch with the patient's blood. HBOCs do not have blood types. They are suitable for patients of all blood types and also there is no problem of cross-matching.

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b) After receiving from donor, real blood need to be stored carefully and has limited shelf life. But, HBOCs can have a longer shelf life, making them more readily available when needed.

c) Donated blood carries risk of transmission of infections from donor. Whereas, HBOCs can be manufactured and sterilized to eliminate the risk of transmitting infectious diseases.

There are many problems associated with HBOCs. They have shorter circulatory half-life compared to natural red blood cells. They get cleared from the bloodstream more rapidly. Hence, they need to be transfused more frequently. They can also cause adverse side effects like oxidative stress, inflammation, and kidney damage.

Due to these safety concerns, HBOCs have not yet been granted regulatory approval for usage. However, research and development efforts are in progress to improve the safety and efficacy of HBOCs for potential clinical applications in near future.

PERFLOUROCARBONS (PFCs)

PFCs are synthetic organic compounds made up of only carbon and fluorine atoms. In these molecules all hydrogen atoms in a hydrocarbon molecule are replaced by fluorine atoms. These compounds can be infused to human blood as oxygen carriers. PFCs exhibit very high solubility for oxygen and they carry oxygen in physical dissolved form. They do not chemically bind to oxygen like hemoglobin does.

PFCs exhibit several advantageous properties due to which they are used as oxygen carriers in blood substitutes. Few of the advantages of PFCs as oxygen carrier are given below:

1. The carbon-fluorine bonds in PFCs create a structure with high gas solubility. They can dissolve large amounts of oxygen and carbon dioxide. Hence, PFCs have a high capacity for carrying oxygen, often exceeding that of human hemoglobin. They can supply oxygen to poorly perfused parts also.
2. PFCs are chemically stable and inert molecules. They are not degraded easily and also do not undergo any chemical reactions. Hence, they are non-

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immunogenic. When they are infused in to blood, they do not elicit immune responses or toxic reactions. Hence, they are highly useful for patients having problems with blood compatibility and cross-matching.

3. Donated blood carries risk of transmission of infections from donor. Whereas, PFCs are synthetic compounds and can be sterilized to eliminate the risk of transmitting infectious diseases.
4. They have low boiling points. They are highly volatile and get evaporated quickly at room temperature. Hence, they can be removed from the blood stream easily.

The mechanism of oxygen transport by PFCs involves the following steps:

a) PFCs in liquid solution form are infused directly into the bloodstream.
b) Due to their high oxygen solubility, PFCs readily pick up oxygen from lungs. PFCs can dissolve oxygen gas directly and carry it in the form of dissolved oxygen. PFCs do not bind oxygen like in case of hemoglobin.
c) The blood containing PFCs loaded with dissolved oxygen circulates through the body, similar to natural blood. The PFCs act as oxygen carriers, delivering oxygen to tissues with low oxygen tension.
d) After delivering oxygen to tissues, picks up carbon dioxide, returns to the lungs through the veins and release carbon dioxide during exhalation.
e) They are again re-oxygenated and ready to carry oxygen.

However, there are few limitations and challenges with the use of PFC-based blood substitutes:

a) They have shorter circulatory half-life compared to natural red blood cells. They get cleared from the bloodstream more rapidly. Hence, they need to be transfused more frequently.
b) PFCs are efficient oxygen carriers, but, they do not transport carbon dioxide as effectively as natural blood. Carbon dioxide removal can be critical for maintaining physiological balance.
c) PFCs are cleared from the body through the lungs. Clearance of PFCs should be carefully monitored to avoid excessive accumulation.

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Thus, PFCs are effective as oxygen-carrying solutions. Research and development is still in progress to assess their safety, efficacy, and get regulatory approval before they are widely adopted for clinical use.

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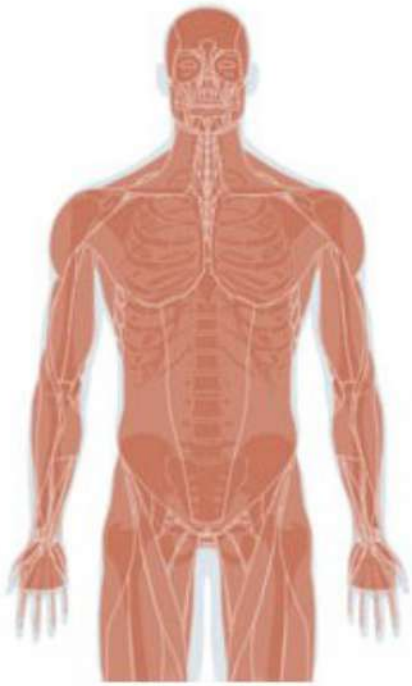
MODULE- V

TRENDS IN BIOENGINEERING (QUALITATIVE):

Muscular and Skeletal Systems as scaffolds (architecture, mechanisms, bioengineering solutions for muscular dystrophy and osteoporosis), scaffolds and tissue engineering, Bioprinting techniques and materials, 3D printing of ear, bone and skin. 3D printed foods. Electrical tongue and electrical nose in food science, DNA origami and Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis. Self- healing Bioconcrete (based on bacillus spores, calcium lactate nutrients and biomineralization processes) and Bioremediation and Biomining via microbial surface adsorption (removal of heavy metals like Lead, Cadmium, Mercury, Arsenic).

MUSCULAR AND SKELETAL SYSTEMS

The muscular and skeletal systems are two interconnected and interdependent systems in the human body. They work together to provide support to the body, enable its movement, give protection to vital organs, and carry out various physiological functions essential for human life.



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The muscular and skeletal systems

The muscular and skeletal systems are briefly described below:

1. **The Skeletal System:** The skeletal system is composed of bones, cartilage, ligaments, and other connective tissues. The main functions of skeletal system are:
    a) Support: The skeleton provides a framework which supports the body, and gives it shape and structure.
    b) Protection: Bones serve as protective coverings for vital organs. The skull protects the brain, the ribcage safeguards the heart and lungs and the vertebrae safeguard the spinal cord.
    c) Movement: The bones, along with joints, muscles, and tendons are responsible for movement of body and body parts.
    d) Blood Cell Production: Within certain bones, the bone marrow produces red blood cells, white blood cells, and platelets.
    e) Mineral Storage: Bones store minerals, particularly calcium and phosphorus, which are essential for various bodily functions. These minerals can be released into the bloodstream as needed to maintain mineral balance.

2. **The Muscular System:**

The muscular system is made up of muscles, tendons, and ligaments. Its main functions include:

a) Movement: Muscles are responsible for the movement in the body. They contract and relax, producing the mechanical energy, and generate force for the movement of bones and joints. Skeletal muscles are attached to bones via tendons, and allow voluntary movements like walking, running, and lifting objects.
b) Posture and Stability: Muscles help maintain proper posture and stability by providing support to the skeleton. They work in coordination to keep the body balanced and aligned.
c) Heat Generation: Muscles generate heat as a byproduct of their metabolic processes. This heat production helps maintain body temperature and contributes to the regulation of body heat.

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d) Protection: Some muscles, such as the abdominal muscles, help protect internal organs and provide support to delicate tissues.

e) Metabolism: Muscles are highly metabolically active tissues. They require energy in the form of calories, and their activity contributes to the overall energy expenditure and metabolism of the body.

Overall, the muscular and skeletal systems work together to enable movement, provide support and protection, maintain posture, and facilitate various physiological functions necessary for human life.

MUSCULAR AND SKELETAL SYSTEMS AS SCAFFOLDS

The muscular and skeletal systems in the human body can be considered as structural frameworks that support and shape the body. They provide support to the body's tissues and organs, allowing for coordinated movement and protection. In this way they play the role of scaffolds in the human body.

The muscular and skeletal systems are able to regenerate when there is a minor injury from incidents like motor vehicle accidents, sport-related injuries. The regenerative ability of these systems is due to constant flow of nutrients and metabolites through complex three-dimensional capillary network of the muscle fibers. However, in the case of severe injury, when muscle mass loss is greater than 20%, tissue repair response is incapable of adequate tissue regeneration. This may lead to extensive and irreversible fibrosis, scarring, and loss of muscle function.

When a natural muscle repair mechanism is not working, we need to search for new muscle regenerative strategies. One such strategy is using scaffolds made of biomaterials as temporary templates or frameworks to facilitate the regeneration and formation of new tissue.

Scaffolds are three-dimensional structures made of biomaterials. They are used in tissue engineering for tissue regeneration. They are designed and used to support and guide the growth of cells, tissues, or organs in human body. The structure of the

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scaffold is designed to mimic the native structure of the tissue. The orientation and alignment of regenerated muscle cells are guided by the structure of scaffold. The structure of regenerated muscle fibers will be similar to structure of scaffold.

Scaffolds are porous structures with interconnected pore networks. The porosity allows for the diffusion of nutrients, oxygen, and waste products. It also allows the infiltration of new cells and integration of regenerated tissue with the previous one. The surface properties of scaffolds are modified to incorporate seed cells, bioactive molecules and growth factors. They provide a temporary support structure for seed cells to adhere, proliferate, and differentiate. This enhances the availability of regenerative cells at the injury site and supports their integration into the damaged muscle tissue. Thus, scaffolds create a favorable microenvironment for muscle regeneration, by stimulating cell proliferation, formation of new blood vessels and by tissue remodeling.

Scaffolds are made from natural polymers like collagen, fibrin and bioplastics like poly lactic acid (PLA). They are biocompatible and biodegradable. This prevents generation of harmful immune responses and eliminates the need for scaffold removal through additional surgeries after tissue regeneration.

Scaffolds are used in a wide range of tissue engineering applications, including the regeneration of bone, cartilage, skin, nerves, and organs.

MUSCULAR DYSTROPHY

Muscular dystrophy is a group of diseases that cause progressive weakness and loss of muscle mass. It is a genetic disorder. It is caused by a mutation in the dystrophin gene. This gene is responsible for production of the dystrophin protein which plays a crucial role in maintaining the structural integrity of muscle fibers. Mutation of dystrophin gene interferes with the production of proteins needed to form healthy muscle and results in muscular dystrophy.

The main sign of muscular dystrophy is progressive muscle weakness. It results in difficulty walking, frequent falls, difficulty rising from a lying or sitting position, trouble running and jumping etc.

Presently, there is currently no cure for muscular dystrophy. The treatment focuses on managing symptoms, improving quality of life, and maintaining mobility and function. Research into gene therapy, stem cell therapy, and other therapeutic approaches is ongoing to develop potential treatments for muscular dystrophy.

BIOENGINEERING SOLUTIONS FOR MUSCULAR DYSTROPHY

Bioengineering is playing an important role in finding new solutions for muscular dystrophy. They aim to improve muscle function, slow down disease progression, and enhance the quality of life. Bioengineering solutions for muscular dystrophy are briefly given below:

1. Gene Therapy: In Gene therapy mutated gene responsible for muscular dystrophy is replaced by a functional gene. Gene therapy can also be used to modify the expression of genes with the genetic defect causing muscular dystrophy. Various strategies are being investigated to introduce corrected genes into muscle cells. This approach can restore the production of essential muscle proteins.
2. Cell Therapy: In cell therapy, the damaged cells of muscle tissue are transplanted by healthy muscle cells. Stem cells can be differentiated into muscle cells and implanted into affected muscles. These cells have the potential to regenerate muscle fibers and improve muscle strength and function.
3. Tissue Engineering: In this approach, stem cells or muscle precursor cells are seeded onto bioengineered scaffolds. These scaffolds are then implanted into the affected muscle tissue. The scaffold provides structural support and guidance for new tissue growth and their integration into the host tissue.

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Tissue engineering strategies can also be used to create functional muscle tissue in the laboratory and transplant it into individuals with muscular dystrophy. Researchers are exploring different scaffold materials, growth factors, and bioreactor systems to optimize the generation of functional muscle tissue for transplantation.

Many of these bioengineering approaches are still in the experimental or early stages of development. Further research and clinical trials are needed to validate their safety and efficacy before they can be widely implemented for the treatment of muscular dystrophy.

OSTEOPOROSIS

Osteoporosis is a skeletal disorder in which there is decrease in bone mineral density and bone mass. This can lead to a decrease in bone strength that can increase the risk of fractures (broken bones). Osteoporosis occurs when the body fails to form enough new bone, or when existing bone is reabsorbed more quickly than it is replaced. It is commonly seen in postmenopausal women and in older men. Fractures happen most often in bones of the hip, vertebrae in the spine, and wrist.

The main causes of osteoporosis are:

a) Hormonal imbalances, particularly a decrease in estrogen levels in women after menopause.

b) Inadequate calcium and vitamin D intake and physical inactivity,

c) Smoking, and excessive alcohol consumption,

d) Taking certain steroidal medications.

Osteoporosis is a chronic condition that requires long-term management. Early diagnosis, lifestyle modifications, and appropriate medical interventions can help reduce the risk of fractures, manage symptoms, and maintain bone health. The main strategies for preventing and managing osteoporosis are:

a) Adequate calcium and vitamin D intake.

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b) Regular weight-bearing and muscle-strengthening exercises.

c) Avoidance of smoking and excessive alcohol consumption.

d) Fall prevention measures, and maintaining a healthy body weight.

e) Taking medications to slow bone loss or stimulate bone formation in certain cases.

BIOENGINEERING SOLUTIONS FOR OSTEOPOROSIS

Bone grafting is a common surgical method used to improve bone regeneration in patients suffering from osteoporosis. Recently, lot of research and developments are happening in bioengineering field to find solution to osteoporosis. Bioengineering offers several solutions for osteoporosis. All these solutions focus on promoting bone formation, enhancing bone strength, and preventing fractures. Main bioengineering solutions for osteoporosis are briefly given below:

1. **Use of scaffolds.**

Bioengineered scaffolds can be used to provide structural support and a three-dimensional environment for bone regeneration. These scaffolds are made from biocompatible and biodegradable materials, such as calcium phosphate ceramics. They are porous materials with well-defined shape and size. Seed cells are attached to scaffolds and allowed proliferate and differentiate to produce new cells. The scaffolds act as a template for new bone growth and provide mechanical support. Once the new bone is generated, scaffolds gradually degrade and dissolved in new bone tissue.

2. Use of Growth Factors and Bioactive Molecules.

Growth factors and bioactive molecules enhance bone formation. Growth factors are the molecules that can stimulate the proliferation and differentiation of bone-forming cells and accelerate bone regeneration. Example for growth factors is morphogenetic proteins (BMPs). Bioactive molecules can also be used to promote bone

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formation and increase bone mineral density. Example for a bioactive molecule is parathyroid hormone (PTH).

These molecules can be delivered locally to the bone defect site. They can also be incorporated into scaffolds.

3. Use of cell therapies:

Stem cells or seed cells can be isolated from various sources like bone marrow and adipose tissue. These cells can be incorporated into scaffolds and implanted into bone defect sites. At this site, they proliferate and differentiate to produce new cells and contribute to new bone formation. Mesenchymal stem cells (MSCs) are highly effective for bone regeneration in osteoporosis.

4. Delivering Drugs to target site:

Bio engineering based modern targeted drug delivering systems can be utilized to delivery drugs directly and specifically to osteoporosis site. These systems can deliver medications, such as bisphosphonates or teriparatide, directly to the bone tissue. The drugs are encapsulated within biomaterial carriers and carried to specific site and released in a controlled manner. These systems enhance the efficacy of drugs and minimizing side effects.

Many of these bioengineering approaches are still in the experimental or preclinical stages of development. Before using these treatments for osteoporosis, further research, optimization, and clinical trials are needed to validate their safety, efficacy, and long-term outcomes.

PRINCIPLES OF 3D PRINTING

3D printing is a modern technique used to create three-dimensional objects. In this technique, a 3D digital model of the object to be created is designed in a computer using computer-aided design (CAD) software. Then the 3D digital model is sliced into thin cross-sectional layers using slicing software. Each layer represents a virtual "slice" of the final object. A set of instructions required to print each layer is provided for the

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3D printer. The 3D printer follows the instructions from the slicer software to print the object layer by layer. Each layer adheres to the one below it. Finally, after printing of all layers, required three-dimensional object is obtained. Depending on the technology and material used, each layer may need time to cool and solidify before the next layer is added. Some objects require support structures to be printed alongside them. These temporary structures prevent overhanging parts from collapsing during the printing process. Support structures are removed after printing is complete. The printed object is subjected to additional post-processing steps, such as cleaning, polishing, painting, or assembly.

In 3D printing, object is obtained by adding material layer by layer. Therefore, this technique is also commonly called as additive manufacturing technique.

Successful printing of a 3D object depends upon:

a) Selection of suitable starting materials required for printing 3D object.
b) Selection of suitable printer for printing 3D object.
c) Selection of suitable printing method for printing 3D object.

Materials used in 3D printing

It is important to choose the right material based on the desired properties of the final object and the capabilities of the 3D printer. These materials are used in various forms, such as filaments, powders, pellets, and liquids. Some of the common types of materials used in 3D printing are:

1. Plastics, Polymers and resins like Nylon, ABS (Acrylonitrile Butadiene Styrene), PETG (Polyethylene Terephthalate Glycol), Polyurethane and biodegradable polymer, Polylactic Acid.
2. Metals like Stainless Steel, Titanium, Aluminum, Cobalt, Chromium etc.
3. Ceramics like Alumna, Zirconia and Porcelain.
4. Composite Materials like Carbon Fiber Reinforced Polymers and Metal Matrix Composites.

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5. Biological and Food Materials are used in bio-printing.

3D Printing Technologies

There are many different 3D printing technologies available to create a 3D object. Each method has its own advantages and disadvantages. Some of the most common 3D printing technologies are:

a) **Fused deposition modelling (FDM):** In this method, printer heats a plastic into molten form. Melted plastic filament is extruded through the printer's nozzle which can move in the x, y, and z directions. The material solidifies as it is deposited layer by layer. Each layer of material deposited, fuses with the layer beneath it. This layer-by-layer deposition process creates a three-dimensional object.

b) **Selective laser sintering (SLS):** This technology uses a high-powered laser to melt powdered material like plastic or metal. Then it is deposited layer by layer to create the final object.

c) **Stereolithography (SLA):** This technology is used to produce highly detailed and accurate parts with high resolution and smooth surface finishes. In this method, a liquid photopolymer resin is used. This resin is sensitive to light and can solidify when exposed to specific wavelengths. The resin is deposited as per the pattern and a UV laser is directed onto the surface of the resin, tracing the pattern. The laser's energy causes the resin to solidify and harden at the desired locations.

There are many advantages with 3D printing techniques compared to other manufacturing techniques. These can be used to create objects with complex geometries that would be difficult or impossible to create using traditional manufacturing methods. They can be used to create prototypes quickly and easily. They can be used to create custom-made objects that are tailored to the individual user's needs.

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3D printing has a wide range of applications across industries like manufacturing, aerospace, healthcare (for creating medical implants and prosthetics), automotive, architecture, fashion, and art. Its ability to create complex shapes with intricate details and its potential for rapid prototyping make it a versatile and transformative technology.

BIOPRINTING

Bio-printing is an innovative technology to fabricate three-dimensional structures that mimic the architecture and functionality of natural tissues and organs. Functional and viable living tissues and organs produced by bio-printing can be used in medical and pharmaceutical fields. They can be used for organ transplantation, disease modelling, drug testing, drug development and other areas of clinical research.

3D bio-printing is an additive manufacturing technique in which compatible biomaterials are deposited by a 3D printer in a layer-by-layer fashion to fabricate 3D tissues and organs. Variety of 3D bio-printing technologies and variety of raw materials are available for fabrication of 3D tissues and organs. The selection of raw material and printing technique for bio-printing process is based on the type of organ being printed and its application.

BIOPRINTING PROCESS

Bio-printing is an advanced technology that combines 3D printing with biology to create complex structures made of living cells. The following steps are involved in a bio-printing process:

1. In the first step, a digital 3D model of the tissue or organ to be printed is created in a computer. Advanced computer-aided design (CAD) software is used to design the structure. It is designed carefully taking into account the desired shape, size, and internal architecture of the organ.

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2. Then the 3D digital model is sliced into thin cross-sectional layers using slicing software. Each layer represents a virtual "slice" of the final object. A set of instructions required to print each layer is provided for the 3D printer.
3. The printer is loaded with bio ink containing cells, biomaterials and bioactive factors required for the printing.
4. The 3D printer follows the instructions from the slicer software to print the object layer by layer.
5. The bio-ink is precisely placed to create the desired 3D structure. Each layer is deposited with precision, and the printed structure gradually takes the shape of desired tissue or organ.
6. After bio-printing, the printed structure is typically cultured under controlled conditions to allow the cells to mature, differentiate, and organize into functional tissue-like structures. This maturation process helps cells to get adjusted to the host environment and to improve overall compatibility and functionality of tissues.

MATERIALS USED IN BIO-PRINTING

Bio-printers use a special type of "ink" in the bio-printing process called as bio-ink. Bio-ink is a printable medium containing variety of materials like living cells, nutrients, growth factors and bioactive molecules. Bio-inks provide natural environment for the living cells and facilitate their survival, growth and differentiation. The composition of bio-ink depends on the type of tissue being printed.

Some of the common materials used in bio-printing ink are briefly described below:

1. **Hydrogels:** Hydrogel is a soft, flexible jelly-like substance that can retain high amount of water. Hydrogels can be synthesized from natural polymers like agar, cellulose, collagen and synthetic polymers like polyacrylamide, polyvinyl alcohol and polyethylene glycol. Hydrogels provide a 3D structure and environment for living cells to survive and grow. They also provide extracellular matrix (ECM) and mechanical support to 3D printed organs and tissues. ECM mimics the natural

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environment of cells in tissues and organs, and enhances cell attachment, differentiation, and overall tissue formation. Hence, hydrogel is used as the major component used in bio-ink for bio-printing of organs and tissues.

2. **Synthetic Polymers:** Synthetic polymers are commonly used in bio-ink to customize its properties. Various polymers used in bio-ink help in controlling mechanical strength, degradation rate, and cell interaction in bio-ink and printed organs. Common examples for the polymers used in bio-ink for this purpose are polyethylene glycol (PEG), polycaprolactone (PCL), and poly(lactic-co-glycolic acid) (PLGA).
3. **Cells:** Cells are the building blocks of tissues and organs. The primary living component used in bio-ink is living cells. Living cells required for bio-printing of organs and tissues are taken from the patient or from other compatible donor. These cells are chosen based on their compatibility with the target tissue or organ and their ability to proliferate and differentiate. Cells are mixed into the hydrogel to create a functional bio-ink.
4. **Bioactive Molecules:** Other bioactive molecules like cytokines, growth factors are also added to bio-ink. These molecules are critical for guiding the development of the printed tissue. Cytokine molecules regulate various cellular processes, including cell growth, differentiation, inflammation, and immune responses. They can enhance the development and functionality of printed tissues by providing important biological cues to the cells. Growth factors are proteins that help cells to grow and multiply. They are essential for the bio-printed structure to develop and mature.
5. **Cross-Linking Agents:** Cross-linking agents are used to solidify the bio-ink and maintain the desired shape after printing is completed. These agents undergo crosslinking chemical reactions with the hydrogel, and create a stable solid structure.
6. **Nanoparticles:** Nanoparticles can be incorporated into bio-inks for various purposes, such as enhancing cell adhesion, delivering drugs, or improving the mechanical properties of the printed organs. Examples include nanoparticles of metals, ceramics, or polymers.

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In addition to the above mentioned materials, other components based on the specificity of the printed organ are used in the bio-ink. Bio-printing is rapidly evolving field, and researchers are constantly experimenting with new materials and combinations to improve the functionality and viability of printed tissues and organs. The choice of bio-ink depends on the specific tissue being printed, the required mechanical properties, the desired cell behaviour, and the ultimate intended application.

BIOPRINTING TECHNIQUES

There are several bio-printing techniques that have been developed to create 3D biomaterials. Few of these techniques are briefly described below:

1. **Fused deposition modelling (FDM):** This is one of the oldest techniques in 3D printing.In this method, printer heats a thermoplastic into molten form. Melted plastic filament is extruded through the printer's nozzle which can move in the x, y, and z directions. The material solidifies as it is deposited layer by layer. Each layer of material deposited, fuses with the layer beneath it. This layer-by-layer deposition process creates a three-dimensional object. Main drawback of this method is, it requires heat and living cells are not compatible with the heat generated in the process.
2. **Stereo-lithography (SLA):** This technology is used to produce highly detailed and accurate parts with high resolution and smooth surface finishes. In this method, a liquid photopolymer resin is used. This resin is photo-curable resin which is sensitive to UV light and can solidify when exposed to light of specific wavelengths. The resin is deposited as per the pattern and a UV laser is directed onto the surface of the resin, tracing the pattern. The laser's energy causes the resin to solidify and harden at the desired locations. The main disadvantage of this method is UV light.
3. **Inkjet Bio-printing:** This technique is similar to the way inkjet printers work, but instead of ink, it deposits tiny droplets of bio-ink (cell-containing material) onto a substrate in a precise pattern. The bio-ink is composed of living cells suspended in a

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supportive hydrogel. These droplets form layers that finally build up into a 3D structure. The main advantages of this method are acceptable cost, compatibility with living materials, and the high speed of construction of bio-ink droplets.

4. **Extrusion-Based Bio-printing:** In this method, bio-ink is extruded through a fine nozzle or syringe to create complex tissue-like structures. The nozzle moves in a programmed path to deposit the bio-ink layer by layer, forming the desired 3D structure. Depending on the viscosity of the bio-ink, different extrusion mechanisms like pneumatic or mechanical systems can be used. In pneumatic method pressurized air is used to push the bio-ink whereas in mechanical method force is applied using a piston.

Still lot more bio-printing methods are available with slight modification in their working principle. These bio-printing techniques are continually evolving, and researchers are working to improve their precision, scalability, and ability to create functional and viable tissues and organs for transplantation and other applications.

3D PRINTING OF EAR

Making new ears or correcting damaged ear through plastic surgery by a surgeon is very difficult. Because, they need to use a person's own cartilage for this which can be absorbed by the body and the place where they take it from can also have issues. Use of Artificial Implants made of silicone or polyethylene for ears is also risky. They may cause infections or rejection by body.

By 3D printing, very detailed ear shapes can be bio-printed. The printed ear tissue must have cells, a scaffold (support structure), and an environment that's similar to natural ear cartilage in the body. Bio-ink used in 3D printing of ear mainly consists of:

1. Hydrogels: Hydrogels 3D network of natural or synthetic polymers which can hold water and provide medium for cells. Hydrogels act as carriers for cells and provide a

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temporary matrix (structure) outside cells, called the extracellular matrix (ECM), which helps in making different tissues.

Cells required for generation of ear are called as Chondrocytes. These cells grow and proliferate best in a highly aqueous hydrogel made of natural polymers like gelatin, collagen, chitosan, alginate, agarose, nano-fibrillated cellulose, silk fibroin, etc. Hydrogels from natural polymers do not possess sufficient mechanical strength. Therefore, to provide mechanically strong extracellular matrix synthetic polymers such as poly caprolactone (PCL), poly glycolic acid (PGA), poly L-lactic acid (PLA), and methacrylate are also used in the ear cartilage bio-printing process.

2. Living Cells: The cell sources commonly preferred for 3D ear printing are primary chondrocytes and stem cells. Primary chondrocytes can provide natural extracellular matrix (ECM). But, they can be obtained only in a small amount from donor sites and have low proliferation rate. Stem cells are available easily and can proliferate easily. But they should be carefully directed to differentiate into chondrocytes in the hydrogel.
3. Bio-active molecules: These molecules play an important role during the period of cells growing into functional tissues. They provide guidance and bioactive cues to chondrocytes and stem cells to differentiate and proliferate. Chondroitin sulfate (CS) and hyaluronic acid (HA) are two examples of bio-active molecules. Growth factors such as Insulin growth factor (IGF), the transforming growth factor beta (TGF-ß) family, and the fibroblast growth factor (FGF) family take an important part in the formation of cartilage tissue.

In the 3D printing of ear cartilage, a multi-head deposition system (MHDS) and inkjet bio-printing methods have been successfully applied. In MHDS method multiple cartridges combining synthetic thermoplastic polymer and living cells in different hydrogels is used. This kind of hybrid bio-printing is required for printing of complex-shaped ear cartilage scaffolds.

Inkjet bio-printing method is also used to fabricate cartilage tissue with strong mechanical properties. In this method, bio-ink made of gelatin methacrylate (PEG-

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GelMA), poly(ethylene glycol) dimethacrylate, and human mesenchymal stem cells (MSCs) is used.

3D PRINTING OF BONE

When the human bone is slightly damaged, it has the ability to self-repair and regenerate. But, when the damage is severe, it requires artificial repair. Ideally, the damaged bone can be transplanted by bones taken from other parts of the patient. But, there is practical difficulty for this due to limited availability of transplantable bones at the donor site of patients and also the possible complications at the donor site after transplantation.

3D printing of bone offers a solution for these problems. 3D printing techniques can be used to fabricate bone structures required in medical field for bone reconstruction, regeneration, and customization. Main significance and applications of 3D printing of bone are:

1. Bone defects caused by trauma, disease, or congenital conditions often require patient specific bone grafts for repair. By 3D printing, custom-made bone grafts that match the patient's needs can be created. The intricate structures of the graft can be created similar to natural bone, promoting bone regeneration and integration.

2. By 3D printing, patient-specific implants can be fabricated. The design and fabrication of implants can be tailored to an individual patient's anatomy, ensuring a better fit and reducing the risk of implant failure.

3. 3D printing of bone can be used in fracture treatment in complex cases. For example, when the reconstruction of highly precise facial bones with restoration of both function and aesthetics is needed, 3D printing of bone is employed.

Materials and methods used in 3D printing of bones

Human bone is a composite material with 60% inorganic components, 30% organic materials, mainly hydroxyapatite and collagen, 5–10% water and 3% lipid.

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Therefore in the fabrication of bones by 3D printing technology, composite materials are commonly used. Main materials used in 3D printing of bone are given below:

1. Inorganic Materials
   a) Metals: Titanium and its alloys which exhibit high strength, low density, low modulus of elasticity, non-biodegradability, and bio-inertness. Magnesium and its alloys which exhibit high strength, low density, good rigidity, good degradability but poor corrosion resistance.
   b) Inorganic biomaterials and bio-ceramics like hydroxyapatite and ß-tricalcium phosphate which are biocompatible.
2. Organic materials
   a) Natural polymers like Gelatin, Collagen, Hyaluronic acid, Chitosan, Poly (lactic-co-glycolic acid) (PLGA), and Polycaprolactone (PCL) which are biocompatible, non-immunogenic, and biodegradable.

3D Printing methods commonly employed for fabrication of bones arefusion deposition modelling, stereo lithography, and selective laser sintering.

In laser methods like stereo lithography and selective laser sintering, the laser is used to locally melt the material powder and the three-dimensional parts were formed through layer-by-layer stacking. Post-printing, 3D material is processed to obtain implantable bone graft. These methods can print parts with complicated shapes. The surface quality of the parts is good with high mechanical properties. However, the equipment is large in size, high in energy consumption, requires a special environment and is high in cost.

In the fusion deposition modelling method, the material is melted by heating in the spray head. The spray head moves along three directions and extrudes the melted material at the same time, stacking layer by layer. In this method the equipment volume is relatively small, simple to use, low in cost and does not need a special environment. However, the surface of the printed product is relatively rough, and the printing speed is slow.

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3D PRINTING OF SKIN

Having healthy skin is really important for a healthy life. When skin is damaged due burns, accidents, non-healing cuts, and deep wounds, it needs treatment. Conventional treatment of skin injuries is through transplantation of the skin taken from other part of patient or from donor.

3D printing is new technology by which skin required for transplant can be created easily in lesser time and cost. It is used to reconstruct the injured skin layer by layer by taking inputs from computer-aided design (CAD) model. Different biomaterials and cells are used as starting materials for 3D bio-printing of skin.

Human skin is one of the largest organs which can sense pressure, temperature and pain. It is made up of several types of cells and has multiple layers with a well-defined structure. There are three layers in the skin:

a) Epidermis: This is the outermost layer of the skin which provides a waterproof barrier and makes our skin tone.
b) Dermis: It contains connective tissue, sweat glands and hair follicles.
c) Hypodermis: This is a deeper subcutaneous tissue that is made of fat and connective tissue.

Therefore, to create a skin-like part by 3D bio-printing, a bio-ink is carefully formulated using specific biomaterials and multiple cells types. Bio-ink used is a mixture of hydrogels of natural polymers like fibrin, collagen and alginate. It also contains perfect mixture of proteins, cells and other biological components. Bio-ink must be biocompatible which accommodate live cells. It should provide high resolution and mechanical stability during printing.

Inkjet 3D printing and laser assist 3D printing methods are commonly used for skin printing. Steps involved in in situ bio-printing of skin is given below:

a) In the first step, markers are placed around the skin damaged area as reference points and then imaging of skin damaged area is created using different scanning technologies like CT, MRI and 3D scanners.

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b) This data is converted into 3D digital model using different software likemimics, 3D slicers etc. and create an accurate 3D model of the damaged skin.
c) Then bio-ink is prepared using required biomaterials and cells.
d) Output code is then provided from the computer to the custom bio-printer to print skin.
e) Printed skin is post-processed, which involves tissue maturation in a bioreactor and is finally tested.
f) This tested skin is easily implanted in the patient body.

3D PRINTED FOOD

3D printed food refers to the edible items created using 3D printing technology. Pizza vending machines that entered into market in the year 2015 can be considered as a primitive 3D food printing process. In this machine, the dough is prepared, extruded, topped with tomato sauce and cheese, and finally sent to the oven – all within the same machine.

Now, several 3D food printers are available in the market and in future every household kitchen will be equipped with its own food 3D printer.

In 3D printing of food items, 3D printers and food-safe materials are used to create intricate and customized food items layer by layer. There are many advantages and potentials of this 3D technology:

a) In 3D printing, shape, texture, and composition of food can be precisely controlled. This enables chefs and food manufacturers to create unique, customized and personalized dishes. For example, chocolates can be created with intricate and detailed shapes, such as flowers, animals, and even people. 3D printed desserts can be made with different types of cakes, cookies, and ice cream. 3D printed candy can be made with different types of sugar, chocolate, and nuts.

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b) The nutritional content of foods can be adjusted to meet specific dietary requirements.
c) Chefs and culinary artists can use 3D printers to craft visually stunning and intricate food designs which are not possible with traditional cooking methods.
d) Food can be produced as per the required quantity and hence, food waste can be reduced.
e) Alternative Ingredients: 3D printing could facilitate the use of alternative and sustainable ingredients, such as insect-based proteins or algae, which might be less palatable in their original forms.

The most commonly used process for 3D printing food material is extrusion method like fused deposition modelling. This method requires inputs in the form of paste and viscous liquids. The raw material is fed into a printer container with syringe-like nozzle. The viscous material is deposited onto a surface layer by layer to create a final 3D food item.

3D Food printers do not actually cook the ingredients. They are mostly used for architecting intricate shapes and designs with computer aided 3D sketches. In some cases, the resulting product can be consumed directly or it is cooked in an external oven (or grill) before consuming.

Main applications of 3D printing of food are:

a) 3D printed foods like chocolates, pastries, pasta, candies, and even more complex items like pizzas or burgers can be designed in varieties of design, shape, and structure.
b) It is used for gourmet dining and in fancy bakeries. Gourmet dining is a type of dining that emphasizes high-quality ingredients, creative dishes, and impeccable service. The dishes are often presented in an artistic and visually appealing way.
c) It is used to design plant-based meat called meat analogues, mimicking the texture of the real meat.

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Technology of 3D printed food is still in its early stages and faces several challenges. With active research and development in future technology will be more accessible.

ELECTRONIC TONGUE (e-TONGUE) AND ELECTRONIC NOSE (e-NOSE) IN FOOD SCIENCE

Electronic tongue and electronic nose are electronic sensing devices used to taste and smell the food items respectively. They mimic the taste and smell functions of human tongue and nose. These devices use number of sensors to detect and identify various compounds in food and beverages.

The human nose can identify many odours. Each type of odour can be due to individual chemical components present. An electronic nose is a system designed to mimic the human sense of smell. It consists of an array of sensors that respond to volatile organic compounds (VOCs) released by foods, beverages, or other substances. Each sensor has a detector, which interacts selectively with chemical component and produces an electrical signal. Produced signal is processed by a signal processor. By analysing the patterns of sensor responses, an electronic nose can identify and classify different odours.

An electronic tongue consists of an array of sensors and can detect chemical compounds related to taste such as sweet, salty, sour, and bitter. In principle, "electronic tongues" function in a similar way to the "electronic nose". When a food or beverage sample is introduced to the electronic tongue, the sensors generate specific responses based on the taste compounds present. These responses are then processed to create a "taste profile" that can be used to assess the quality, authenticity, and composition of the sample.

There are many advantages with electronic tongue and nose. They are non-destructive, which means that they do not damage the food being analysed. They are also small in size, rapid and sensitive, which makes them ideal for real-time monitoring of food quality. And also, they can be used to analyse a wide variety of food products.

Due to these sensing abilities electronic tongue and nose find applications in the food industry for the food quality control, food safety assessment, and in the development of new food products. The main applications of electronic tongue and nose are given below:

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1. Assessment of quality of food products, such as milk, wine, and meat based on taste and aroma profiles.
2. Detection of adulteration in food products.
3. Detection of spoilage or contamination in perishable products.
4. Monitoring the ripeness or spoilage of fruits and vegetables.
5. Monitoring the freshness of packaged goods.
6. Determining the authenticity and origin of food products.
7. Optimizing flavour formulations in the food and beverage industry.
8. They can be used to detect the presence of foodborne pathogens which helps to prevent food poisoning outbreaks.
9. They can be used to develop new food products with specific flavor profiles. This information can be used to create more appealing and nutritious food products.
10. They can be used in analysis and quality control for determining the nature and extent of food adulteration.

Both electronic tongues and noses are still under development, but they have the potential to revolutionize food science in future.

DNA ORIGAMI

"Origami" refers to a traditional Japanese art form in which various shapes and designs are created just by folding paper without using cuts or glue. The word "origami" comes from the Japanese words "ori," meaning "fold," and "kami," meaning "paper." By origami technique range of materials from simple animals and objects to intricate and complex designs can be created. Origami is also used as a way to study geometry and spatial relationships. Origami has a rich history and cultural significance in Japan, and it has gained popularity worldwide as a recreational activity, educational tool, and even a source of inspiration for various fields, including science and engineering.

In recent years, the concept of origami has been extended beyond paper to other materials and scales. This has led to the development of new field like "DNA origami."

DNA origami is a modern application of the origami concept. In this technique, similar to folding of paper, DNA strands are folded to create different nanostructures.

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Majority of biological materials present in living cells are nanoscale materials. They are synthesized by a well guided self-assembly of smaller atoms and molecules. This process of building a nanoscale material by self-assembly of atoms and molecules is called as bottom-up synthesis. There is a well-defined order and precision in the self-assembly. However, in the synthetic methods used to prepare nanoscale materials, such a precision and controllability is not possible.

DNA origami technology is an effective DNA nanotechnology technique used in bottom-up fabrication of well-defined nanostructures. In this technique, DNA strands are self-assembled at nanoscale to create complex and precise 2D and 3D nanostructures.

In DNA origami, a long DNA strand and hundreds of short DNA strands are folded to create nanostructures. A long DNA strand contains around 7,000 nucleotides and is called as scaffold. Short DNA strands are called as staples. Scaffold is folded into specific shapes by attaching staple strands at specific positions along the scaffold. Staple strands guide the folding process. Each staple has multiple binding sites that bind and bring together distant regions of the scaffold. This causes the scaffold to bend and twist into the desired shape. The geometries of the resulting structures can be programmed with the staple sequences. Computer-aided design with universal synthesis protocols can be used to automatically fabricate well-defined nanostructures by DNA origami. caDNAno (also written as **CADnano)** is the first- generation (1G), most routinely used DNA origami design software.

DNA origami objects can be fabricated in varieties of dimension, geometry and shape. Starting from simple single layer structure to multilayer structures, and flexible wireframe structures to rigid polyhedral structures with user-defined asymmetry, cavities and curvatures can be fabricated.

A typical planar DNA origami structure with approximately 200 staples with unique sequences and positions, can exhibit surface area of 8,000–10,000 nm^2. Various types of materials can be placed on these high surface area, DNA origami structural frameworks at specific locations. Materials like metal particles, silica, lipids, polymer

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coatings, and nano systems for nano-photonic and nano-electronic devices can be placed over DNA origami structures. These engineered dynamic DNA origami structures find various innovative applications. Few of their applications are given below:

1. **Nanofabrication:** DNA Origami structures have well defined geometry at nano scale. These have been used as templates or frameworks for the assembly of diverse nano materials.
2. **Nano-photonics and nano-electronics:** DNA origami- templated nanostructures exhibit tuneable optical or electronic properties. Hence, they find applications in nano-photonicsand nano-electronics.
3. **Bio-catalysis:** DNA origami nanostructures provide an excellent platform for spatially organizing enzymes which can catalyse many bio-chemical reactions.
4. **Computing:** DNA is an information-carrying molecule. Hence, DNA origami structures are used for creating molecular computation and signalling networks.

DNA BIOCOMPUTING

In the present day electronic computing, digital circuits composed of transistors which are made of silicon semiconducting materials are used. Information is processed and stored in binary states (0s and 1s) using electrical voltage. Computation and data storing is based on the movement of charge carriers (electrons) from one state to other state under the influence of external field. The size of the transistors used in electronic computing is very close to minimum level and further size reduction is not possible. Therefore, practically it is not possible to further reduce size of the computer and increase its processing speed. Hence, a new way of computing is required to further reduce size of the computer and improve its speed. One very promising approach is the bio computing device using deoxyribonucleic acid (DNA). The actual DNA computing work started with the storing of data on strands of DNA in 1990s with Leonard Adelman presented the first prototype of a DNA computer, called the Testtube-100 (TT-100). In 2019, researchers from Microsoft and the University of Washington

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demonstrated the first fully automated system to store and retrieve data in manufactured DNA. They have managed to store one gigabyte in DNA.

Deoxyribonucleic acid (DNA) is the biomolecule which carries genetic information. In DNA, genetic coding is represented by four nucleotide bases: adenine (A), thymine (T), cytosine (C), and guanine (G). When chained together, these four bits can hold an incredible amount of data. The entire human genome is encoded in a very small space and packed into a single nucleus of a cell. Hence, DNA can be utilized for computation due to its ability to store and process information.

In DNA Bio-computing the complementary base pairing of nucleotides (adenine with thymine, cytosine with guanine) is used to represent binary information (0s and 1s). Operations in DNA computing involve encoding and manipulating information using DNA strands and biochemical reactions. DNA strands are used to encode the input, output, and intermediate data for the computational process. The sequences of nucleotide bases in the DNA strands represent specific pieces of information. The basic mechanism of DNA bio-computing is the strand displacement reaction. In a strand displacement reaction, a single-stranded DNA molecule (the "input strand") binds to a double-stranded DNA molecule (the "target strand"). The input strand then displaces one of the strands of the target strand, forming a new double-stranded DNA molecule. This process is similar to the base pairing of A-T and C-G in DNA replication and transcription. This strand displacement reaction can be used to perform a variety of logic operations. For example, if the input strand has a sequence that matches a specific sequence on the target strand, then the input strand will only be able to displace that strand. This can be used to create a NOT gate, which inverts the input signal.

The main advantages of DNA computing are:

1. It is cheaper technology because DNA required for computing will be inexpensive to create.

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2. DNA is easy to synthesize. DNA naturally wants to reproduce, by harnessing this natural tendency in an artificial environment, DNA required for bio-computing can be produced easily.
3. DNA can perform countless calculations in parallel. DNA can perform numerous operations simultaneously due to the parallel nature of biochemical reactions. While, number of parallel computations that can be made by electronic computing is limited, DNA computing has almost no limit. This makes it ultrafast and incredibly powerful for scenarios like machine learning.
4. DNA molecules can store vast amounts of information in a small space. Hence, they have very high data storage density. Hence, size of the computer and data storage devices can be reduced.
5. DNA reactions occur at the molecular level, requiring minimal energy. Hence, they are highly energy efficient.
6. DNA computing has the potential to solve for complex problems compared to traditional computers.

There are few limitations and challenges which need to be addressed before shifting from present day electronic computing to DNA Biocomputing. DNA computing is still a developing field with no universally accepted standards or practices. Designing and programming of DNA-based systems is currently more complex than traditional programming. DNA reactions occur at a slower pace compared to electronic computation, and hence need longer processing times. And also, biological processes are susceptible to errors, which can lead to inaccuracies in computation.

BIOIMAGING AND ARTIFICIAL INTELLIGENCE FOR DISEASE DIAGNOSIS

Bioimaging techniques are used to diagnose diseases by visually examining biological structures and processes within living organisms. These techniques provide detailed insights into the anatomical and functional aspects of tissues and organs. These are used by doctors to identify abnormalities, conditions, and diseases. Various

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imaging modalities are employed to capture detailed information about tissues, organs, and cells. Few of the commonly used bioimaging techniques and their applications are briefly given below:

1. **X-ray Imaging:** This technique uses X-rays to create images of the internal structures of the body, commonly used for detecting bone fractures and lung issues.
2. **Angiography:** Angiography is utilized to discover heart artery blockages, new vessel development, and arrangement of catheters and stents.
3. **Dual X-beam Absorptiometry (DEXA):** It is also called as bone densitometry which is utilized for osteoporosis tests.
4. **Computed Tomography (CT):** CT scans involve the use of X-rays to create cross-sectional images of the body, providing detailed insights into various delicate and hard tissues.
5. **Magnetic Resonance Imaging (MRI):** MRI utilizes strong magnetic fields and radio waves to generate detailed images of soft tissues, such as the brain, muscles, and organs.
6. **Ultrasound:** Ultrasound imaging uses high-frequency sound waves to create images of internal body structures. It is often used for visualizing pregnancies and abdominal organs.
7. **Positron Emission Tomography (PET):** PET scans involve the injection of a radioactive tracer to visualize metabolic processes within the body, helping in cancer detection and neurological studies.

Doctors use bio-imaging techniques to diagnose diseases. After image acquisition, the doctor interprets the images to identify any abnormalities. This process involves careful observation of the structures and tissues depicted in the images. The doctor uses their expertise to recognize patterns, anomalies, and any indications of disease. They compare the images to their knowledge of normal anatomy and known pathological conditions. They look for specific patterns, such as the appearance of

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tumors, inflammation, blockages, or structural abnormalities. Based on the observed patterns and findings, doctors create a list of possible diagnoses. This is known as a "differential diagnosis," where multiple potential conditions are considered. Doctors combine the imaging results with the patient's clinical history, symptoms, and laboratory test results to narrow down the list of possible diagnoses. In complex cases or when a definitive diagnosis is challenging, doctors may consult with other medical specialists or radiologists to gain additional insights and expertise. Once a diagnosis is made or strongly suspected, the doctor develops a treatment plan. This plan may involve further tests, procedures, or treatments tailored to the specific disease or condition.

There are several drawbacks in this manual interpretation of bioimaging results by doctors in disease diagnosis like:

1. Interpretation of bioimaging results may vary among different radiologists or medical professionals. This can lead to inconsistencies in diagnosis and treatment recommendations.
2. Even experienced radiologists can make errors. They may miss subtle details in complex images. This can lead to misdiagnosis and delayed treatment.
3. Analyzing complex bioimaging data can be time-consuming, especially for large datasets. This can lead to delays in diagnosis and treatment.
4. Specialized expertise in certain types of imaging may not be readily available in all healthcare facilities, particularly in remote or underserved areas.

Majority of these problems associated with disease diagnosis through manual interpretation of complex bioimaging data can be overcome by using new technology that is Artificial Intelligence (AI).

Artificial Intelligence with machine learning and deep learning, has revolutionized disease diagnosis by processing and analyzing large volumes of medical data with exceptional accuracy. AI technologies can play a crucial role in enhancing the

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accuracy, speed, and accessibility of disease diagnosis, ultimately leading to improved patient outcomes. Benefits of AI in disease Diagnosis are:

1. **Standardized Interpretation:** AI and deep learning algorithms are trained on large datasets of medical images. During training, the AI learns to recognize patterns associated with different diseases or conditions. Hence, they provide standardized and consistent interpretations of bioimaging results. This can eliminate variability in interpretation of results among radiologists and doctors.
2. **Speed and Efficiency:** AI can process images much faster than humans. This reduces diagnosis time leading to quicker treatment decisions.
3. **Error Reduction:** AI can help identify subtle details and anomalies that might be missed by human eyes, thus reducing human errors in diagnosis.
4. **Pattern Recognition:** AI can analyze vast amounts of imaging data to identify complex patterns that might not be apparent to human observers. This can aid in disease detection in early stages.
5. **Remote Consultations:** AI-powered systems can be used in healthcare facilities in remote or underserved areas.
6. **Assisted Diagnosis:** AI can highlight potential areas of interest, anomalies, or patterns in the images. This is useful in more accurate and efficient diagnosis.
7. **Predictive Analytics:** AI can analyze historical patient data along with bioimaging results to predict disease progression, treatment responses, and potential outcomes.
8. **Continuous Learning:** AI systems can continuously learn from new data and update their algorithms, improving accuracy over time.
9. **Resource Optimization:** AI can help prioritize and allocate resources by identifying cases that require immediate attention, thus improving patient care and workflow efficiency.
10. **Data Integration:** AI can integrate bioimaging data with other patient information, such as electronic health records, to provide a comprehensive view of the patient's health. It provides data-driven insights for personalized treatment planning.

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Some of the examples of using of AI technology in disease diagnosis are:

a) AI-powered algorithms are being used to identify breast cancer on mammograms with a higher degree of accuracy than human radiologists.
b) AI is being used to detect diabetic retinopathy using data of eye scan. This is the main leading cause of blindness.
c) AI is being used to identify Alzheimer's disease in brain scans.
d) AI is being used to predict the risk of heart attack and stroke.
e) AI is being used to personalize cancer treatment for individual patients.

These are just a few examples of how AI is being used for disease diagnosis today. As AI technology continues to develop, we can expect to see even more innovative and effective ways to use AI to diagnose diseases and improve patient outcomes.

Even though AI technology possesses so many advantages, there are few drawbacks:

a) AI may struggle with complex cases that require a deep understanding of medical context.
b) AI lacks the clinical judgment, empathy, and understanding that doctors possess.
c) AI's performance heavily relies on the quality and diversity of the training data.

Hence, AI technology cannot be used as a replacement for doctors. However, it can complement their expertise and helps them to enhance patient care. Doctors and AI can collaborate to enhance disease diagnosis using bio-imaging techniques. AI aids doctors by providing efficient and accurate analyses of medical images, enabling more timely and precise treatment decisions.

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SELF-HEALING BIOCONCRETE (BIOMINERALIZATION PROCESSES BASED ON BACILLUS SPORES, AND CALCIUM LACTATE NUTRIENTS)

Concrete is a common construction material, used in the construction of buildings, dams, storage tanks, seaports, roads, bridges, tunnels, subways and other infrastructures. It is prepared by mixing appropriate amount of cement, water, and aggregates (coarse and fine). Cement binds the aggregates and fills the voids between coarse and fine particles. Concrete as a material is known for its availability, simple preparation, low cost, high compressive strength, durability, compatibility with aggregates and iron bar.

One of the major disadvantages of concrete is its high tendency to form cracks. Cracks are one of the main causes of concrete deterioration and decrease in durability. Cracks are formed in hardened states due to weathering, drying shrinkage, thermal stress, error in design and detailing, chemical reaction, constant overload and external load.

Cracks formation can be a serious risk to concrete lifespan in the long term. The direct cost of crack repair and maintenance has been estimated to be higher than that of concrete production. Therefore, it is essential to prevent crack formation at early stage.

There are two types of treatments used to fill cracks:

1. **Passive treatment:** In this treatment, cracks are detected and sealants in the form of chemical mixtures and polymers are either injected or sprayed into the cracks. These can heal only the surface cracks. These sealers usually comprise chemical materials such as epoxy resins, chlorinated rubbers, waxes, polyurethane, acrylics and siloxane. Chemical sealers suffer from major limitations like poor weather resistance, moisture sensitivity and low heat resistance, unsustainability.
2. **Active treatment:** Active treatment techniques are also known as self-healing techniques. In this concept, certain microorganisms and nutrients are incorporated

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into the concrete mixture during construction. After the construction, over a period of time, when a crack is formed, self-healing is activated immediately and the concrete material to repairs itself. This type of concrete which is capable of self-healing itself through biological process is called as self-healing bio-concrete. It is an innovative and sustainable solution in the field of construction materials. It integrates biological processes into concrete to enhance its durability and longevity.

Among all the techniques available for healing cracks in concrete, biological healing process is the most promising way. Calcium carbonate is one of the most suitable fillers to seal cracks in the concrete. It is similar to cement and hence has high compatibility with concrete. In bio-mineralization process, bacteria consume calcium source and produces calcium carbonate as the metabolic product. There are many bio-mineralization processes available to produce calcium carbonate using different calcium sources, bacteria and conditions.

One common method is metabolic conversion of organic compound such as calcium lactate to calcium carbonate using Bacillus bacteria. Bacillus bacteria, particularly Bacillus subtilis, are commonly used in self-healing bio-concrete. These bacteria are known for their ability to form spores, which are highly resistant to harsh environmental conditions. These bacteria consume organic compounds as a source of energy for their growth.

In self-healing bio-concrete, Calcium lactate and Bacillus subtilis spores are mixed with concrete mixture. Over the period of time, the concrete may develop micro-cracks due to factors such as shrinkage, thermal expansion, or external forces. Water enters the cracks and reactivates the dormant Bacillus spores. These bacteria then start consuming the calcium lactate nutrients and oxygen. Aerobic oxidation of calcium lactate leads to the production of calcium carbonate and carbon dioxide as metabolic products. Metabolic conversion of calcium lactate to calcium carbonate in the presence of oxygen is shown in the below equation.

$$Ca(C_6H_{10}O_6) + 6\,O_2 \rightarrow CaCO_3 + 5\,CO_2 + 5\,H_2O$$

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The produced calcium carbonate functions as a mineral binder. It starts to fill the cracks and gaps within the concrete structure. This bio-mineralization process helps to repair the cracks and restore the concrete's integrity. Water produced in the reaction can react with available calcium oxide in the concrete matrix and can further contributes to the increase of autogenous healing of concrete. The healing process can take time, depending on the size and extent of the cracks. Smaller cracks can be healed relatively quickly, while larger cracks might take longer to repair.

The main advantage of this method is consumption of oxygen by bacteria and production of calcium carbonate without formation of ammonium, or chloride ions. Chemicals like oxygen, ammonium, and chloride are capable of penetrating into concrete and can cause corrosion of reinforcement iron bars. Thus in this method, cracks in the concrete are healed by calcium carbonate produced in higher concentration without causing corrosion of reinforcement iron bars.

There are several benefits of using self-healing bio-concrete:

a) Self-healing concretes can improve the overall durability and longevity of concrete structures.
b) Self-healing concretes reduce the cost of frequent repairs and maintenance.
c) Self-healing concretes repair the cracks automatically at an early stage, and prevent any accidents due to sudden collapse of concrete structures.

BIOREMEDIATION via MICROBIAL SURFACE ADSORPTION (REMOVAL OF HEAVY METALS LIKE LEAD, CADMIUM, MERCURY, ARSENIC)

Heavy metals like lead, cadmium, mercury, and arsenic are known to be toxic to humans and the environment. These metals have ability to accumulate in living organisms, and cause adverse health effects even at low concentrations. They can cause neurological damage, cardiovascular diseases, kidney dysfunction, reproductive problems and cancer. Human activities, such as industrial processes, mining, and

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improper waste disposal, are responsible for release of these metals into the air, water, and soil. From there, these pollutants can then enter the food chain, affecting human health and ecosystems.

Efforts are being made to prevent the release and remove of heavy metals. Few of them are:

a) Imposition of strict regulations on emissions.
b) Development of proper waste disposal practices.
c) The development of technologies for removing heavy metals from contaminated sites.

Among the various technologies that have been developed to remove heavy metals from contaminated sites, bioremediation is an important one. Bioremediation means clean-up of polluted environment through transformation of toxic heavy metals into less toxic form using microorganisms. Bioremediation is used to remove toxic heavy metals from contaminated soil, groundwater, and surface water.

Microbial surface adsorption is one of the commonly used techniques for bioremediation. In microbial surface adsorption, certain microorganisms like bacteria, fungi, or algae which exhibit specific affinity towards heavy metal are chosen. Some of the microorganisms that have been shown to be effective in removing heavy metals through microbial surface adsorption are:

a) Bacteria: Bacillus subtilis, Pseudomonas aeruginosa, and Escherichia coli.

b) Yeast: Saccharomyces cerevisiae and Candida albicans.

c) Fungi: Aspergillus niger and Penicilliumchrysogenum.

The selected microorganisms are introduced to the contaminated site where they grow and colonize. Then, they bind the heavy metals on to their cell surfaces. Binding between metal and cell surface can be through a variety of mechanisms as described below:

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i) The bio-sorption of heavy metals is mainly achieved through cell wall functional groups. The cell wall is negatively charged as a result of functional groups such as phosphate, carboxyl, carbonyl, sulfhydryl and hydroxyl groups. These can bind toxic metals to form insoluble substances.

ii) Cations such as Na^+ and K^+ present in cell walls of microorganisms can exchange toxic heavy metals like Lead, Cadmium, Arsenic and Mercury on the outer layer.

iii) The cell wall is composed of organic macromolecules, such as polysaccharides, polypeptides and proteins, which can adsorb heavy metals via electrostatic forces, Van der Waals' forces, and covalent bonds.

iv) Heavy metals may get deposited on the micro-fibrous porous structure of the cell wall and embedded in the cell wall before entering the cell.

Removal of heavy toxic metals from source by bioremediation occurs through combination of more than one mechanism described above. The heavy metals adsorbed on the surface of microorganisms are either removed along with the microorganisms or are stabilized on the microorganism surfaces, reducing their mobility and potential for harm.

The microbial surface adsorption is particularly effective for heavy metal removal from soil and water, as well as for treating industrial wastewater contaminated with various pollutants. However, it might be limited by factors such as the availability of suitable microorganisms and the concentration of contaminants in the environment.

BIO-MINING

In traditional mining, ores containing metals are physically extracted from the Earth's crust and then metals are extracted from ores through various chemical processes. Traditional mining procedures are not environmental friendly. They result in deforestation, soil erosion, habitat destruction, water pollution, and the release of greenhouse gases and other waste materials.

Bio-mining is a biotechnological process in which microorganisms such as acidophilic bacteria or archaea are used to extract metals from their ores. These

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microorganisms are capable of breaking down and solubilizing metal compounds in the ores. Examples for the acidophilic bacteria are acidithiobacillus, acidiphiliumspp, At. thiooxidans etc. Examples for the archaea are ferroplasmaacidiphilum, acidiplasmacupricumulans, metallosphaeraspp etc.

Bio-mining is also usedto process electronic wastes, and to selectively recover metals from waste water.

Bio-mining is used mainly to extract copper from their copper sulfide ores. It is also commonly used in extraction and recovery of other base metals such as gold, silver, cobalt, nickel and zinc. About 15% of copper, 5% of gold and smaller amounts of nickel, silver, cobalt and zinc are produced globally using bio-mining technology. Bio-mining is also used to extract metals from low-grade ores that would not be economically viable to mine using traditional methods.

In contrast to traditional mining, bio-mining is much more environmentally benign ('green') approach. It operates at atmospheric pressure and at normal temperatures (20–50°C) and hence at lower cost. Microorganisms used in bio-mining processes are autotrophs i.e. they fix carbon dioxide, much in the same way as green plants.

Main steps involved in bio-mining process are:

1. In the first step, the mineral ore is crushed into smaller particles. This increases its surface area and makes it more accessible to microbial interactions.
2. Specific microorganisms like acidophilic bacteria or archaea, are introduced to the crushed ore.
3. These microorganisms grow and colonize in the medium. Then, they adhere to the surface of the mineral particles. They use their metabolic processes to oxidize the minerals and generate acid as by-product.
4. The acid generated by the micro-organisms dissolves ores and releasing metal ions into the surrounding solution.

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5. The metal-rich solution is known as a leachate. Various separation and purification techniques are used to process and recover valuable metals from leachate.

The adhesion of microbes on to the mineral surface which is called as microbial surface adsorption is the main step in bio-mining. Microbial cells can form biofilms on the mineral particles, creating a protective environment where they can efficiently carry out their metabolic activities. Acids produced as metabolic by-products of the microorganisms, aid in the breakdown of minerals and subsequent release of metals in to solution.

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